LS-CAT 21-ID

Final Design Report

Life Sciences Collaborative Access Team

Synchrotron Research Center

Northwestern University

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Table of Contents

INTRODUCTION **1**

1. LAYOUT UPDATE **2**

Criterion 1.1: Layout CAD drawings 2

Criterion 1.2: Changes from PDR layout 3

Criterion 1.3: Survey and alignment plan 4

Criterion 1.4: Utility distribution 4

Criterion 1.5: Ray tracings 4

2. COMPONENT DESIGN UPDATE **4**

Criterion 2.1: Drawings 4

Criterion 2.2: Vacuum Compliance 6

Criterion 2.3: Beamline Component Designs 7

3. SCHEDULE **7**

Criterion 3.1: Installation 7

Criterion 3.2: APS Procurement 7

Criterion 3.3: APS Support 7

4. SAFETY **7**

Criterion 4.1: PSS 7

Criterion 4.2: BLEPS 10

Criterion 4.3: Shielding 11

Criterion 4.4: Ozone 11

Criterion 4.5: Other Hazards 11

Criterion 4.6: Fire Hazards 12

5. SPECIAL OPERATING REQUIREMENTS UPDATE **12**

Criterion 5.1 12

APPENDIX A: LS-CAT 21-ID LAYOUT DRAWINGS **13**

Enclosure Dimensions 15

Beamline Components 17

Plan and Elevation Phase I 19

Plan and Elevation Phase II 21-ID-E 21

Plan and Elevation Phase II 21-ID-F 23

Plan and Elevation Phase II 21-ID-G 25

Electrical Distribution, Ground Level 27

Electrical Distribution, Roof 29

Cable Trays and Raceways, Interior 31

Cable Trays and Raceways, Exterior 33

Labyrinths 35
PSS 37
Guardrails and Stairs 39
Cranes 41
BLEPS 43
APPENDIX B: PHASE I COMPONENT LOCATIONS 45
APPENDIX C: SURVEY TARGETS 51
APPENDIX D: RAY TRACINGS 53
Horizontal Bremsstrahlung 55
Vertical Bremsstrahlung 57
Horizontal Synchrotron 59
Vertical Synchrotron (Inboard) 61
Vertical Synchrotron (Outboard) 63
APPENDIX E: COMPONENT DRAWINGS 65
Bremsstrahlung Collimator 1 67
Inboard ID Fixed Mask 69
Outboard ID Fixed Mask 71
Inboard ID Apertures 1 & 2 73
Sharma Beamstop 75
K5-30 Collimator 77
Bremsstrahlung Collimator 2 79
Outboard ID Apertures 1 & 2 81
Kohzu Monochromator 83
P4-60 Shutter 85
P4-40 Shutter 87
Beryllium Window 89
K5-50 White Beam End Stop 91
P8-50 Monochromatic Shutter 93
APPENDIX F: IDT MASKS – HEAT LOAD CALCULATION 95
APPENDIX G: IDT MASKS – COOLANT FLOW CALCULATION 101
APPENDIX H: IDT MASKS – FINITE ELEMENT ANALYSIS 107
APPENDIX I: WORK BREAKDOWN SCHEDULE 115
APPENDIX J: BLEPS REQUIREMENTS DOCUMENT 127
APPENDIX K: SHIELDED BEAMPIPE DRAWINGS 135
Shielded Pipe Stubs 137
Shielded Transport Pipes 139
Shielded Transport Flange Covers 141
APPENDIX L: BERYLLIUM WINDOW COATING 143

Table of Figures

Figure 1: Schematic of LS-CAT 21-ID layout....2
Figure 2: Water Circuit 1 (21-ID-B) 9
Figure 3: Water Circuit 2 (21-ID-C) 9
Figure 4: Water Circuit 3 (21-ID-C) 9
Figure 5 : Water Circuit 4 (21-ID-D) 9

Table of Tables

Table 1: LS-CAT 21-ID Layout Drawings....2
Table 2: LS-CAT ID Build-out Phases....3
Table 3: Drawing numbers of supplied ray tracings 4
Table 4: Components, locations, and drawing numbers....5
Table 5: U33S (Phase I) Operational Limits for IDT Masks and Apertures 6
Table 6: U30S (Phase II) Operational Limits for IDT Masks and Apertures....6
Table 7: PSS Shutters 8
Table 8: PSS Logic, Front End Shutter 8
Table 9: PSS Logic, Station Shutters....8
Table 10: Shielded Pipe Drawing Summary....11

Introduction

The Life Sciences Collaborative Access Team (LS-CAT) is a consortium of academic and research institutions comprised of Michigan State University, Northwestern University, the University of Michigan, Wayne State University, the Van Andel Institute, the University of Illinois at Urbana-Champaign, Vanderbilt University, and the University of Wisconsin at Madison. LS-CAT has selected Northwestern University as the managing agent to design, build, instrument, and operate macromolecular crystallography facilities at Argonne National Laboratory's Advanced Photon Source.

The investigators participating in the LS-CAT are involved in cutting-edge structural biology/macromolecular crystallography with special emphasis in four major areas:

1. membrane proteins
2. multi-component systems in cellular signaling and protein transport
3. the protein/nucleic acid complexes involved in genome replication, regulation, and expression
4. enzymes that are novel targets for mechanistic studies and for biotechnology or drug design.

Many of the researchers associated with LS-CAT have already reached the technical barriers of small crystal size (maximum dimensions of 20-80 microns) or large asymmetric units (300-500 kDa), which make access to a dedicated synchrotron source essential to their scientific progress. Crystals of membrane proteins and of many protein-nucleic acid complexes are notoriously weak diffractors that can only be studied using very intense x-ray sources.

The beamlines incorporate the 1 milliradian canted undulator design. The general approach is to use the outboard undulator (a 3.3 cm device) for a fully tunable end station known herein as the Phase I beamline. The inboard undulator (a 3.0 cm device) will be exploited similar to the Quadriga beamline concept at the ESRF known herein as the Phase II beamline. The two most upstream beamlines will be fixed wavelength facilities using semi-transparent diamond monochromators as their monochromator crystals. The final beamline will be a moderately tunable station using a large offset monochromator system with a single Bragg deflecting crystal. A cartoon schematic of the monochromator deflections is given in Figure 1. Currently, funding has been secured to build out the entire insertion device beamline. Although the bending magnet beamline is currently descoped, ample space has been left in the layout for its development.

For ease of reference, this document follows the outline for the review criteria given in ANL/APS/TB14, Section 6: Criterion 1.1 through 5.1 appear here as section headings.

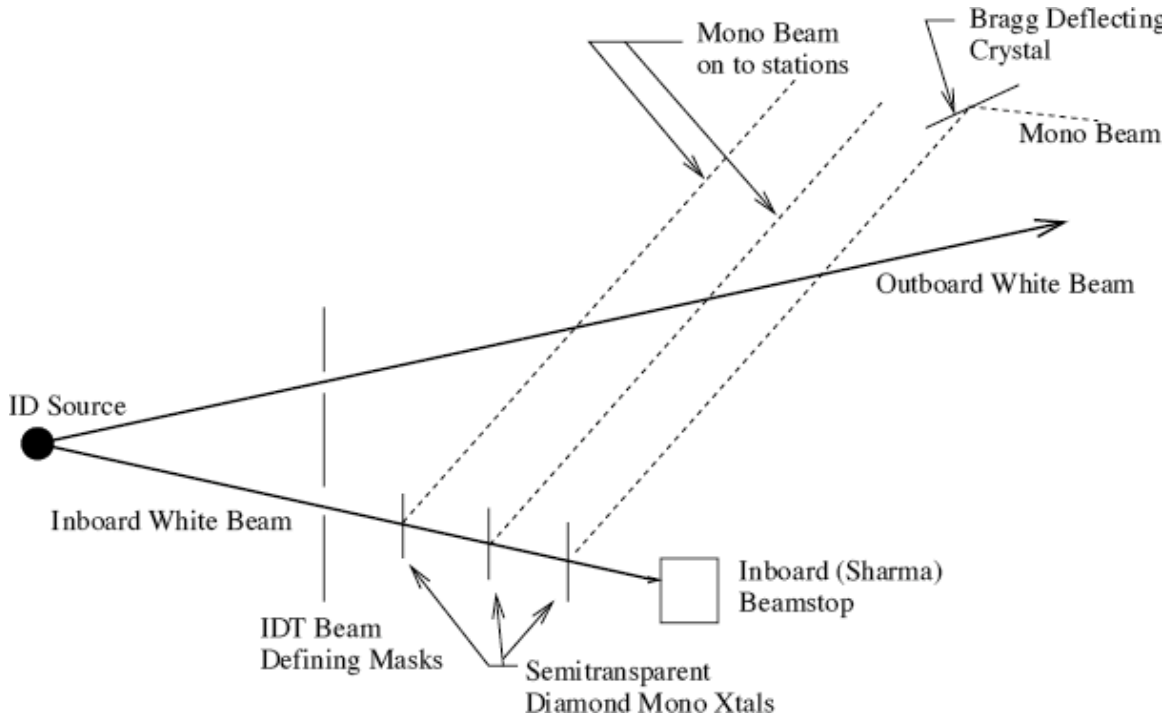


Figure 1: Schematic of LS-CAT 21-ID layout

1. Layout Update

Criterion 1.1: Layout CAD drawings

Plan and elevation layouts of the beamline with clearly identified beamline components, beamline support facilities, and aisle ways have been provided to the Intralink Commonspace in lieu of being uploaded to the Design Exchange as required by ANL/APS/TB-14. These layouts can be found in Commonspace/LS_CAT/AutoCAD Layouts as shown in Table 1 and are reproduced in Appendix A.

Table 1: LS-CAT 21-ID Layout Drawings

Drawing	Description
21_1-5_100000_08.dwg	Enclosure Dimensions
21_1-5_200000_09.dwg	Beamline Components
21_1-5_210000_09.dwg	Plan and Elevation Phase I
21_1-5_220000_09.dwg	Plan and Elevation Phase II 21-ID-E
21_1-5_230000_09.dwg	Plan and Elevation Phase II 21-ID-F
21_1-5_240000_09.dwg	Plan and Elevation Phase II 21-ID-G
21_1-5_310000_08.dwg	Electrical Distribution, Ground Level
21_1-5_320000_08.dwg	Electrical Distribution, Roof
21_1-5_410000_08.dwg	Cable Trays and Raceways, Interior

21_1-5_420000_08.dwg	Cable Trays and Raceways, Exterior
21_1-5_500000_08.dwg	Labyrinths
21_1-5_600000_08.dwg	PSS
21_1-5_700000_08.dwg	Guardrails and Stairs
21_1-5_800000_08.dwg	Cranes
21_1-5_900000_08.dwg	BLEPS

Criterion 1.2: Changes from PDR layout

There is a minor nomenclature change between the FDR and PDR: the identification of the build-out phases for the FDR is identical to that used in the CDR and not the PDR. That is, Phase I refers to the ID beamline based on the upstream undulator that provides the outboard beam, Phase II refers to the ID beamline based on the downstream undulator that provides the inboard beam, and Phase III refers to the bending magnet beamline that is currently not in the scope of the LS-CAT project. See Table 2.

Table 2: LS-CAT ID Build-out Phases

Experimental Enclosure	Source	Description	Phase
21-ID-D	Upstream Undulator	Fully Tunable 5-20 keV	I
21-ID-E	Downstream Undulator	MAD around 12.6 keV and 13.5 keV	II
21-ID-F	Downstream Undulator	Fixed Energy 12.6 +/- 0.2 keV	II
21-ID-G	Downstream Undulator	Fixed Energy 12.6 +/- 0.2 keV	II
21-BM	Bending Magnet	Not in current scope	III

Various minor design changes have been made. However, the number of enclosures, their nominal size and purpose, and the scope of the project are unchanged from the PDR.

The layout changes include:

1. 21-ID-F and 21-ID-G are now sitting at an angle of 27.5° from the ID centerline instead of 30°.
2. 21-ID-C no longer has a secondary access door on the outboard wall.
3. 21-ID-G has been reshaped to move the door away from the main thoroughfare and to reclaim space left from the now removed 21-ID-C secondary access door.

Criterion 1.3: Survey and alignment plan

All major components are supplied with locating holes for standard tooling balls to aid the survey and alignment process. The APS survey group will perform all survey and alignment work. Individual components are sent to the APS survey group for fiducialization as they are received. Appendix B shows the location of all beamline components and the offset, if any, from the component centerline and the nominal ID centerline. Note that the inboard and outboard undulator centerlines are ±0.5 mrad with respect to the nominal ID centerline.

Prior to enclosure construction, a number of survey targets were placed at various locations in the areas to be enclosed to aid the alignment process. These target locations are listed in Appendix C.

Criterion 1.4: Utility distribution

Utility distribution has been specified in the appropriate layout drawings. The utilities are included in the enclosure contract. See Table 1 for details.

Criterion 1.5: Ray tracings

An attempt has been made to bring the LS-CAT 21-ID ray tracings into line with the various draft guidelines. All ray tracings can be found on the Intralink Commonspace/LS_CAT/RayTraces, as indicated in Table 3, and included in Appendix D for reference.

Table 3: Drawing numbers of supplied ray tracings

Title	Drawing Number
Horizontal Bremsstrahlung	21-1.4.1-110000-05
Vertical Bremsstrahlung	21-1.4.1-120000-05
Horizontal Synchrotron	21-1.4.1-210000-04
Vertical Synchrotron / Inboard	21-1.4.1-221000-04
Vertical Synchrotron / Outboard	21-1.4.1-222000-04

2. Component Design Update

Criterion 2.1: Drawings

All drawings, including assembly drawings and/or performance specifications have been provided for each beamline component on the Intralink Commonspace in the LS-CAT area. See Table 4. Appendix E contains B size versions of these drawings for reference.

Table 4: Components, locations, and drawing numbers

Component	Location (meters from ID center)	Assembly Drawing
Bremsstrahlung Collimator 1	34.6	21_1-8-1-2_100000
Inboard ID Fixed Mask	46.0	LSC-101-11-0003
Outboard ID Fixed Mask	46.4	LSC-101-11-0011
Inboard ID Aperture 1	46.7	LSC-101-11-0012
Inboard ID Aperture 2	47.1	LSC-101-11-0012
21-ID-F Monochromator	48.0	N/A
21-ID-G Monochromator	50.3	N/A
21-ID-E Monochromator (1st xtal)	53.9	N/A
Sharma Beamstop	56.2	21_1-8-1-18_100000
K5-30 Collimator	56.7	K5300000-00
Bremsstrahlung Collimator 2	57.1	21_1-8-1-21_100000
Outboard ID Aperture 1	58.3	LSC-101-11-0013
Outboard ID Aperture 2	58.7	LSC-101-11-0013
Kohzu Monochromator	60.0	110550-0001
P4-60 Shutter	61.2	P4600000-00
P4-40 Shutter	62.2	21_1-8-1-32_100000_02
Beryllium Window	63.3	B-2510146-200F
K5-50 White Beam End Stop	75.2	K5500000-00
P8-50 Monochromatic Shutter	—	21_1-7-1-37_110000_00

The Phase II monochromator designs are not complete at this time. See Section 2.3.

LS-CAT critical components are based on APS designs and for the most part rely on existing APS thermal analysis. The exceptions are the fixed masks and apertures (LSC-101-11-00xx) that are based on the L5-92 design. Documentation provided by the manufacturer can be found in Appendices F, G, and H. Appendix H demonstrates that these components can operate safely when subjected to 1100 Watts total power. Table 5 and Table 6 list the U33S and U30S undulator operational limits (respectively) based this power incident on the IDT masks and apertures. Note that Appendix F uses 60 periods for the U33S calculations instead of the more generally accepted value of 62 periods for heat load calculations. The operational limits tables here include the end poles as though they were full strength, namely, 62 poles for U33S and 69 poles for U30S, and reflect, therefore, a slightly weaker magnetic field than assumed in Appendix F.

Note that the Phase II beamline is intended to operate between 12.6 and 13.5 KeV and hence, as is evident in Table 6, the IDT masks will not limit the ability to operate at high currents. The Phase II monochromators will likely require limits that are more stringent.

Table 5: U33S (Phase I) Operational Limits for IDT Masks and Apertures

Current (mA)	Deflection Parameter	1st Harmonic (KeV)	Gap (mm)
100	2.810	2.850	10.407
110	2.563	3.291	11.325
120	2.361	3.723	12.143
130	2.191	4.147	12.888
140	2.048	4.553	13.562
150	1.924	4.946	14.184
200	1.507	6.603	16.620

Table 6: U30S (Phase II) Operational Limits for IDT Masks and Apertures

Current (mA)	Deflection Parameter	1st Harmonic (KeV)	Gap (mm)
100	2.316	4.213	10.013
110	2.122	4.770	10.792
120	1.960	5.310	11.499
130	1.827	5.812	12.124
140	1.713	6.287	12.698
150	1.616	6.727	13.217
200	1.288	8.478	15.236

The differences between other LS-CAT critical components and their APS counterparts are summarized here:

1. The P8-50 monochromatic shutter has been modified to include a vacuum spool; the original APS drawing number is P8-500000-01. Note that the APS drawing tree has a minor inconsistency between the part and assembly drawings.
2. The P4-40 shutter has been modified to include a different offset flange to match our Kohzu monochromator offset. The original drawing number is P4400000-00.
3. The Sharma beamstop mount flange has been modified to correctly position the device at 56.2 meters and to include a more robust bypass tube for our outboard beam; the original APS drawing number is 31050403-400010.

Criterion 2.2: Vacuum Compliance

Figure 1.2 lists the various vacuum zones. The LS-CAT vacuum systems are intended to be fully compliant at the HV* level ($\leq 10^{-6}$ with no in-vacuum water joints) of the APS vacuum policy as stated in APS TU-18. All main pumps are ion pumps, typically Varian Noble Diode 500 l/s models. Turbo and oil-less roughing pumps are incorporated where appropriate. Windowless operation is not in the scope of this review: a separate review

of the LS-CAT vacuum systems will be solicited before requesting the removal of the commissioning window.

Criterion 2.3: Beamline Component Designs

Component designs for the Phase I beamline are unchanged from the PDR. Differences between the APS and LS-CAT designs are outlined in Section 2.1.

For Phase II, the details of the 21-ID-F and 21-ID-G monochromators are being negotiated with the vendor and the 21-ID-E monochromator specifications are being prepared. All three monochromators will be either Laue (21-ID-F and 21-ID-G) or Bragg (21-ID-E) oriented diamond (111) crystals. Cooling will be accomplished using a water cooled copper block attached thermally to the diamond crystals with an In/Ga eutectic.

3. Schedule

The work breakdown schedule (WBS) is provided in Appendix I.

Criterion 3.1: Installation

The installation and commissioning activities are shown in Appendix I items 1.9 (Install Beamline Components) and 1.6.6 (Shielding Verification).

Criterion 3.2: APS Procurement

Component procurement is included in section 1.8 of the WBS (Appendix I). Procurement is nearly complete for Phase I and does not involve the APS beyond the enclosure contract.

Criterion 3.3: APS Support

Further support from the APS is outlined in the WBS under section 1.7 (PSS) and 1.9 (Install Beamline Components). LS-CAT has been communicating with the relevant APS groups to coordinate activities.

4. Safety

Criterion 4.1: PSS

LS-CAT has been working with the PSS group to develop the PSS User Requirements Document. This document will be issued by the PSS group in due course. In the meantime, the PSS requirements are outlined here.

The 21-ID-D station is capable of running white beam and for this purpose is protected by a P4-40 shutter. It is, however, anticipated that mostly 21-ID-D will be run as a monochromatic station and hence a P4-60 shutter will be installed in 21-ID-C to reduce scatter from the direct beam. This P4-60 is therefore part of the BLEPS and serves no purpose as far as the PSS is concerned. It is noted here that should the 21-ID-D shielding be verified in monochromatic mode then it will need further verification in white beam mode when the P4-60 is removed.

The locations of the shutters can be found in Table 7 while Tables 8 and 9 show the station access as a function of shutter position. There are no modes to consider and each

station shutter secures one and only one station. The front end shutter requires that 21-ID-A, 21-ID-B, and 21-ID-C all be secure before being allowed to open.

Table 7: PSS Shutters

Designation	Location	Protects	Type
21-ID-SD	21-ID-C	21-ID-D	P4-40
21-ID-SE	21-ID-C	21-ID-E	P8-50
21-ID-SF	21-ID-B	21-ID-F	P8-50
21-ID-SG	21-ID-B	21-ID-G	P8-50

Table 8: PSS Logic, Front End Shutter

FE Shutter	21-ID-A	21-ID-B	21-ID-C
Open	No Access	No Access	No Access
Closed	Access Allowed	Access Allowed	Access Allowed

Table 9: PSS Logic, Station Shutters

Shutter	21-ID-D	21-ID-E	21-ID-F	21-ID-G
21-ID-SD Open	No Access	—	—	—
21-ID-SD Closed	Access Allowed	—	—	—
21-ID-SE Open	—	No Access	—	—
21-ID-SE Closed	—	Access Allowed	—	—
21-ID-SF Open	—	—	No Access	—
21-ID-SF Closed	—	—	Access Allowed	—
21-ID-SG Open	—	—	—	No Access
21-ID-SG Closed	—	—	—	Access Allowed

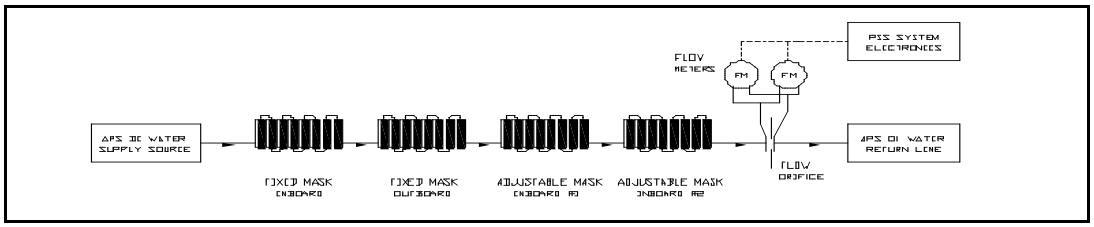


Figure 2: Water Circuit 1 (21-ID-B)

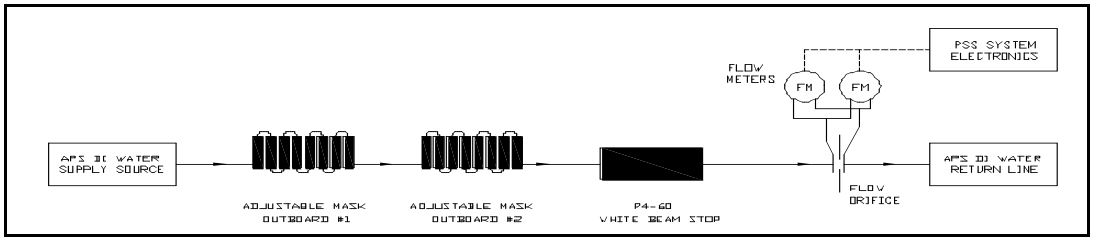


Figure 3: Water Circuit 2 (21-ID-C)

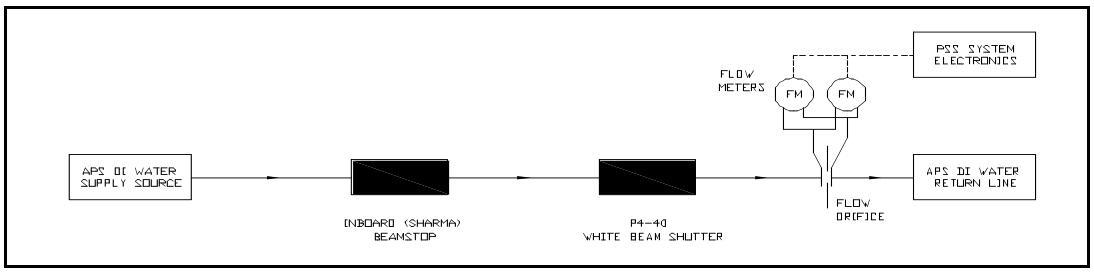


Figure 4: Water Circuit 3 (21-ID-C)

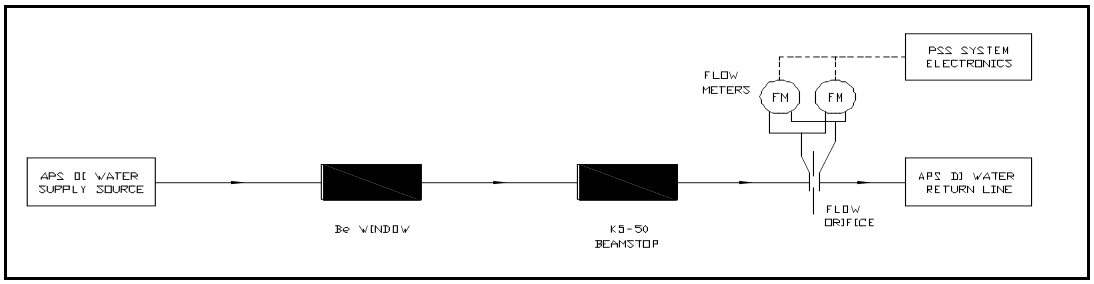


Figure 5 : Water Circuit 4 (21-ID-D)

Water flow diagrams for PSS cooling systems are shown in Figures 2–5. These circuits will tie into the APS Process Water system. Both the PSS and BLEPS components are included in this design to simplify maintenance.

Criterion 4.2: BLEPS

The LS-CAT 21-ID BLEPS will be installed by LS-CAT according to the requirements document included here as Appendix J. For reference, the approach taken by LS-CAT is outlined below.

LS-CAT will implement two independent systems to protect and monitor beamline components on the sector 21 insertion device beamline (21-ID). The Beamline Equipment Protection System (BLEPS) will determine if components are in an acceptable state to make them safe for operations and if they are not, act to prevent such operations. The Beamline Equipment Monitoring System (BLEMS) will monitor numerous beamline variables for the purpose of preventing and alerting conditions which could trip the BLEPS or otherwise compromise the performance of the beamline. These two systems will be functionally independent but, for convenience, will share common user interfaces and logging.

The BLEPS will monitor a set of system states in the form of discrete (relay) contacts and, based on these states, perform actions necessary for keeping beamline components safe. The first priority is that beamline components are safe from x-ray beam damage. For this, vacuum pressures, coolant flows and gate valve positions will be ascertained. If an unsafe condition exists, the BLEPS will immediately revoke the relevant shutter permit. The second priority of the BLEPS is to isolate a vacuum leak should one occur. If vacuum pressure in a given zone rises above a defined set-point, the BLEPS will first revoke the relevant shutter permit. Then, once it is verified that the shutter is closed, the system will close gate valves adjacent to the aberrant vacuum zone.

Physically, the BLEPS will consist of a single programmable logic controller (PLC) networked via stand-alone Ethernet to a series of remote I/O units. The logic will be made as simple as possible for debugging and maintenance purposes. The program will reside solely on the central PLC so remote units will be "dumb." Programming, control, and monitoring will be accessible via the beamline network through a separate Ethernet connection. The central PLC and remote I/O units will be built from modular PLC bases to allow for a maximum of flexibility and expandability.

Wiring of BLEPS contacts will conform to specifications described in APS document 4104-913053-00, "Beamline Equipment Protection (BE-EPS) Interface Control Document for the Generation-3 Personnel Safety System (PSS)." For fail-safe operation, a closed output contact will be interpreted as a granted permit.

The BLEMS will monitor numerous continuous (analog) system variables including vacuum pressures, coolant flows, and temperatures. The PLC program will convert the raw analog input values into meaningful numbers and determine whether or not these values fall into an acceptable range. The BLEMS will inform staff of potential conditions which could trip the BLEPS, allowing diagnosis and correction of problems before the BLEPS is required to revoke a shutter or gate valve permit.

Like the BLEPS, the BLEMS will consist of a central PLC which is linked via stand-alone Ethernet to remote I/O units. A separate Ethernet connection will allow programming and monitoring from the beamline network. The BLEMS will be required to take no external action so the programming of this system will be minimal. Again, modular PLC bases will be used for the central and remote units to maximize flexibility.

Shared hardware and software (EPICS) interfaces will be developed for accessing both systems simultaneously. For the BLEPS, the interfaces will provide over-ride capability and direct control of vacuum zone gate valves. For the BLEMS, the interfaces will provide graphical displays of monitored variables.

Criterion 4.3: Shielding

No changes in the shielding have been made since the PDR. The downstream wall of 21-ID-A has 100 mm thickness in the square meter about the beam (50mm thickness for the rest of the downstream wall) instead of 100 mm thickness for the entire downstream wall as called for in the LS-CAT Technical Specifications. This change is consistent with the APS shielding requirements as called for in ANL/APS/TB-44.

The enclosures and white beam transport are of standard APS design and are being installed by an APS approved contractor under direct control by the APS.

The shielded pipes and pipe stubs that make up beamline from the station shutter to the experimental enclosure (and through 21-ID-F for the case of 21-ID-G) are non-standard shielding components. Table 10 lists the drawings that make up the design. These can be found in the LS-CAT area of the IntraLink Commonspace and are also included in Appendix K.

Table 10: Shielded Pipe Drawing Summary

Drawing Number	Description
21_1-5_201000_09	Shielded pipe stubs
21_1-5_241000_09	Mono beam transport assembly
21_1-5_242000	Mono beam transport flange cover assembly

Criterion 4.4: Ozone

Each enclosure shall be fitted either with roof exhaust fans or to the building HVAC system to mitigate any ozone hazard. The locations and type of the ventilation systems are indicated in drawing 21_1-5_500000 (see Table 1.1.) All x-ray beam paths shall be in either vacuum or helium to reduce ozone generation: No white beam path in air shall be allowed.

Criterion 4.5: Other Hazards

The entire beamline shall be grounded. Ion pumps and gauges shall be grounded directly to their respective controller. The ion pumps procured by LS-CAT include an interlock that prevents operation if the ion pump is not properly grounded.

LS-CAT does not currently plan to use chemical, electrical, high power lasers, biohazards, or any other hazards that would impact beamline design with the following exception: LS-CAT intends to operate auto-filling liquid nitrogen cryostats in 21-ID-C, 21-ID-D, 21-ID-E, 21-ID-F, and 21-ID-G.

Auto-fill LN_2 Safety analysis

The normal oxygen level in the atmosphere is 20.8% while a value of 19.5% or lower is too low to allow entry (29 CFR 1915).

Consider the case of the 21-ID-D enclosure with a volume of approximately 145 cubic meters. Although the details depend on the LN_2 temperature, the barometric pressure, and the room temperature, 13.6 liters of LN_2 produces roughly 9.5 m^3 of gas and would reduce the oxygen concentration in 21-ID-D to 19.5%.

A spill of 13.6 liters in 21-ID-D, while large, is not out of the question. The other experimental enclosures are smaller and therefore even more susceptible to an oxygen deficiency hazard. The smallest enclosure, 21-ID-F, encloses 53 m^3, hence a spill (with the door closed) as small as 5 liters could produce a hazard.

Mitigating the oxygen deficiency hazard is that all LS-CAT enclosures either are attached to the APS HVAC system or have roof fans ensuring air circulation with the experimental hall. A hazard remains if an LN_2 auto-fill failure provides a constant influx of nitrogen that exceeds the ventilation capabilities of the enclosure. LS-CAT, therefore, will require oxygen sensors in all enclosures except 21-ID-A, which does not contain cryogenic equipment. Note that 21-ID-B and 21-ID-C may share a sensor system, as these enclosures are contiguous.

Criterion 4.6: Fire Hazards

LS-CAT does not currently plan to use flammable liquids or gasses in the experimental stations or along the beamline. There are no enclosed areas planned that will be occupied during normal operations.

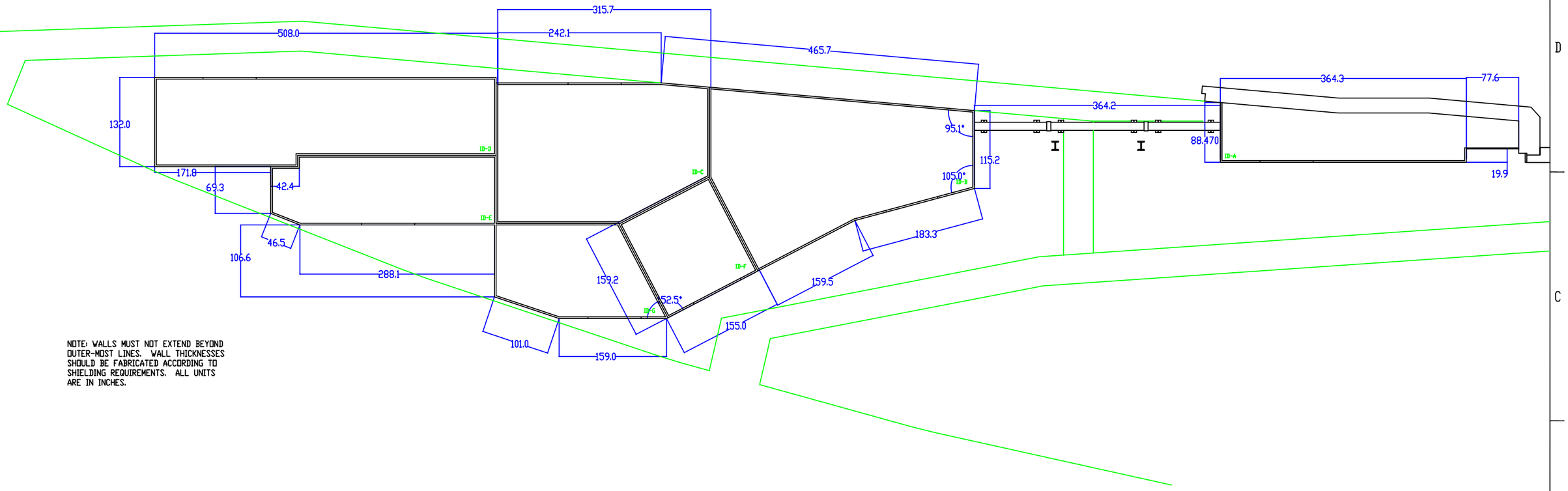
5. Special Operating Requirements Update

Criterion 5.1

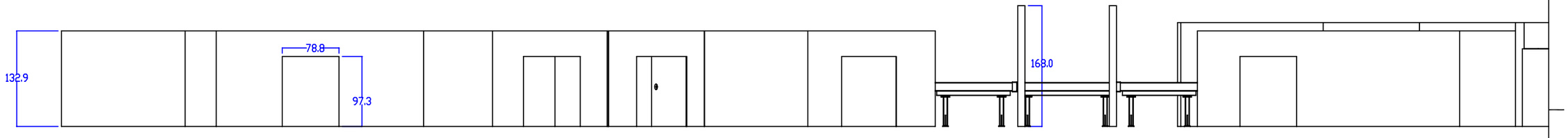
There are no operating requirements beyond the scope of ANL/APS/TB-9.

Appendix A: LS-CAT 21-ID Layout Drawings

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



NOTE: WALLS MUST NOT EXTEND BEYOND OUTER-MOST LINES. WALL THICKNESSES SHOULD BE FABRICATED ACCORDING TO SHIELDING REQUIREMENTS. ALL UNITS ARE IN INCHES.



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±.1 XX ±.03 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 2-22-06
CHECKED BY: — DATE:
DESIGNED BY: — DATE:
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:

MODEL NAME: 21_SECTOR.ASM
RELEASE LEVEL: — MODEL VER: — DRVG VER:
ELECTRONIC FILE NAME: — MATERIAL:

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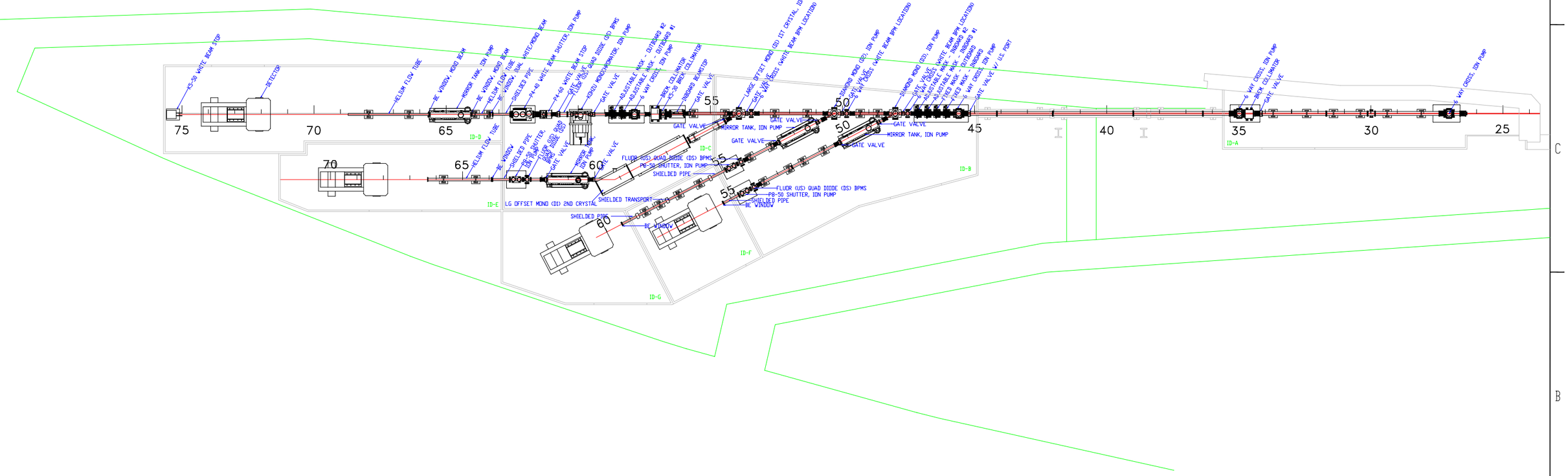
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
HUTCH WALLS

SIZE: D — FSCM No: — DRAWING NUMBER: 21-1.5-100000-08 — REV: 8.0
SCALE: 1/75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE
	9.0	MV FIXED MASK #1, INCR. BELLOWS DIA MV BEAMS 0.0134" OUTBOARD	BKS		5-11-06



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±.1, XX ±.03, XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE 5-11-06		
CHECKED BY:	DATE		
DESIGNED BY:	DATE		
RESPONSIBLE ENGINEER:	DATE		
GROUP LEADER:	DATE		
APPROVED BY:	DATE		
MODEL NAME: 21_SECTOR.ASM	RELEASE LEVEL:	MODEL VER:	DRWG VER:
ELECTRONIC FILE NAME:	MATERIAL:		

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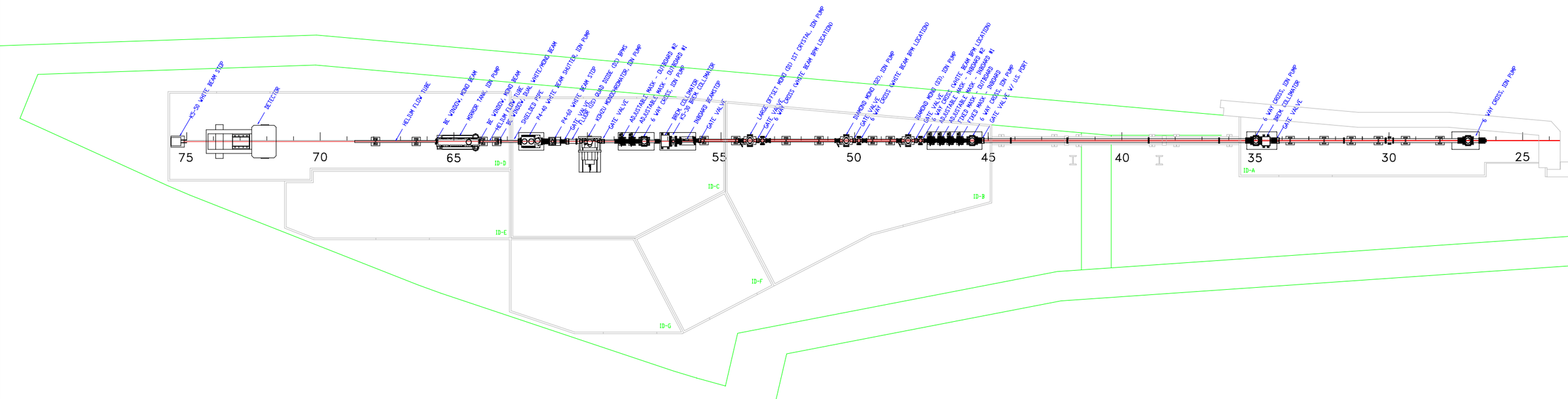
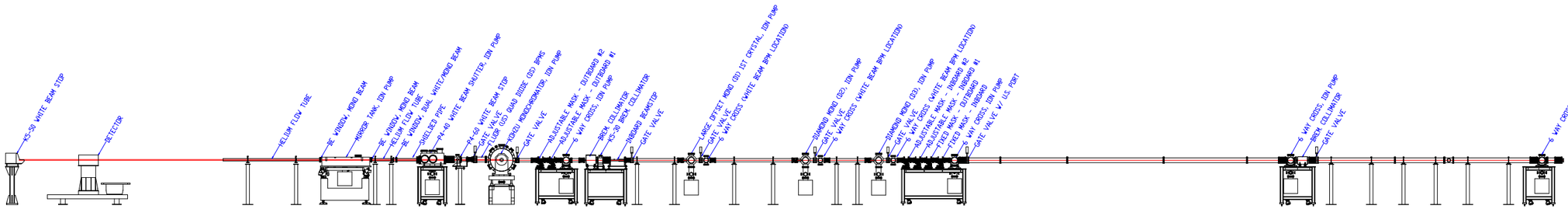
TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
COMPONENT LAYOUT - ALL

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-200000-09	9.0

SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE
	9.0	MV FIXED MASK #1, INCR. BELLOWS DIA MV BEAMS 0.0134" OUTBOARD	BKS		5-11-06





ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS X ±.1 XX ±.03 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE: 5-11-06	
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RESPONSIBLE ENGINEER:	DATE:	
GROUP LEADER:	DATE:	
APPROVED BY:	DATE:	
RELEASE LEVEL:	MODEL VER:	DRVG VER:
MATERIAL:		

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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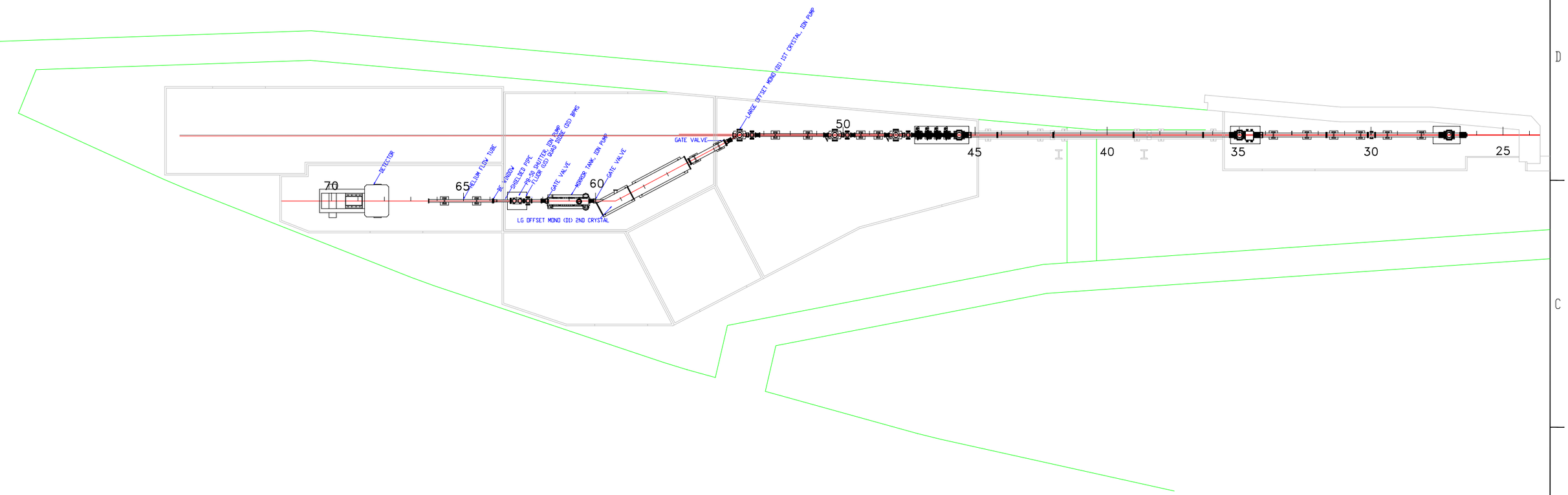
LIFE SCIENCES COLLABORATIVE (LS-CAT)

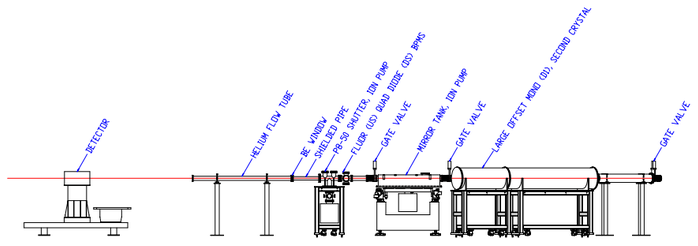
TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
COMPONENTS-PHASE 1

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-210000-09	9.0

SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE
	9.0	MV FIXED MASK #1, INCR. BELLOWS DIA MV BEAMS 0.0134" OUTBOARD	BKS		5-11-06





ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS .X ±.1 .XX ±.03 .XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE 5-11-06	
CHECKED BY:	DATE	
DESIGNED BY:	DATE	
RESPONSIBLE ENGINEER:	DATE	
GROUP LEADER:	DATE	
APPROVED BY:	DATE	
MODEL NAME: 21_SECTOR.ASM	RELEASE LEVEL:	MODEL VER: / DRWG VER:
ELECTRONIC FILE NAME:	MATERIAL:	

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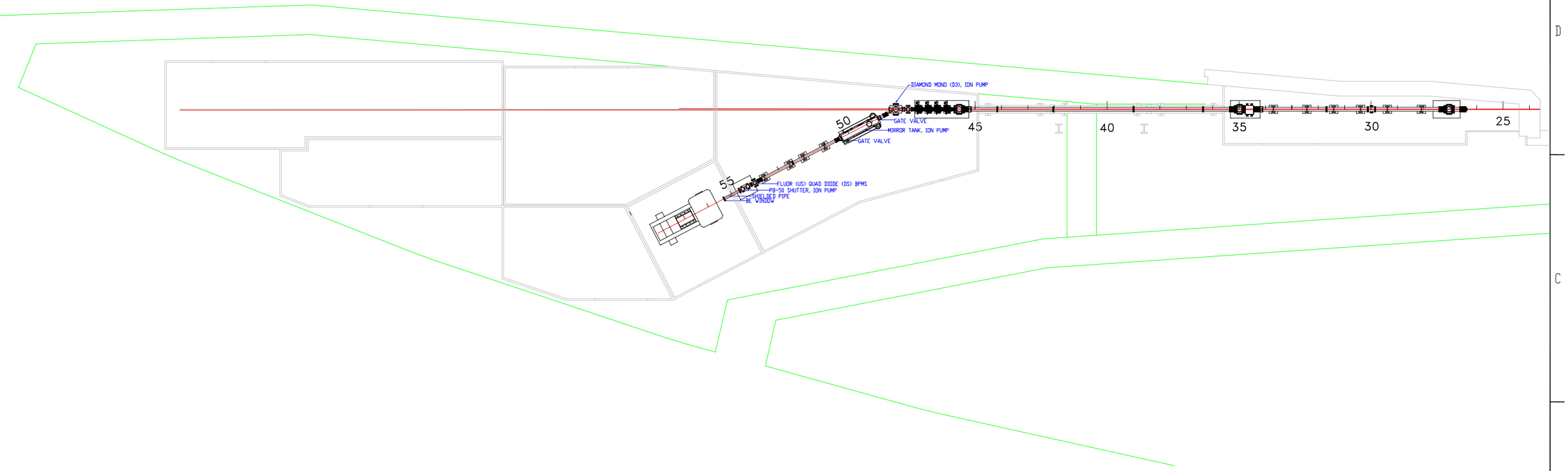
LIFE SCIENCES COLLABORATIVE (LS-CAT)

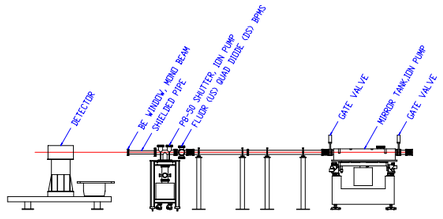
TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
COMPONENTS - PHASE 2, TO E STATION

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-220000-09	9.0

SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE
	9.0	MV FIXED MASK #1, INCR. BELLOWS DIA MV BEAMS 0.0134" OUTBOARD	BKS		5-11-06





ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS .X ±.1 .XX ±.03 .XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 5-11-06
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DESIGNED BY: — DATE:
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

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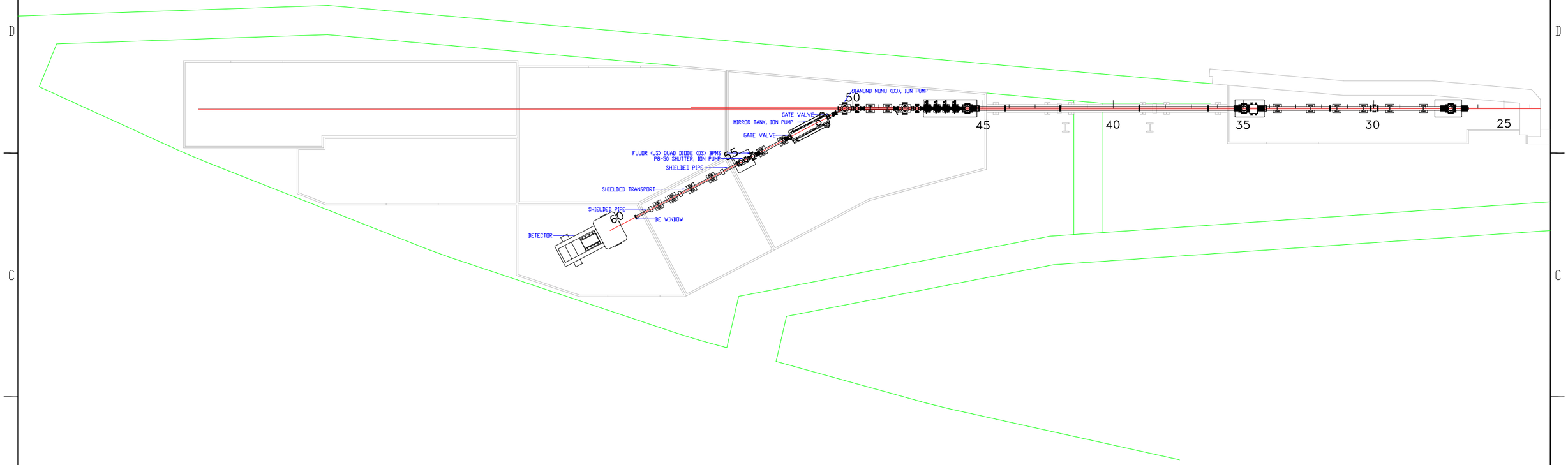
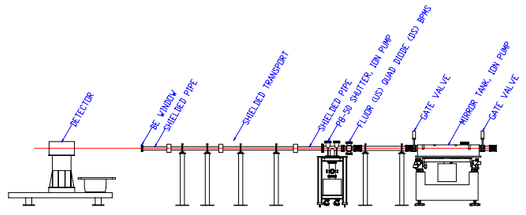
TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
COMPONENTS - PHASE 2, TO F STATION

SIZE: D — FSCM No: — DRAWING NUMBER: 21-1.5-230000-09 — REV: 9.0
SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE
	9.0	MV FIXED MASK #1, INCR. BELLOWS DIA MV BEAMS 0.0134" OUTBOARD	BKS		5-11-06





ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS .X ±.1 .XX ±.03 .XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE: 5-11-06	
CHECKED BY:	DATE:	
DESIGNED BY:	DATE:	
RESPONSIBLE ENGINEER:	DATE:	
GROUP LEADER:	DATE:	
APPROVED BY:	DATE:	
RELEASE LEVEL:	MODEL VER:	DRWG VER:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:
MATERIAL:

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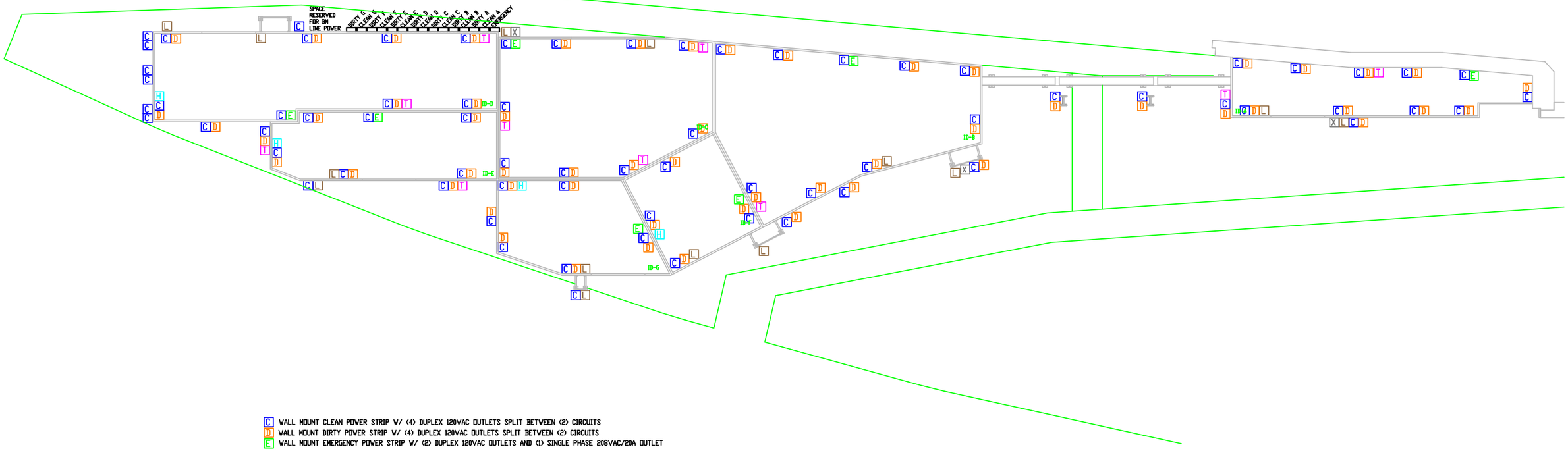
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
COMPONENTS - PHASE 2, TO G STATION

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-240000-09	9.0

SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



[C] WALL MOUNT CLEAN POWER STRIP W/ (4) DUPLEX 120VAC OUTLETS SPLIT BETWEEN (2) CIRCUITS
[D] WALL MOUNT DIRTY POWER STRIP W/ (4) DUPLEX 120VAC OUTLETS SPLIT BETWEEN (2) CIRCUITS
[E] WALL MOUNT EMERGENCY POWER STRIP W/ (2) DUPLEX 120VAC OUTLETS AND (1) SINGLE PHASE 208VAC/20A OUTLET
[L] LIGHT SWITCH - 3 WAY
[T] (1) DIRTY SINGLE PHASE 208VAC/30A OUTLET MOUNTED ON ADJACENT [D] POWER STRIP
[X] EXHAUST FAN SWITCH
[H] WALL MOUNT POWER STRIP W/ (2) SINGLE PHASE 208VAC/30A OUTLETS AND (1) 3 PHASE, 5 WIRE 208VAC/30A OUTLET

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: X ±.1, XX ±.03, XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE: 2-23-06	
CHECKED BY:	DATE:	
DESIGNED BY:	DATE:	
RESPONSIBLE ENGINEER:	DATE:	
GROUP LEADER:	DATE:	
APPROVED BY:	DATE:	
RELEASE LEVEL:	MODEL VER:	DRWG VER:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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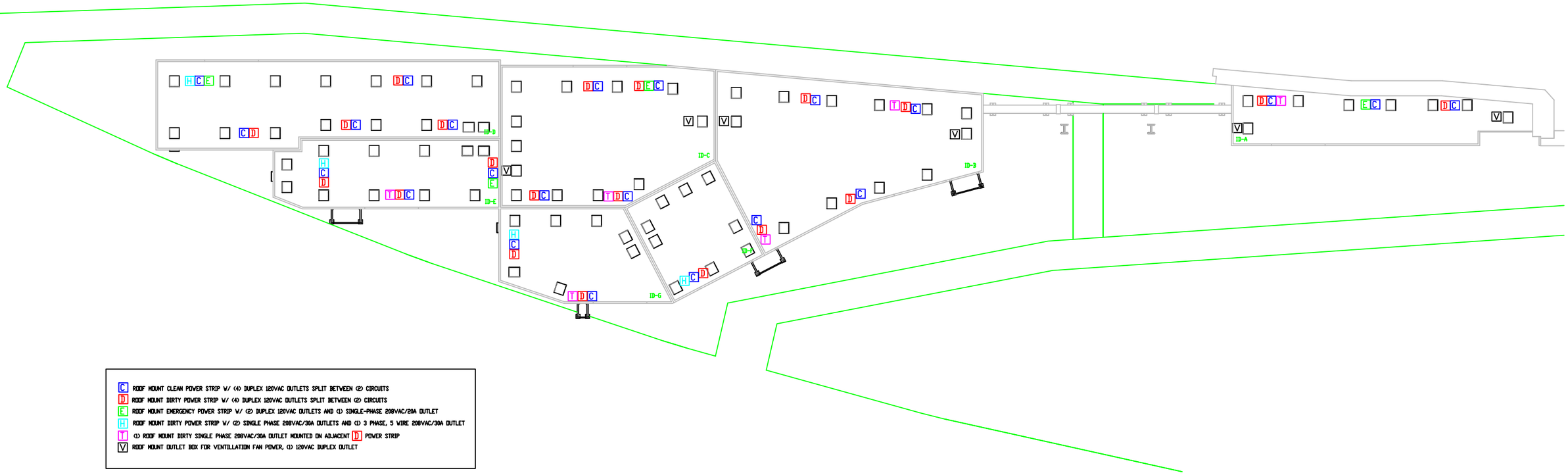
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
GROUND LEVEL ELECTRICAL

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-310000-08	8.0

SCALE: 1:75 | DO NOT SCALE DRAWING

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



- C ROOF MOUNT CLEAN POWER STRIP W/ (4) DUPLEX 120VAC OUTLETS SPLIT BETWEEN (2) CIRCUITS
- D ROOF MOUNT DIRTY POWER STRIP W/ (4) DUPLEX 120VAC OUTLETS SPLIT BETWEEN (2) CIRCUITS
- E ROOF MOUNT EMERGENCY POWER STRIP W/ (2) DUPLEX 120VAC OUTLETS AND (1) SINGLE-PHASE 208VAC/20A OUTLET
- H ROOF MOUNT DIRTY POWER STRIP W/ (2) SINGLE PHASE 208VAC/30A OUTLETS AND (1) 3 PHASE, 5 WIRE 208VAC/30A OUTLET
- T (1) ROOF MOUNT DIRTY SINGLE PHASE 208VAC/30A OUTLET MOUNTED ON ADJACENT D POWER STRIP
- V ROOF MOUNT OUTLET BOX FOR VENTILLATION FAN POWER, (1) 120VAC DUPLEX OUTLET

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS X ±.1 XX ±.25 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 2-23-06
CHECKED BY: — DATE:
DESIGNED BY: — DATE:
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GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

MODEL NAME: 21_SECTORASM
ELECTRONIC FILE NAME:

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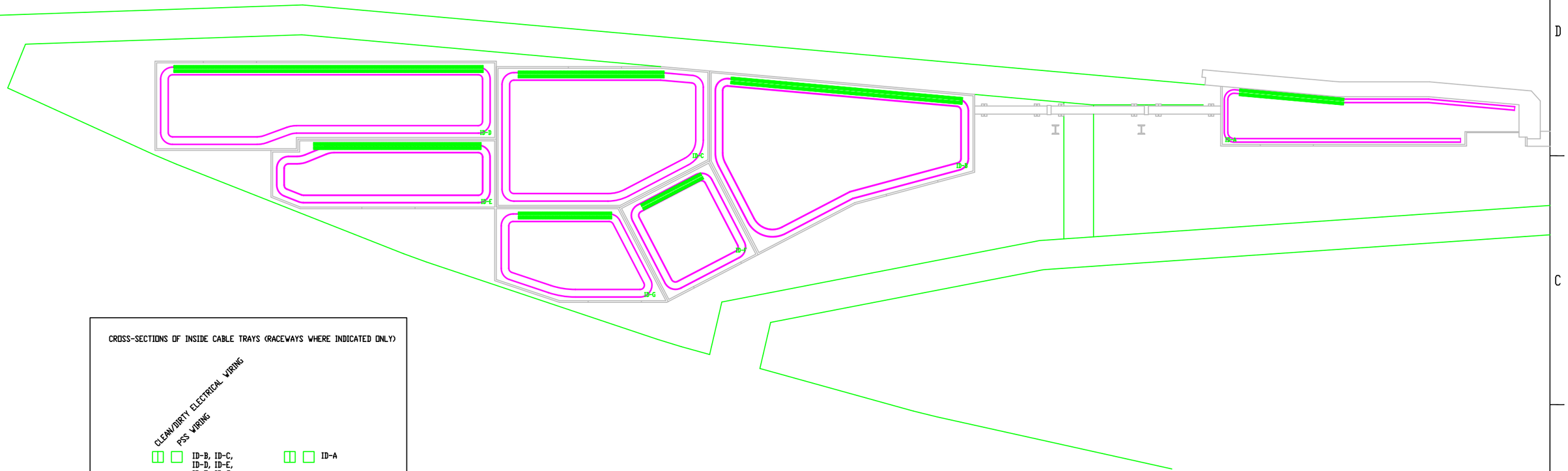
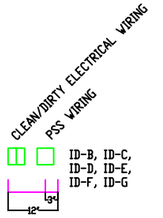
TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
ROOF ELECTRICAL

SIZE: D — FSCM No: — DRAWING NUMBER: 21-LS-320000-08 — REV: 8.0
SCALE: 1:75 — DO NOT SCALE DRAWING

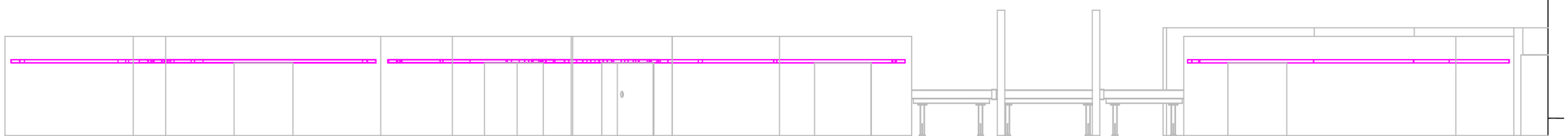
REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



CROSS-SECTIONS OF INSIDE CABLE TRAYS (RACEWAYS WHERE INDICATED ONLY)



CABLE TRAYS SHOULD RUN JUST ABOVE TOP OF DOOR OPENINGS



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±.1 XX ±.25 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 2-23-06
CHECKED BY: — DATE:
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APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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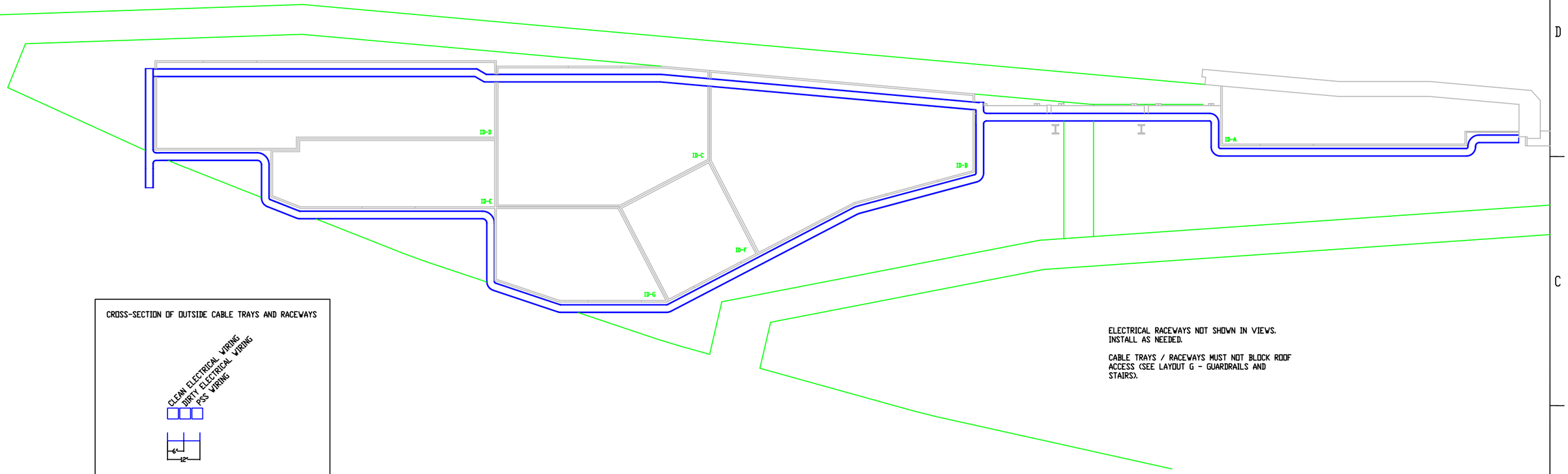
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
CABLE TRAYS - INSIDE

SIZE: D — FSCM No: — DRAWING NUMBER: 21-LS-410000-08 — REV: 8.0
SCALE: 1:75 — DO NOT SCALE DRAWING

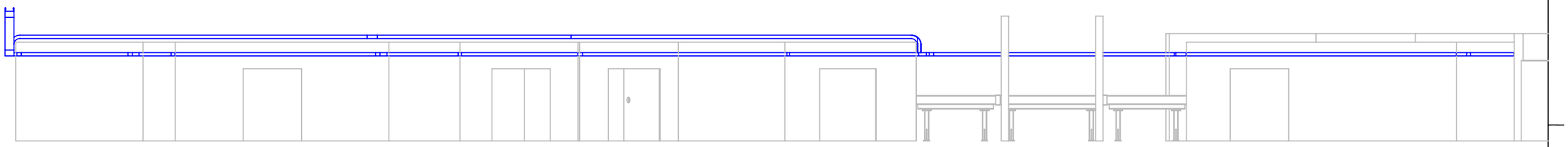
REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



CROSS-SECTION OF OUTSIDE CABLE TRAYS AND RACEWAYS

ELECTRICAL RACEWAYS NOT SHOWN IN VIEWS.
INSTALL AS NEEDED.

CABLE TRAYS / RACEWAYS MUST NOT BLOCK ROOF ACCESS (SEE LAYOUT G - GUARDRAILS AND STAIRS).



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

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DECIMALS X ±.1 XX ±.03 XXX ±.010
ANGULAR ±.5°

DRAWN BY:	B.STILLWELL	DATE: 2-23-06
CHECKED BY:		DATE:
DESIGNED BY:		DATE:
RESPONSIBLE ENGINEER:		DATE:
GROUP LEADER:		DATE:
APPROVED BY:		DATE:
RELEASE LEVEL:		MODEL VER: DRWG VER:
MATERIAL:		

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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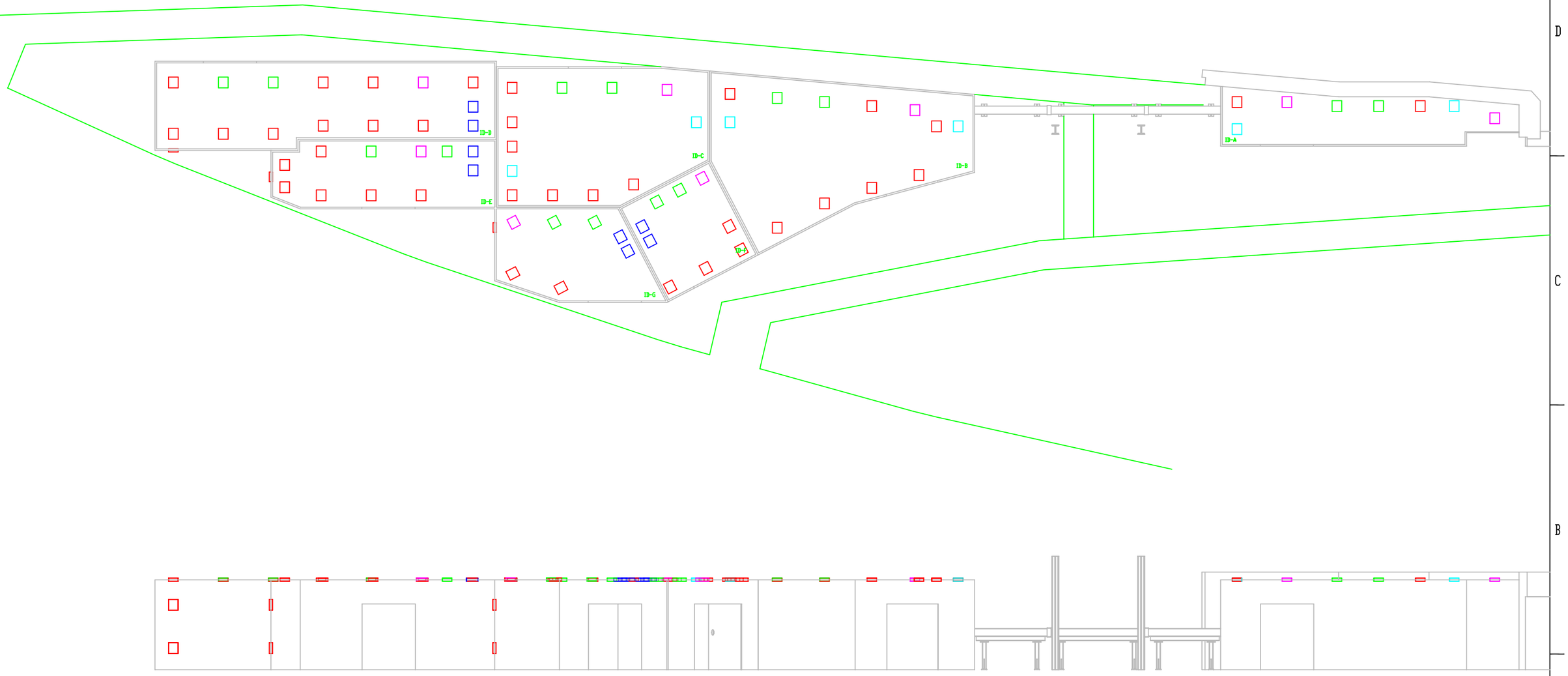
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
CABLE TRAYS - OUTSIDE

SIZE: D | FSCM No: | DRAWING NUMBER: 21-LS-420000-08 | REV: 8.0
SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



- ROOF MOUNT LABRYNTH ELECTRICAL/PSS ACCESS
- ROOF OR WALL MOUNT LABRYNTH UTILITY EXCLUDED
- ROOF MOUNT LABRYNTH HVAC
- ROOF MOUNT LABRYNTH VENTILATION
- ROOF MOUNT LABRYNTH WATER / AIR

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

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TOLERANCES
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ANGULAR ±.5°

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GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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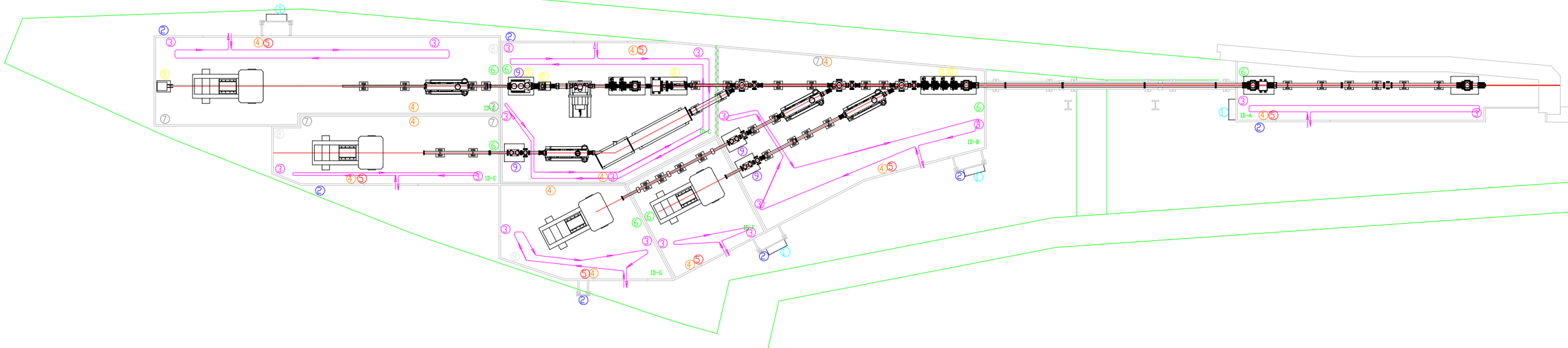
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
LABYRINTHS

SIZE: D — FSCM No — DRAWING NUMBER: 21-LS-500000-08 — REV: 8.0
SCALE: 1:75 — DO NOT SCALE DRAWING

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



0 WATER-COOLED MASK OR BEAMSTOP
1 PSS PANEL
2 DOOR CONTROL BUTTON
3 SEARCH BOX
4 EMERGENCY STOP
5 EMERGENCY DOOR OPEN
6 STROBE AND SPEAKER
7 CONVEX MIRROR
8 SEARCH BOX CONTINGENCY LOCATION
9 SHUTTER
INTERLOCKED PERSONNEL BARRIER

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

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ANGULAR ±.5°

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RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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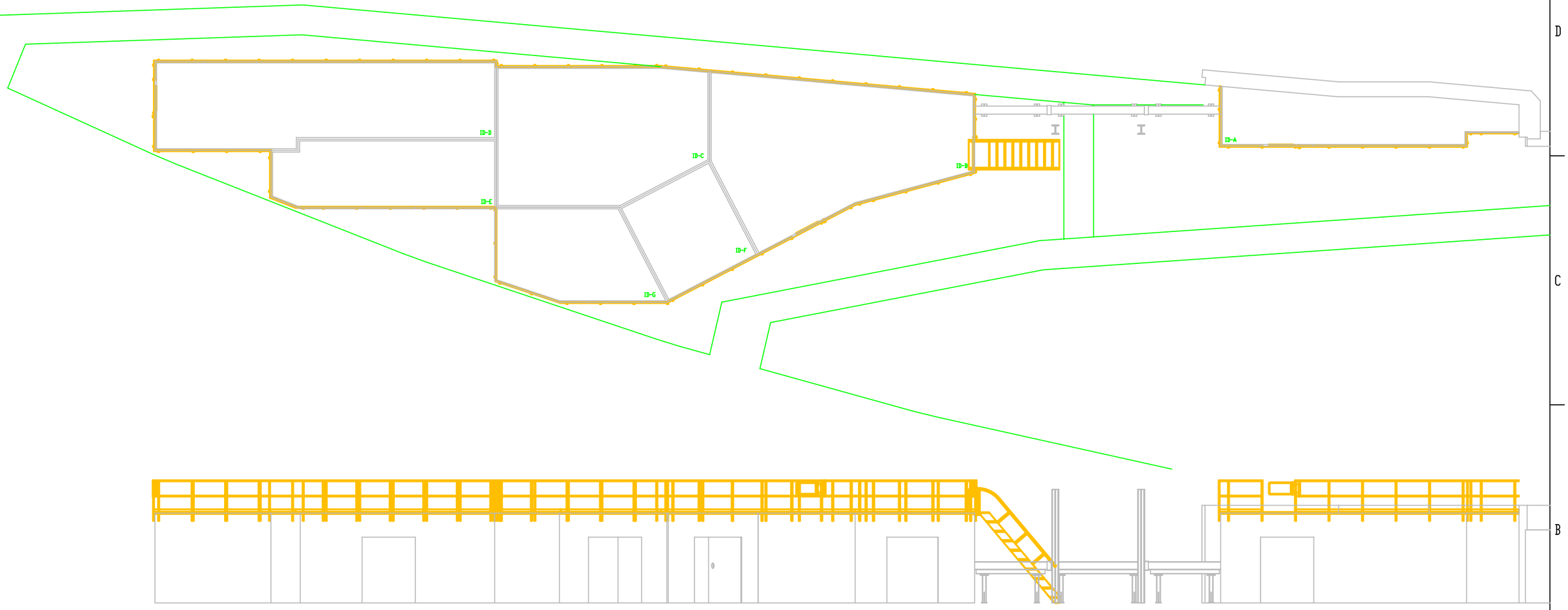
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
PERSONNEL SAFETY SYSTEM (PSS)

SIZE: D — FSCM No: — DRAWING NUMBER: 21-1.5-600000-08 — REV: 8.0
SCALE: 1:75

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL MACHINING ARE IN INCHES TOLERANCES
DECIMALS X ±.1 XX ±.03 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 2-23-06
CHECKED BY: — DATE:
DESIGNED BY: — DATE:
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

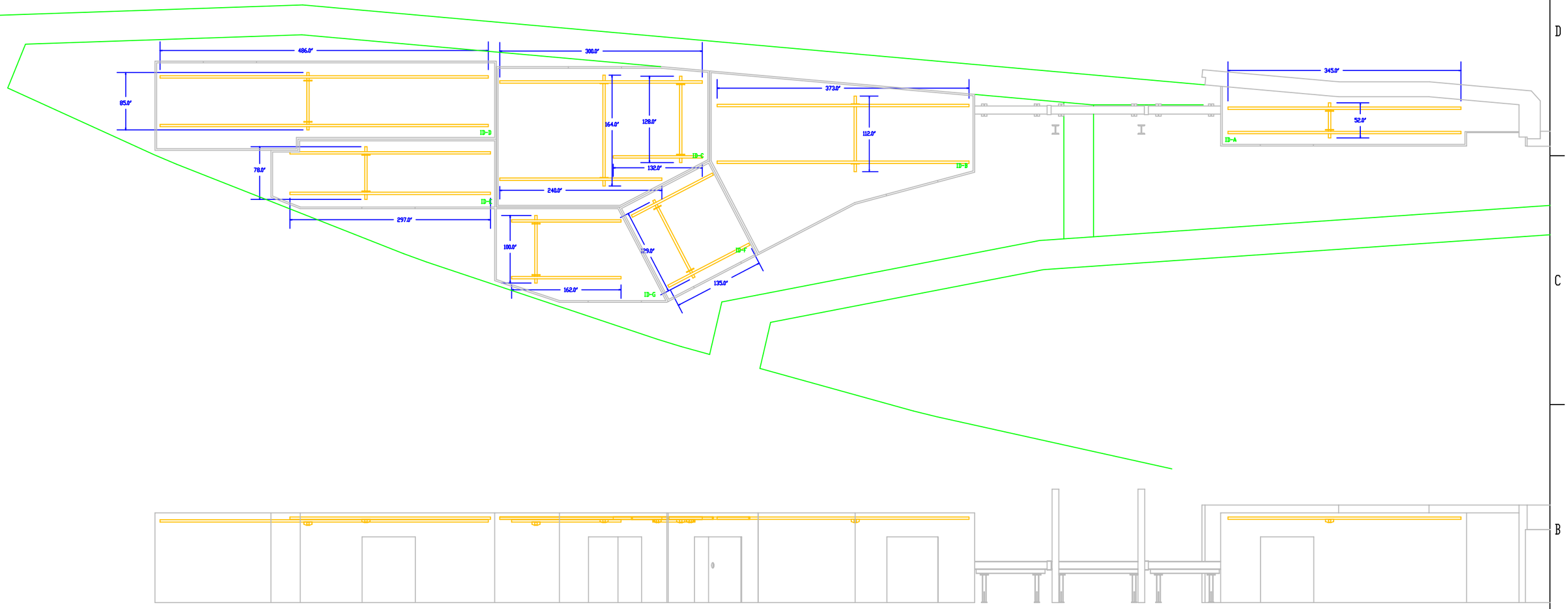
MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
GUARDRAILS AND STAIRS

SIZE: D — FSCM No: — DRAWING NUMBER: 21-I.S-700000-08 — REV: 8.0
SCALE: 1:75 — DO NOT SCALE DRAWING

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS X ±.1 XX ±.03 XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 2-23-06
CHECKED BY: — DATE:
DESIGNED BY: — DATE:
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL:

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

THIS DRAWING IS THE PROPERTY OF
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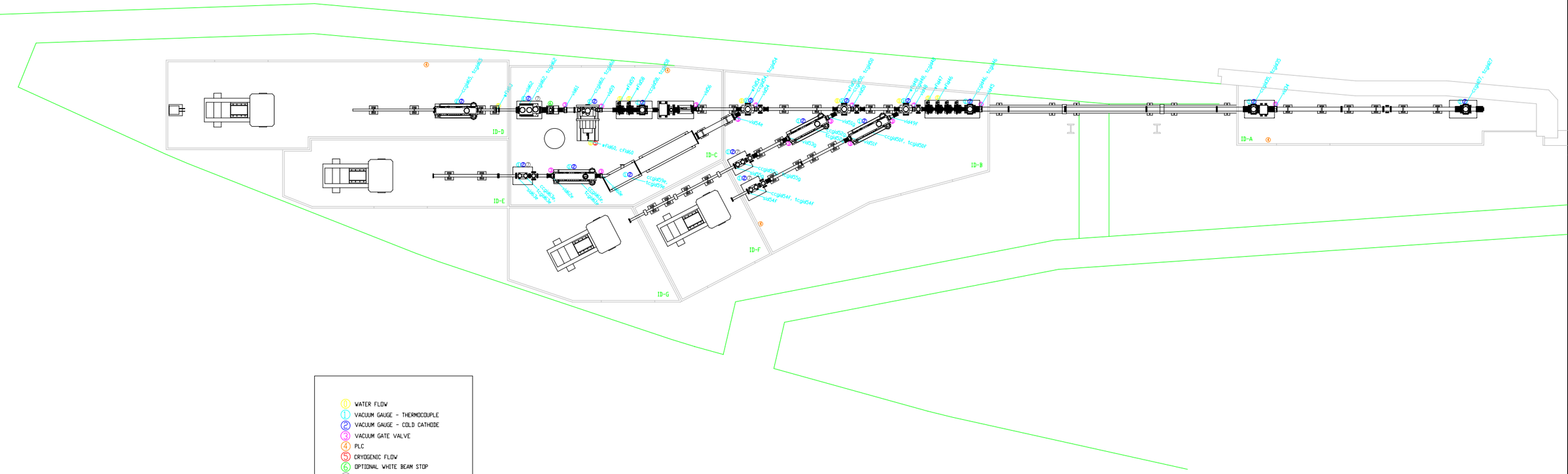
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
CRANES

SIZE: D — FSCM No: — DRAWING NUMBER: 21-LS-800000-08 — REV: 8.0
SCALE: 1:75 — DO NOT SCALE DRAWING

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



4 WATER FLOW
1 VACUUM GAUGE - THERMOCOUPLE
2 VACUUM GAUGE - COLD CATHODE
3 VACUUM GATE VALVE
4 PLC
5 CRYOGENIC FLOW
6 OPTIONAL WHITE BEAM STOP
7 SHUTTER

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS .X ±.1 .XX ±.03 .XXX ±.010 ANGULAR ±.5°

DRAWN BY: B.STILLWELL	DATE: 2-23-06	
CHECKED BY:	DATE:	
DESIGNED BY:	DATE:	
RESPONSIBLE ENGINEER:	DATE:	
GROUP LEADER:	DATE:	
APPROVED BY:	DATE:	
RELEASE LEVEL:	MODEL VER:	DRWG VER:
MATERIAL:		

MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:

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NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
BEAMLINE EQUIPMENT PROTECTION SYSTEM (BLEPS)

SIZE	FSCM No	DRAWING NUMBER	REV.
D		21-1.5-900000-08	8.0

SCALE: 1:75

Appendix B: Phase I Component Locations

Z0 (m)	Z1 (m)	US X Offset mm	DS X Offset mm	Description	LengthMeters	LengthInchesDe	LengthInches16	FlangeUS	US Flange Type	FlangeDS	DS Flange Type	Table	Stand (US)	Stand (DS)
26.355	26.418			Adapter	0.063	2.480	2 8/16	4.5	R	8	NR			
26.418	26.619			Bellows	0.201	7.913	7 15/16	8	R	8	NR			
26.619	26.873			Nipple	0.254	10.000	10	8		8				
26.873	27.206			Cross	0.333	13.110	13 2/16	8	R	8	NR	TID27		
27.206	27.406			Bellows	0.200	7.874	7 14/16	8	R	8	NR			
27.406	29.844			Nipple	2.438	95.984	96	8	R	8	NR		PSID28	PSID29
29.844	30.123			Cross	0.279	10.984	11	8	R	8	NR			
30.123	31.685			Nipple	1.562	61.496	61 8/16	8	R	8	NR		PSID30	PSID31
31.685	34.123			Nipple	2.438	95.984	96	8	R	8	NR			
34.123	34.194			Valve	0.071	2.795	2 13/16	8	R	8	NR		PSID32	PSID34
34.194	34.394			Bellows	0.200	7.874	7 14/16	8	NR	8	NR			
34.394	34.814			Bremsstrahlung Collimator	0.420	16.535	16 9/16	8	R	8	NR	TID35		
34.814	35.148			Cross	0.334	13.150	13 2/16	8	R	8	NR	TID35		
35.148	35.343			Bellows	0.195	7.677	7 11/16	8	R	8	NR			
35.343	36.359			Nipple	1.016	40.000	40	8	R	8	NR		PSBID36	PSBID40
36.359	38.991			Nipple	2.632	103.622	103 10/16	8	R	8	NR		PSBID36	PSBID40
38.991	42.039			Nipple	3.048	120.000	120	8	R	8	NR		PSBID36	PSBID40
42.039	44.153			Nipple	2.114	83.228	83 4/16	8	R	8	NR		PSBID41	PSBID44
44.153	45.169			Nipple	1.016	40.000	40	8	R	8	NR		PSBID41	PSBID44
45.169	45.239			Valve	0.070	2.756	2 12/16	8	NR	8	NR		PSBID41	PSBID44
45.239	45.434			Bellows	0.195	7.677	7 11/16	8	NR	8	TH		PSBID41	PSBID44
45.434	45.768			Cross	0.334	13.150	13 2/16	8	R	6	NR	TID46		
45.768	45.920			Bellows	0.152	5.984	6	6	R	6	NR			
45.920	46.170			Fixed Mask 1	0.250	9.843	9 13/16	6	??	6	??	TID46		
46.170	46.285			Bellows	0.115	4.528	4 8/16	6	R	6	NR			
46.285	46.535			Fixed Mask 2	0.250	9.843	9 13/16	6	??	6	??	TID46		
46.535	46.633			Bellows	0.098	3.858	3 14/16	6	R	6	NR			
46.633	46.883			Heatload Mask Inboard 1	0.250	9.843	9 13/16	6	??	6	??	TID46		
46.883	46.998			Bellows	0.115	4.528	4 8/16	6	R	6	NR			
46.998	47.248			Heatload Mask Inboard 2	0.250	9.843	9 13/16	6	??	6	??	TID46		
47.248	47.400			Bellows	0.152	5.984	6	6	R	6	NR			
47.400	47.670			Cross (White BPM)	0.270	10.630	10 10/16	6	R	6	NR	TID48		
47.670	47.740			Valve	0.070	2.756	2 12/16	6	NR	6	NR	TID48		
47.740	48.265			Diamond Mono D3	0.525	20.669	20 11/16	6	R	6	NR	TID48		
48.265	48.418			Bellows	0.153	6.024	6	6	R	6	NR			
48.418	49.554			Nipple	1.136	44.724	44 12/16	6	R	6	NR		PSID48	PSID49
49.554	49.707			Bellows	0.153	6.024	6	6	R	6	NR			
49.707	49.977			Cross (White BPM)	0.270	10.630	10 10/16	6	R	6	NR	TID50		
49.977	50.047			Valve	0.070	2.756	2 12/16	6	NR	6	NR	TID50		
50.047	50.575			Diamond Mono D2	0.528	20.787	20 13/16	6	R	6	NR	TID50		
50.575	50.724			Bellows	0.149	5.866	5 14/16	6	R	6	NR			
50.724	53.163			Nipple	2.439	96.024	96	6	R	6	NR		PSID51	PSID53
53.163	53.315			Bellows	0.152	5.984	6	6	R	6	NR			
53.315	53.585			Cross (White BPM)	0.270	10.630	10 10/16	6	R	6	NR	TID54		
53.585	53.655			Valve	0.070	2.756	2 12/16	6	NR	6	NR	TID54		
53.655	54.184			Si Mono D1	0.529	20.827	20 13/16	6	R	6	NR	TID54		
54.184	54.336			Bellows	0.152	5.984	6	6	R	6	NR			
54.336	55.765			Nipple	1.429	56.260	56 4/16	6	R	6	NR			

55.765	55.835			Valve	0.070	2.756	2 12/16	6 NR		6 NR			PSID55	PSID56
55.835	55.988			Bellows	0.153	6.024	6	6 R		6 NR				
55.988	56.010			Adapter	0.022	0.866	14/16	6 TH		8 NR	TID56			
56.010	56.493	0.00	27.43	Sharma Beamstop	0.483	19.016	19	8 R		8 R	TID56			
56.493	56.695			Bellows	0.202	7.953	7 15/16	8 R		8 NR				
56.695	56.881	28.35		K5-30 Collimator	0.186	7.323	7 5/16	8 NR		8 NR	TID56			
56.881	57.302		28.65	Bremsstrahlung Collimator	0.421	16.575	16 9/16	8 R		6 NR	TID56			
57.302	57.451			Bellows	0.149	5.866	5 14/16	6 R		6 NR				
57.451	57.705	28.73		Nipple	0.254	10.000	10	6 R		6 NR	TID58			
57.705	58.038		28.35	Cross	0.333	13.110	13 2/16	6 R		6 NR	TID58			
58.038	58.191			Bellows	0.153	6.024	6	6 R		6 NR				
58.191	58.441	29.10	29.22	Heatload Mask Outboard 1	0.250	9.843	9 13/16	6 ??		6 ??	TID58			
58.441	58.556			Bellows	0.115	4.528	4 8/16	6 R		6 NR				
58.556	58.806	29.28	29.40	Heatload Mask Outboard 2	0.250	9.843	9 13/16	6 ??		6 ??	TID58			
58.806	58.958			Bellows	0.152	5.984	6	6 R		6 NR				
58.958	59.384	29.48		Nipple	0.426	16.772	16 12/16	6 R		6 NR	KID60			
59.384	59.454			Valve	0.070	2.756	2 12/16	6 R		6 NR	KID60			
59.454	60.413			Kohozu Monochromator	0.959	37.756	37 12/16	6 R		8 R	KID60			
60.413	60.716			Fluorescence Screen	0.303	11.929	11 15/16	8 R		8 R	KID60			
60.716	60.786			Valve	0.070	2.756	2 12/16	8 NR		8 NR	KID60			
60.786	60.808			Adapter	0.022	0.866	14/16	8 NR		6 TH	KID60			
60.808	60.843		30.42	Quad Diode BPM	0.035	1.378	1 6/16	6 NR		6 NR	KID60			
60.843	60.995			Bellows	0.152	5.984	6	6 R		6 NR				
60.995	61.426	30.50	30.71	P4-60	0.431	16.969	16 15/16	6 R		8 R	TID61			
61.426	61.626			Bellows	0.200	7.874	7 14/16	8 R		8 NR				
61.626	62.577	30.81		P4-40	0.951	37.441	37 7/16	8 R		8 R	TID62			
62.577	63.239			Nipple (Shielded)	0.662	26.063	26 1/16	8 R		6 NR	TID62			
63.239	63.345		31.67	Be Window (Water Cooled)	0.106	4.173	4 3/16	6 ??		6 ??	TID62			
63.345	64.014			He Flow Pipe	0.669	26.339	26 5/16	6		6			PSID63	PSID64
64.014	64.033	32.01		Be Window	0.019	0.748	12/16	6		6	TID65			
64.033	65.642			Mirror	1.609	63.346	63 6/16	6		6	TID65			
65.642	65.662		32.83	Be Window	0.020	0.787	13/16	6		6	TID65			
75.220		37.61		K5-50										

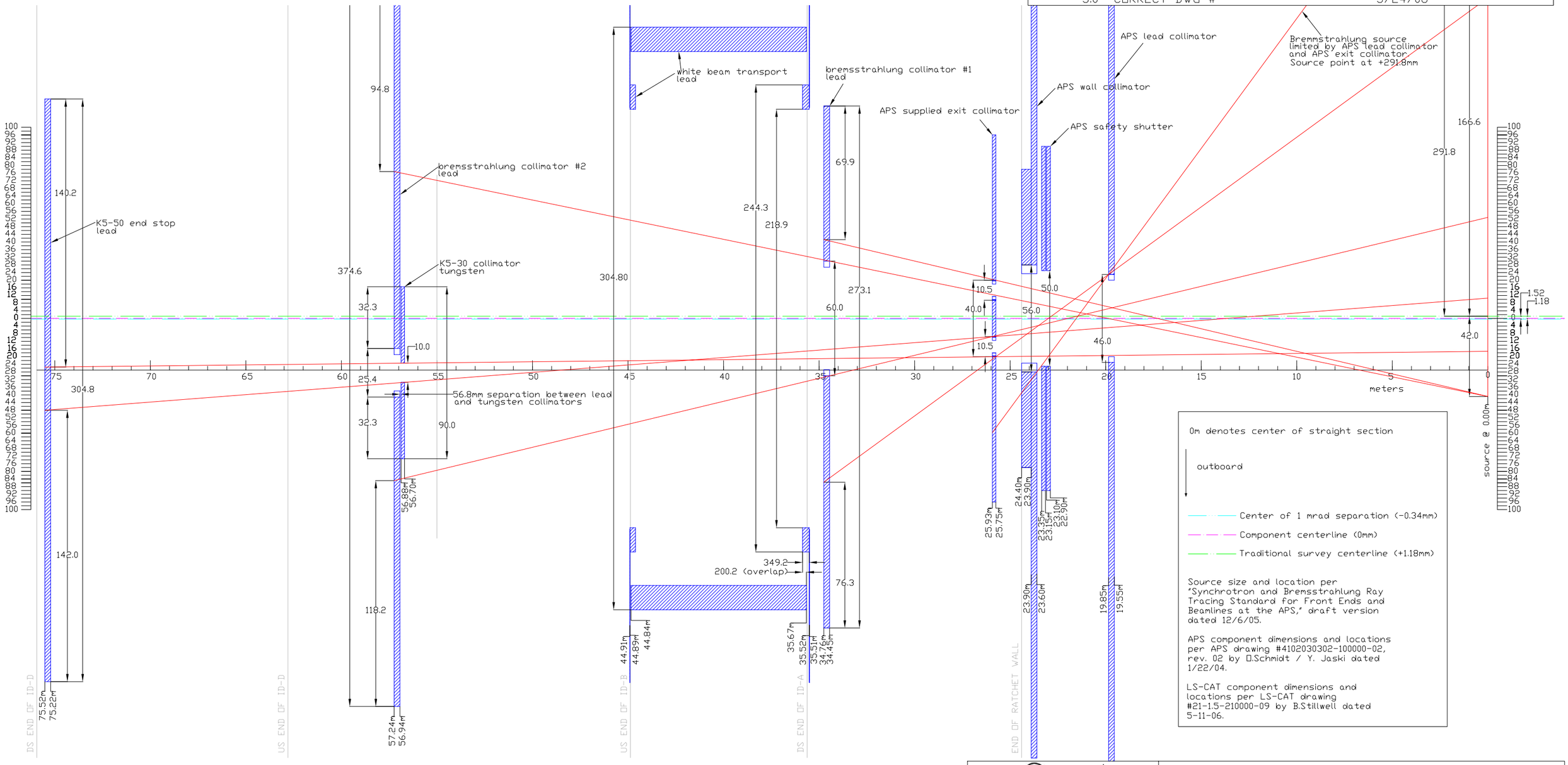
Appendix C: Survey Targets

Target Number	Distance from Nominal Source Point	Branch	Offset (m)
1	25	D	0.38
2	30	D	0.38
3	35	D	0.38
4	40	D	0.38
5	45	D	0.38
6	50	D	0.38
7	55	D	0.38
8	60	D	0.38
9	65	D	0.38
10	70	D	0.38
11	75	D	0.38
12	61.832	D	0.38
13	54.608	D	0.38
14	55.923	E	-0.38
15	56.69	E	0
16	56.684	E	-0.38
17	58.541	E	-0.38
18	59.578	E	-0.38
19	59.795	E	0
20	59.943	E	0
21	59.938	E	-0.38
22	60.833	E	-0.38
23	61.521	E	0
24	61.857	E	-0.38
25	63.02	E	-0.38
26	64.706	E	0
27	64.847	E	0.38
28	66.888	E	0.38
29	65.949	E	0

Target Number	Distance from Nominal Source Point	Branch	Offset (m)
30	68.781	E	0.38
31	70.786	E	0.38
32	70.481	E	0
33	54.756	E	-0.38
34	54.147	E	0
35	48.044	F	0
36	48.529	F	0.38
37	50.71	F	0
38	50.732	F	0.38
39	53.259	F	0.38
40	54.227	F	0
41	54.387	F	0.38
42	55.24	F	0.889
43	56.56	F	0.889
44	58.21	F	0.889
45	60.445	F	0
46	59.48	G	0
47	61.18	G	0.889
48	61.441	G	0
49	62.59	G	0.889
50	50	D	0.38
51	50.324	G	-0.38
52	51.099	G	0
53	51.636	G	-0.38
54	53	G	-0.38
55	54.347	G	0
56	54.629	G	-0.38
57	58.796	G	0

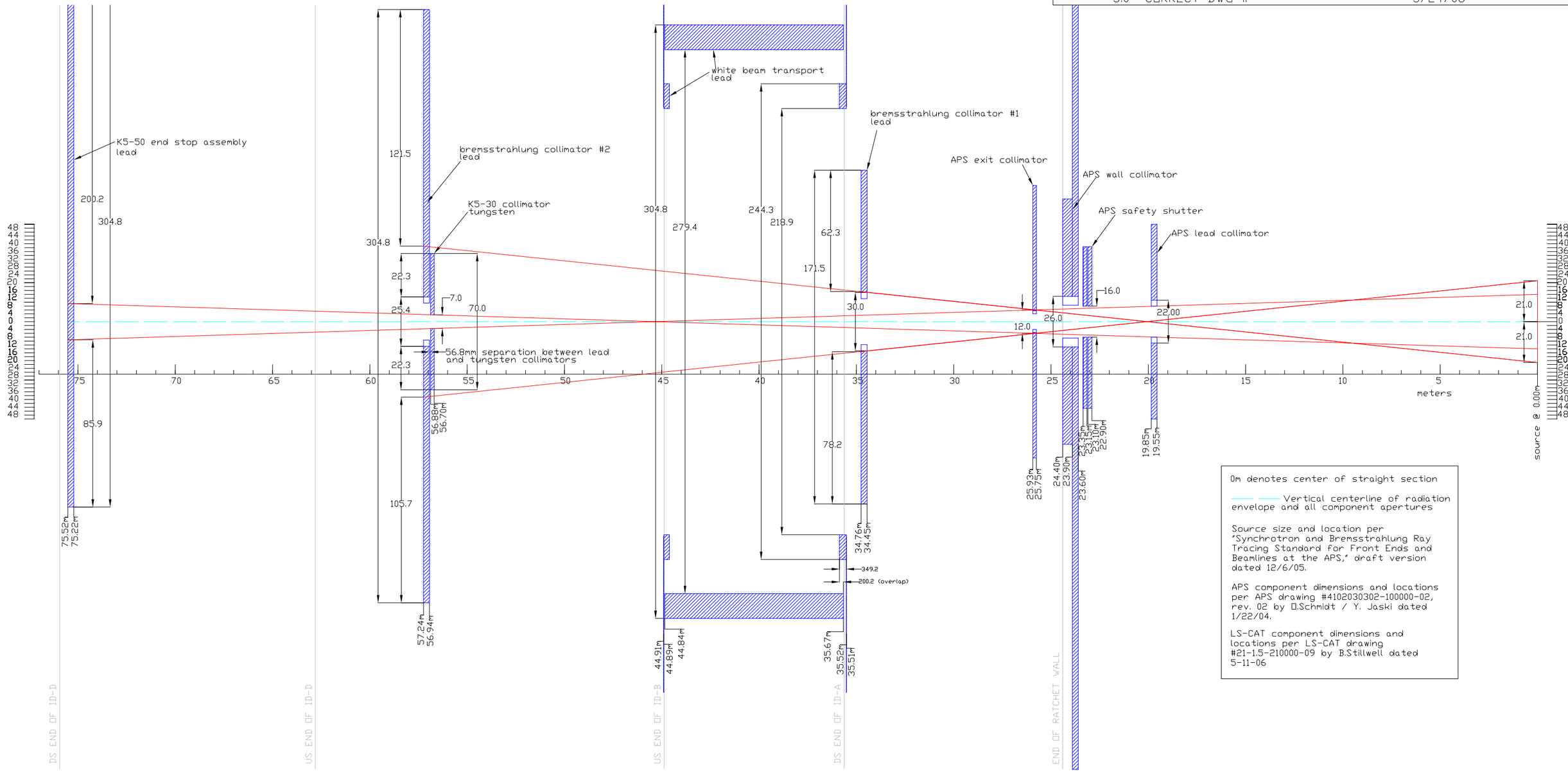
Appendix D: Ray tracings

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	4.0	MOVE SOURCE, ADD FE MASKS, WBT	3/23/06	
	5.0	CORRECT DWG #	5/24/06	



Dimensions are mm unless noted	NORTHWESTERN UNIVERSITY LIFE SCIENCES COLLABORATIVE (LS-CAT)		
5/24/06	EXTREMAL RAY TRACE 21-ID BREMSSTRAHLUNG PLAN VIEW (HORIZONTAL PROFILE)		
	SIZE B	FSCM NO.	DWG NO. 21-1.4.1-110000-05 REV 5.0
	SCALE Z:X=1:100	B. STILLWELL	SHEET 1 OF 1

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	4.0	MOVE SOURCE, ADD FE MASKS, WBT	3/23/06	
	5.0	CORRECT DWG #	5/24/06	



0m denotes center of straight section

——— ——— Vertical centerline of radiation envelope and all component apertures

Source size and location per "Synchrotron and Bremsstrahlung Ray Tracing Standard for Front Ends and Beamlines at the APS," draft version dated 12/6/05.

APS component dimensions and locations per APS drawing #4102030302-100000-02, rev. 02 by D.Schmidt / Y. Jaski dated 1/22/04.

LS-CAT component dimensions and locations per LS-CAT drawing #21-1.5-210000-09 by B.Stillwell dated 5-11-06

Dimensions are mm unless noted

5/24/06

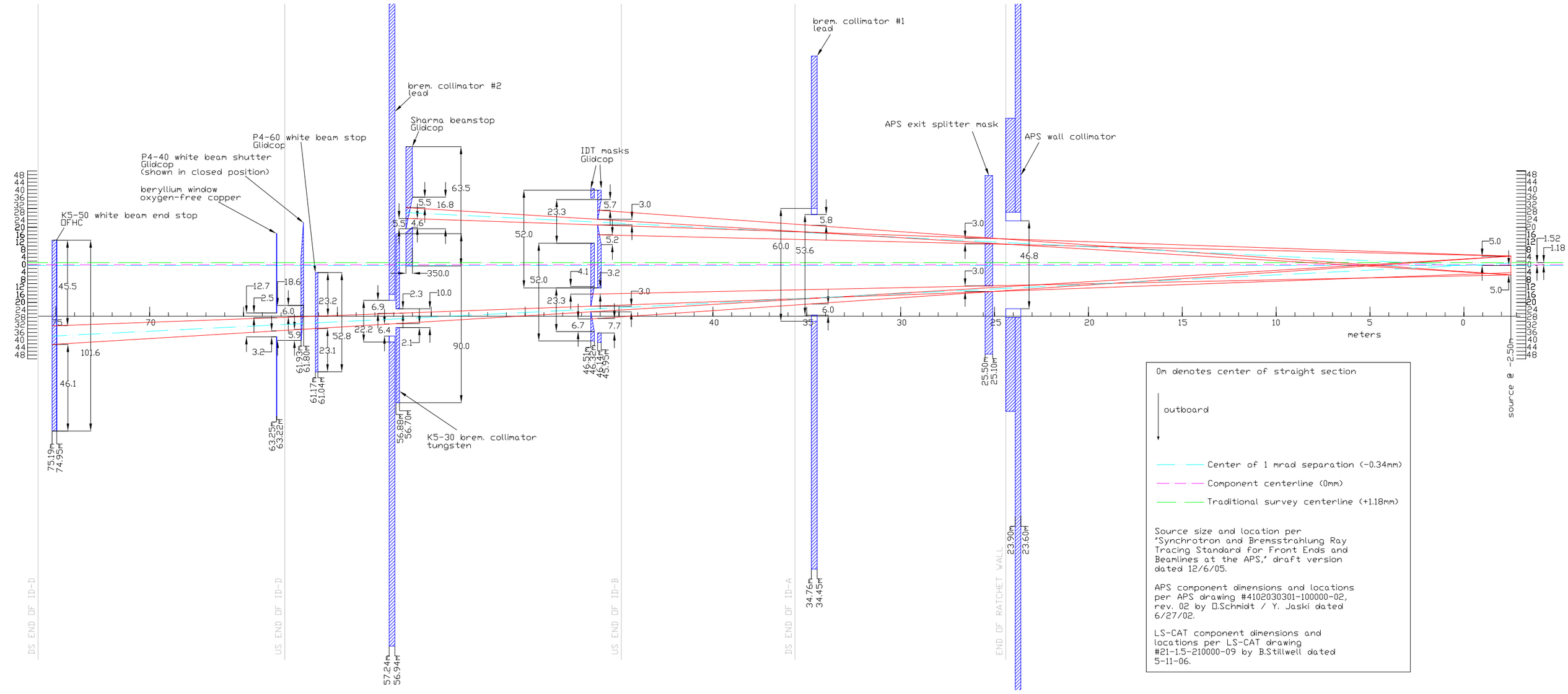
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

EXTREMAL RAY TRACE
21-ID BREMSSTRAHLUNG
ELEVATION VIEW (VERTICAL PROFILE)

SIZE	FSCM NO.	DWG NO.	REV
B		21-1.4.1-120000-05	5.0

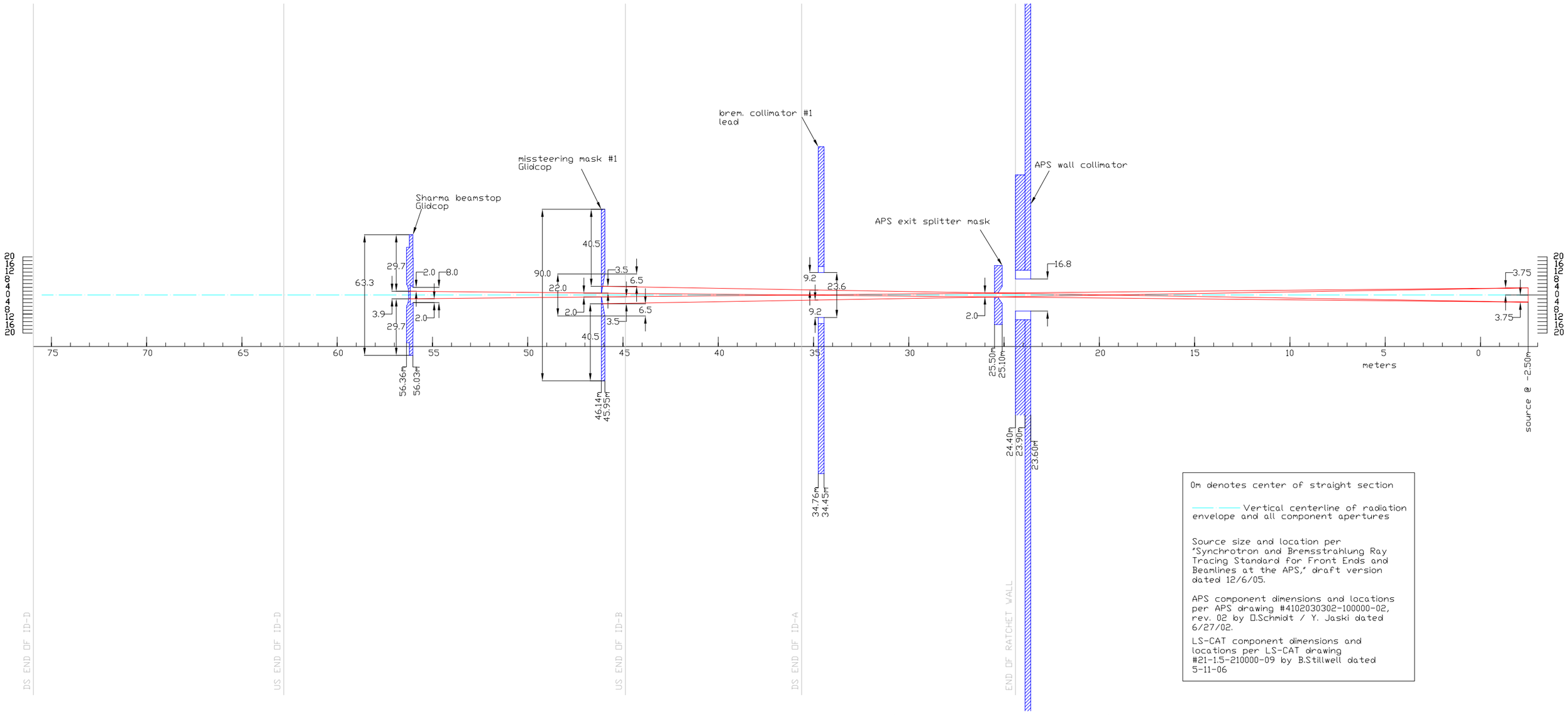
SCALE Z:Y=1:100 | B. STILLWELL | SHEET 1 OF 1

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	2.1	CORRECTED ROTATED COMPONENTS	2/9/06	
	3.0	MOVE SOURCE AND 2ND MASK, ADD BREM COLL	3/23/06	
	4.0	CORRECT DWG #	5/24/06	



Dimensions are mm unless noted	NORTHWESTERN UNIVERSITY LIFE SCIENCES COLLABORATIVE (LS-CAT)			
	EXTREMAL RAY TRACE 21-ID INBOARD AND OUTBOARD WHITE SYNCHROTRON BEAMS PLAN VIEW (HORIZONTAL PROFILE)			
	SIZE B	FSCM NO.	DWG NO. 21-1.4.1-210000-04	REV 4.0
5/24/06	SCALE Z:X=1:100	B. STILLWELL	SHEET 1 OF 1	

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	3.0	MOVE SOURCE, ADD BREM COLL	3/23/06	
	4.0	CORRECT DWG #	5/24/06	



0m denotes center of straight section

Vertical centerline of radiation envelope and all component apertures

Source size and location per "Synchrotron and Bremsstrahlung Ray Tracing Standard for Front Ends and Beamlines at the APS," draft version dated 12/6/05.

APS component dimensions and locations per APS drawing #4102030302-100000-02, rev. 02 by D.Schmidt / Y. Jaski dated 6/27/02.

LS-CAT component dimensions and locations per LS-CAT drawing #21-1.5-210000-09 by B.Stillwell dated 5-11-06

NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

EXTREMAL RAY TRACE
21-ID INBOARD WHITE SYNCHROTRON BEAM
ELEVATION VIEW (VERTICAL PROFILE)

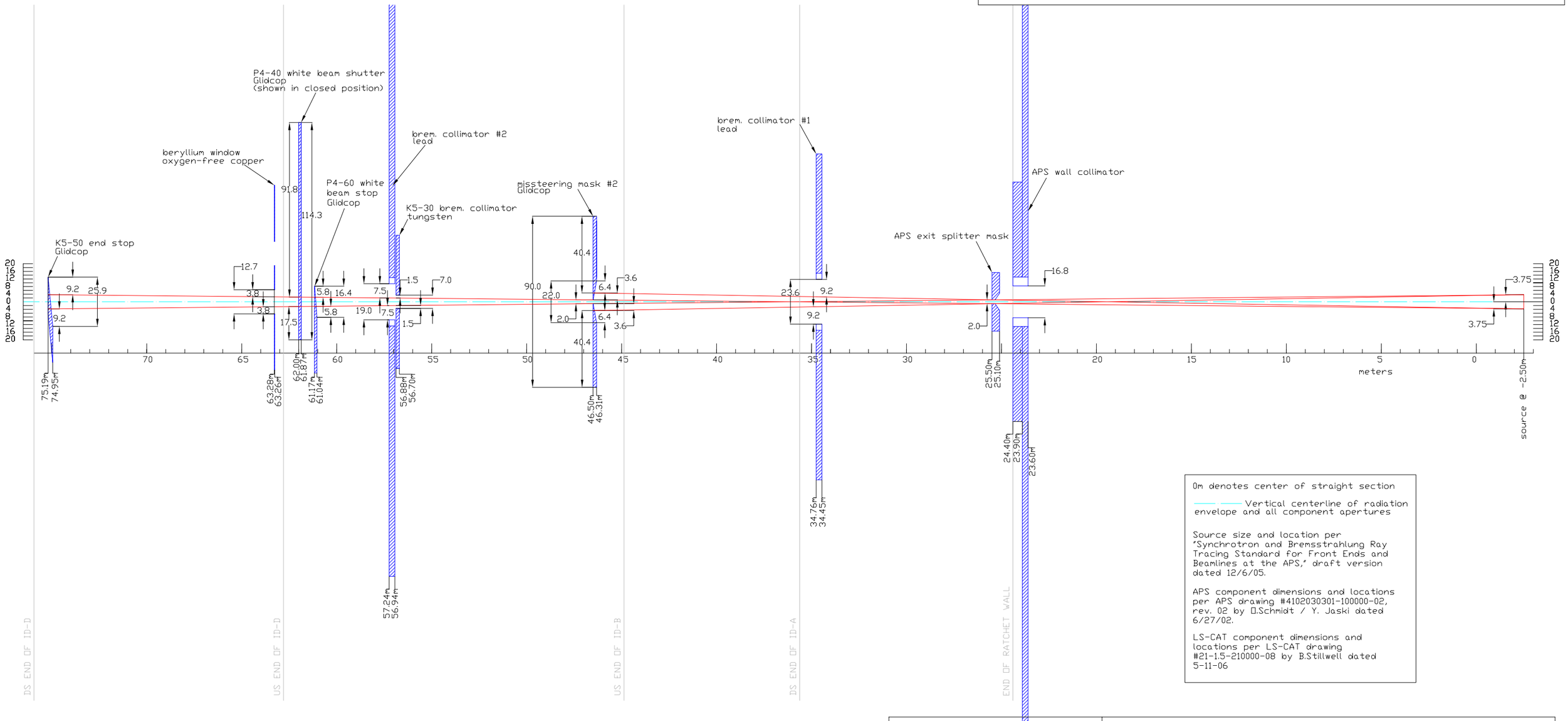
Dimensions are mm unless noted

5/24/06

SIZE	FSCM NO.	DWG NO.	REV
B		21-1.4.1-221000-04	4.0

SCALE Z:Y=1:100 | B. STILLWELL | SHEET 1 OF 1

REVISIONS				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	3.0	MOVE SOURCE AND 2ND MASK, ADD BREM COLL	3/23/06	
	4.0	CORRECT DWG #	5/24/06	



0m denotes center of straight section

——— ——— Vertical centerline of radiation envelope and all component apertures

Source size and location per "Synchrotron and Bremsstrahlung Ray Tracing Standard for Front Ends and Beamlines at the APS," draft version dated 12/6/05.

APS component dimensions and locations per APS drawing #4102030301-100000-02, rev. 02 by D.Schmidt / Y. Jaski dated 6/27/02.

LS-CAT component dimensions and locations per LS-CAT drawing #21-1.5-210000-08 by B.Stillwell dated 5-11-06

NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

EXTREMAL RAY TRACE
21-ID OUTBOARD WHITE SYNCHROTRON BEAM
ELEVATION VIEW (VERTICAL PROFILE)

Dimensions are mm unless noted

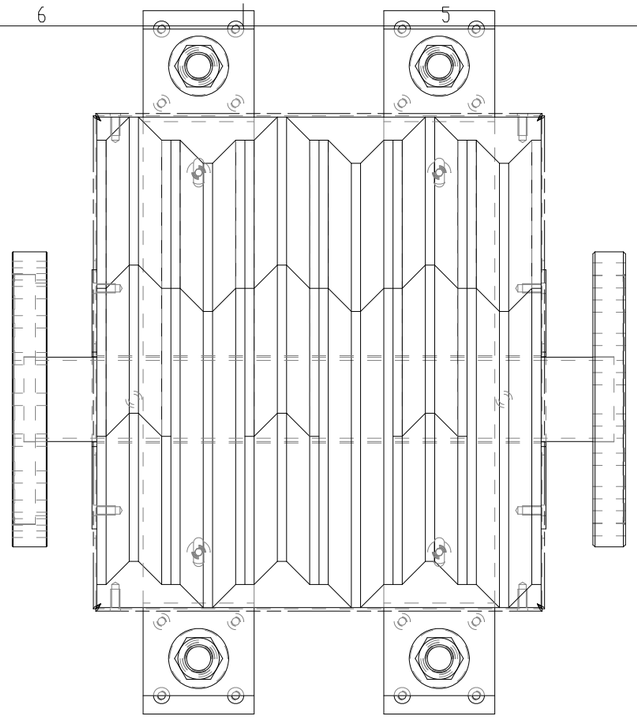
5/24/06

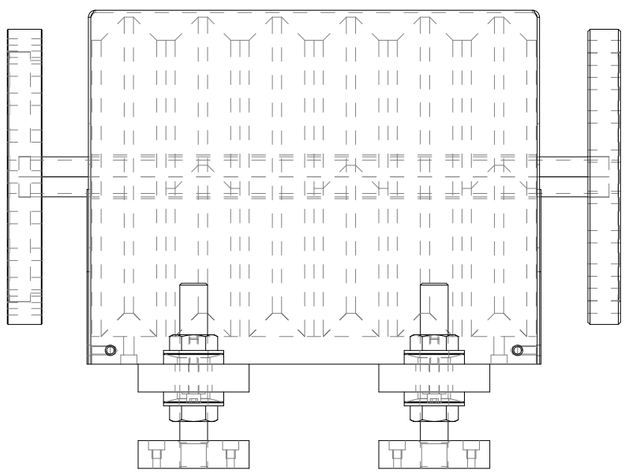
SIZE	FSCM NO.	DWG NO.	REV
B		21-1.4.1-222000-04	4.0

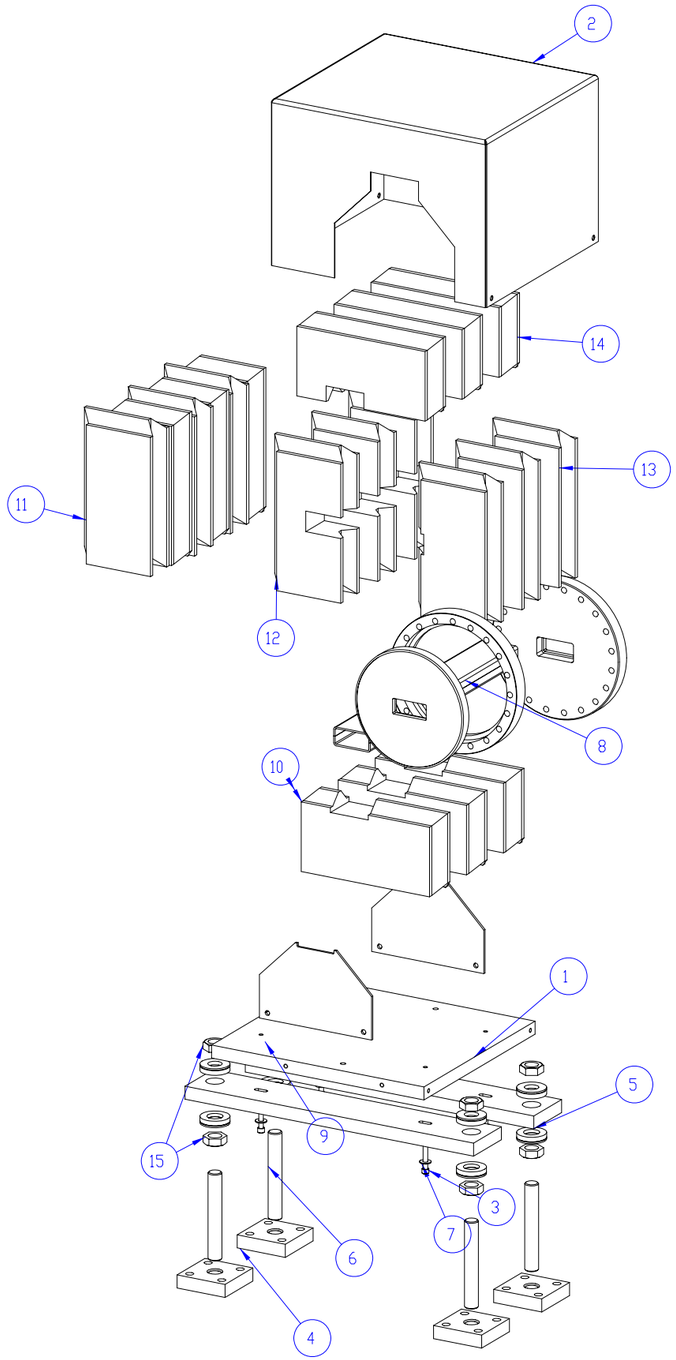
SCALE X:Y=1:100 | B. STILLWELL | SHEET 1 OF 1

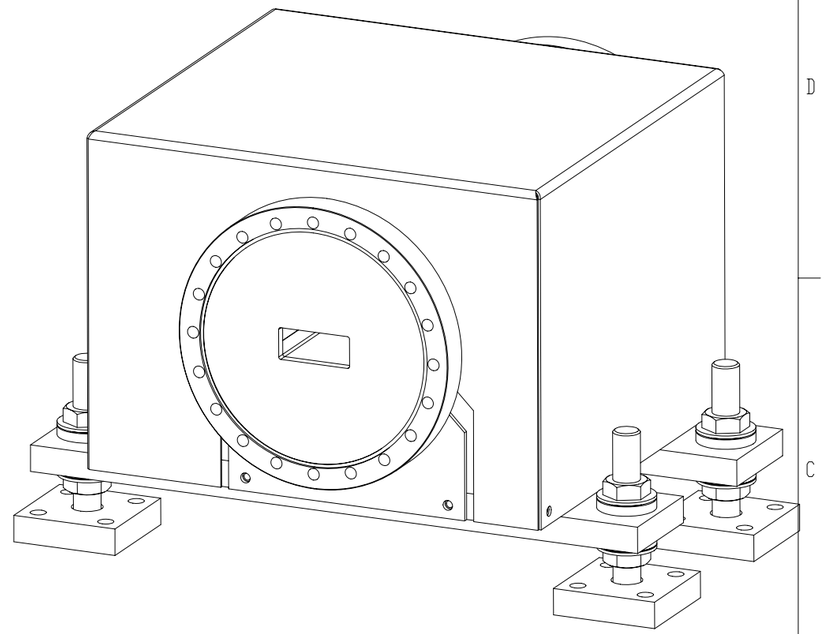
Appendix E: Component Drawings

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE









THIS DRAWING BASED ON A DESIGN BY D.SCHMIDT / ARGONNE NATIONAL LABORATORY
REF. DWG # 4102030110-100000

2 SOURCE OR EQUIVALENT
Mayco Industries
18 West Oxmoor Rd
Birmingham, AL 35209
800-749-6061

1 SOURCE OR EQUIVALENT
McMASTER-CARR
P.O. BOX 4355
CHICAGO, IL 60680-4355
630-833-0300

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
15	91078A240	NUT HEX 3/4-16 27/64 THK UNF-2B	STEEL ZINC PLATED GRADE 2	8
14	21_IDA_BREM_COLL_100009	CHEVRON LEAD BRICK 2" X 4" X 8" - TOP CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
13	21_IDA_BREM_COLL_100010	CHEVRON LEAD BRICK 2" X 4" X 8" - RIGHT CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
12	21_IDA_BREM_COLL_100008	CHEVRON LEAD BRICK 2" X 4" X 8" - LEFT CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
11	CHEVRON_LEAD_BRICK	CHEVRON LEAD BRICK 2" X 4" X 8"	LEAD GRD "C" ASTM B29; QQ-L-201	6
10	21_IDA_BREM_COLL_100007	CHEVRON LEAD BRICK - BOTTOM CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
9	91131A090	WASH SPHR .75 ID X 1.63 OD X .38 THK	12L14 STEEL	8
8	21_IDA_BREM_COLL_101000	TUBE ASSEMBLY	304 SST	1
7	92141A029	WASH FLT .25 NOM X .63 ODX .05 THK	18-8 SST	4
6	21_IDA_BREM_COLL_100006	THREADED STUD	STEEL, ZINC PLT	4
5	21_IDA_BREM_COLL_100005	SUPPORT BAR	6061-T6 ALUMINUM	2
4	21_IDA_BREM_COLL_100004	MOUNTING FOOT	6061-T6 ALUMINUM	4
3	92196A546	SHCS 1/4-20 UNC-2A X 1.50 LG	18-8 SST	4
2	21_IDA_BREM_COLL_100002	COVER	ALUM FLAT SHEET .063 THK	1
1	21_IDA_BREM_COLL_100001	BASE PLATE	ALUM ALLOY 6061-T6	1

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±.1, XX ±.03, XXX ±.010 ANGULAR ±.5°
SURFACE ROUGHNESS
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX. SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1 DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5

MODEL NAME: 21_IDA_BREM_COLL_100000
ELECTRONIC FILE NAME: N/A

DRAWN BY: B.STILLWELL 08-24-05
CHECKED BY:
DESIGNED BY: B.STILLWELL 08-24-05
RESPONSIBLE ENGINEER:
GROUP LEADER:
APPROVED BY:
RELEASE LEVEL: WIP
VERSION:
MATERIAL:

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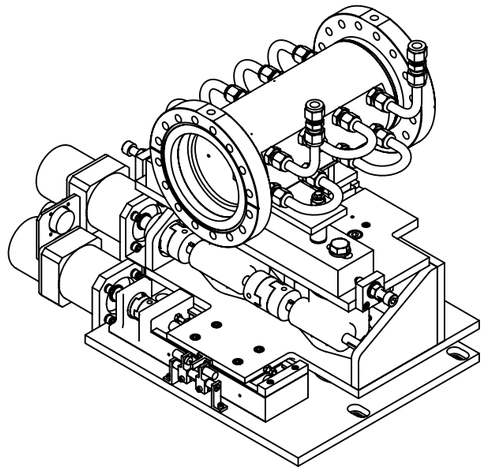
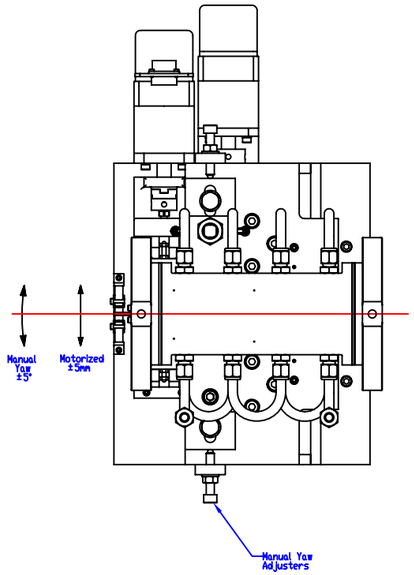
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE
Bitmap Image

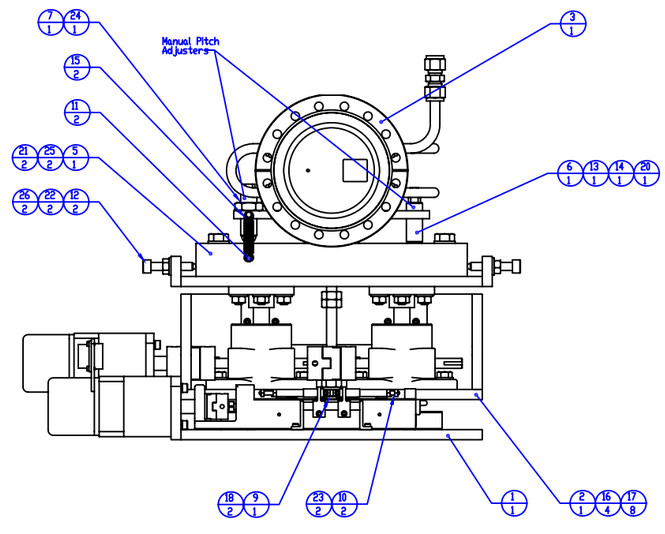
TITLE: (LS-CAT)
BREMSSTRAHLUNG COLLIMATOR #1
TOP LEVEL ASSEMBLY

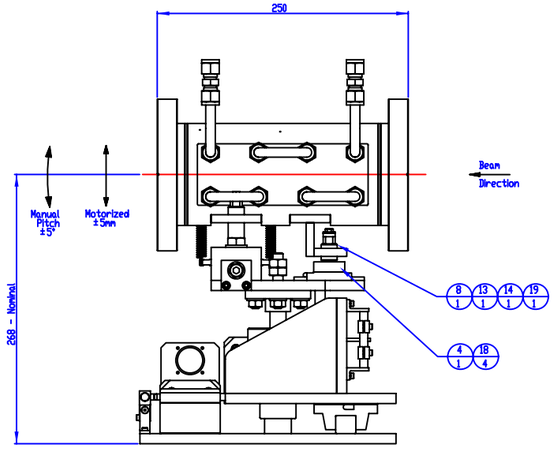
SIZE D | FSCM No | DRAWING NUMBER: 21-1.8.1.2-100000 | REV: 0.0
SCALE: 1:2 | DO NOT SCALE DRAWING | SHEET 1 OF 1

Item No.	Qty	Part Number	Description	Remarks
1	1	NEC-100-11-0011	Linear Slide Assembly	
2	1	NEC-100-11-0009	Jack Assembly	
3	1	LSC-101-11-0002	Fixed Aperture Assembly	
4	1	NEC-100-11-1017	Vee Plate	
5	1	NEC-100-11-1020	Adjustable Block	
6	1	NEC-100-11-1021	Domed Adjustment Screw (Bored)	
7	1	NEC-100-11-1022	Domed Adjustment Screw	
8	1	NEC-100-11-1023	Domed Adjustment Screw (Short)	
9	1	NEC-100-11-1027	Limit Switch Striker	
10	2	NEC-100-11-1028	Mechanical Stop	
11	2	ID-30-1-0-0-09	Spring Pin - Short	
12	2	NEC-100-11-1029	Mtg Plate Adjustment Screw	
13	2	401-2004 F	M4 Spherical Washer - Female	WDS
14	2	401-2004 M	M4 Spherical Washer - Male	WDS
15	2	LEM-120D-3	Extension Spring	Lee
16	4	SHCS-050-010	M5 x 10 Lg Hex Skt Hd Cap Screw	
17	8	SHCS-060-012	M6 x 12 Lg Hex Skt Hd Cap Screw	
18	6	SHCS-030-010	M3 x 10 Lg Hex Skt Hd Cap Screw	
19	1	HHS-040-0375	M4 x 37.5 Lg Hex Hd Screw	
20	1	HHS-040-075	M4 x 75 Lg Hex Hd Screw	
21	2	HHS-100-045	M10 x 45 Lg Hex Hd Screw	
22	2	NUT-080	M8 Hexagon Nut	
23	2	NUT-040	M4 Hexagon Nut	
24	1	FTN-NUT-160	M16 (Fine Series) Hex Thin Nut	
25	2	WAS-100	M10 Plain Washer	
26	2	WAS-080	M8 Plain Washer	









Material: See Notes

Tolerances UOS
Linear Dimensions: Upto 250 ± 0.25, Over 250 ± 0.5
Angular Dimensions: ± 0.25°
Surface Finish: N7

Scale: 1:2 (0, 50, 100)
All Dims in mm's UOS

Approx. Mass: 22594.06g

Drn. PB, Date. 11/7/15

Chkd. PB, Date. 11/7/15

Appvd. PBH, Date. 12/7/15

Used On: LSC

Instrument Design Technology Ltd
Daresbury Innovation Centre
Keckwick Lane, Daresbury
Runcorn, WA4 4FS, UK

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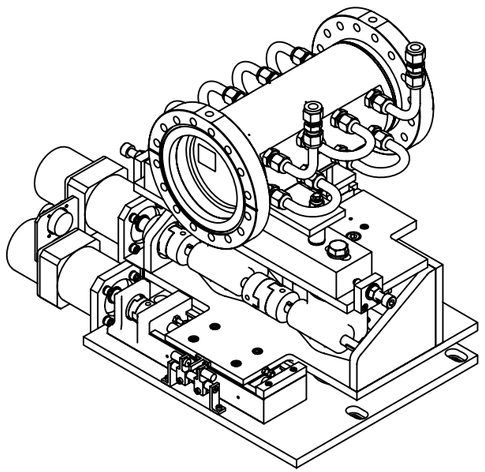
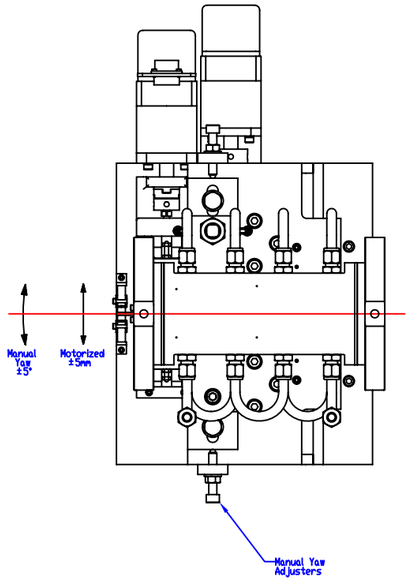
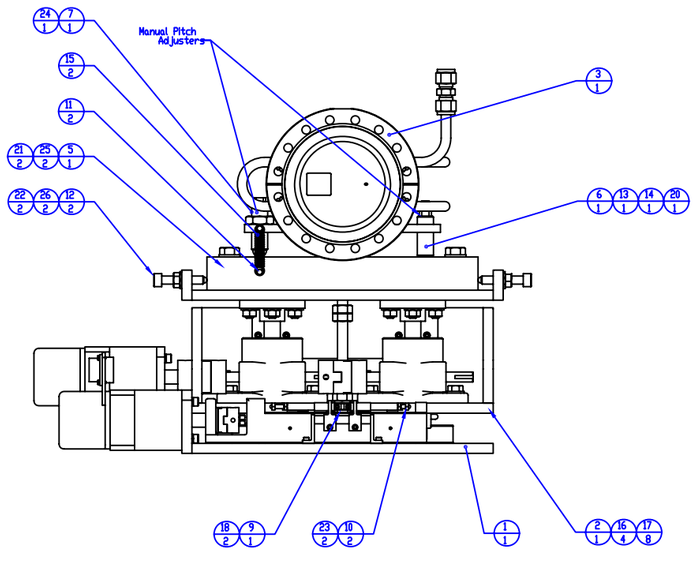
Title: Fixed Aperture Assembly Left Bypass Hole

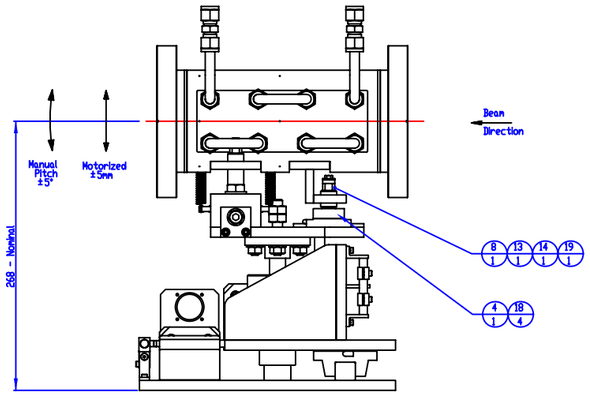
Size: A0, Drawing No: LSC-101-11-0003, Sheet 1 of 1, Issue: A

Item No.	Qty	Part Number	Description	Remarks
1	1	NEC-100-11-0011	Linear Slide Assembly	
2	1	NEC-100-11-0009	Jack Assembly	
3	1	LSC-101-11-0008	Fixed Aperture Mask Assy	
4	1	NEC-100-11-1017	Vee Plate	
5	1	NEC-100-11-1020	Adjustable Block	
6	1	NEC-100-11-1021	Domed Adjustment Screw (Bored)	
7	1	NEC-100-11-1022	Domed Adjustment Screw	
8	1	NEC-100-11-1023	Domed Adjustment Screw (Short)	
9	1	NEC-100-11-1027	Limit Switch Striker	
10	2	NEC-100-11-1028	Mechanical Stop	
11	2	ID-30-1-0-0-09	Spring Pin - Short	
12	2	NEC-100-11-1029	Mtg Plate Adjustment Screw	
13	2	401-2004 F	M4 Spherical Washer - Female	WDS
14	2	401-2004 M	M4 Spherical Washer - Male	WDS
15	2	LEM-120D-3	Extension Spring	Lee
16	4	SHCS-050-010	M5 x 10 Lg Hex Skt Hd Cap Screw	
17	8	SHCS-060-012	M6 x 12 Lg Hex Skt Hd Cap Screw	
18	6	SHCS-030-010	M3 x 10 Lg Hex Skt Hd Cap Screw	
19	1	HHS-040-0375	M4 x 37.5 Lg Hex Hd Screw	
20	1	HHS-040-075	M4 x 75 Lg Hex Hd Screw	
21	2	HHS-100-045	M10 x 45 Lg Hex Hd Screw	
22	2	NUT-080	M8 Hexagon Nut	
23	2	NUT-040	M4 Hexagon Nut	
24	1	FTN-NUT-160	M16 (Fine Series) Hex Thin Nut	
25	2	WAS-100	M10 Plain Washer	
26	2	WAS-080	M8 Plain Washer	



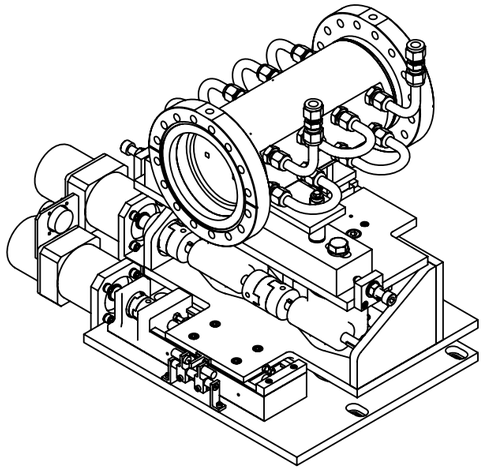


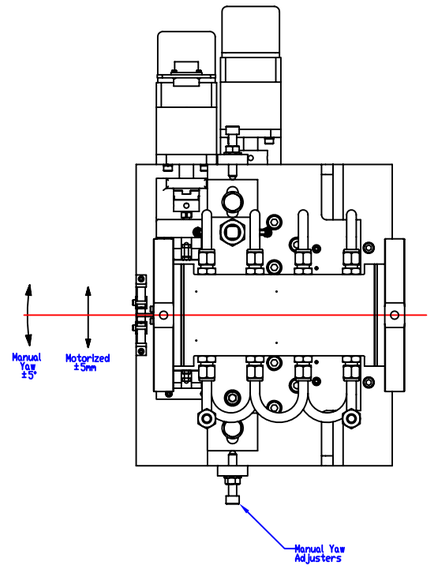


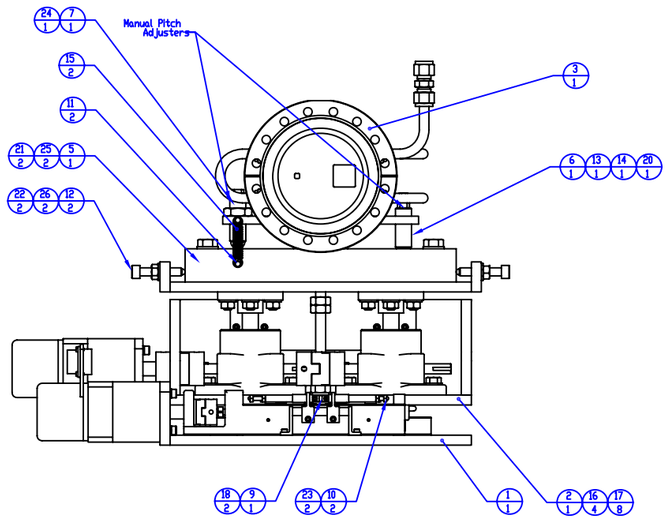


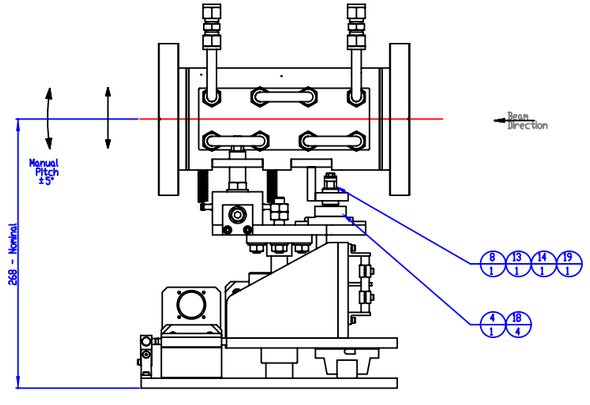
Material	See Notes
Tolerances UOS Linear Dimensions	Angular Dimensions
Upto 125 ± 0.25 Over 125 ± 0.5	± 0.25 °
Surface Finish	N7
Scale	1:2
	0 50 100
	All Dims in mm's UOS
Approx. Mass 35294.86g	Instrument Design Technology Ltd Daresbury Innovation Centre Keckwick Lane, Daresbury Runcorn, WA4 4FS, UK
Drn. PB Date. 12/7/05	Property of Instrument Design Technology Ltd. This document is to be published in confidence to the receiving party. It is not to be published nor its contents otherwise disseminated without the express written consent of Instrument Design Technology Ltd © Copyright Instrument Design Technology Ltd All rights reserved
Chd. PB Date. 12/7/05	Title Fixed Aperture Assembly Right Bypass Hole
Appd. PKM Date. 12/7/05	Size A0 · Drawing No LSC-101-11-0011 · Sheet1 of1 · Issue A
Used On LSC	

Item No.	Qty	Part Number	Description	Remarks
1	1	NEC-100-11-0011	Linear Slide Assembly	
2	1	NEC-100-11-0009	Jack Assembly	
3	1	LSC-101-11-0009	Power Limiting Mask Assy	
4	1	NEC-100-11-1017	Vee Plate	
5	1	NEC-100-11-1020	Adjustable Block	
6	1	NEC-100-11-1021	Domed Adjustment Screw (Bored)	
7	1	NEC-100-11-1022	Domed Adjustment Screw	
8	1	NEC-100-11-1023	Domed Adjustment Screw (Short)	
9	1	NEC-100-11-1027	Limit Switch Striker	
10	2	NEC-100-11-1028	Mechanical Stop	
11	2	ID-30-1-0-0-09	Spring Pin - Short	
12	2	NEC-100-11-1029	Mtg Plate Adjustment Screw	
13	2	401-2004 F	M4 Spherical Washer - Female	WDS
14	2	401-2004 M	M4 Spherical Washer - Male	WDS
15	2	LEM-120D-3	Extension Spring	Lee
16	4	SHCS-050-010	M5 x 10 Lg Hex Skt Hd Cap Screw	
17	8	SHCS-060-012	M6 x 12 Lg Hex Skt Hd Cap Screw	
18	6	SHCS-030-010	M3 x 10 Lg Hex Skt Hd Cap Screw	
19	1	HHS-040-0375	M4 x 37.5 Lg Hex Hd Screw	
20	1	HHS-040-075	M4 x 75 Lg Hex Hd Screw	
21	2	HHS-100-045	M10 x 45 Lg Hex Hd Screw	
22	2	NUT-000	M3 Hexagon Nut	
23	2	NUT-040	M4 Hexagon Nut	
24	1	FTN-NUT-160	M16 (Fine Series) Hex Thin Nut	
25	2	WAS-100	M10 Plain Washer	
26	2	WAS-080	M8 Plain Washer	



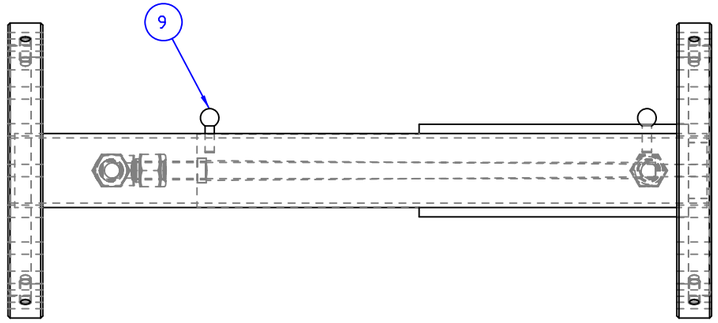


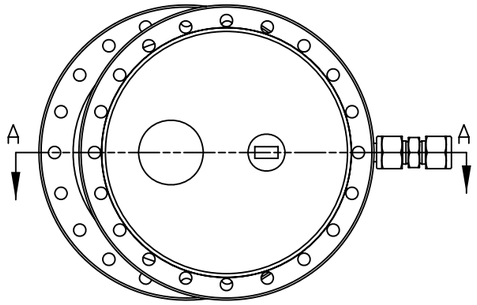




Material: See Notes			
Tolerances: UOS Linear Dimensions Upto 125 ± 0.25 Over 125 ± 0.5		Angular Dimensions ± 0.25 °	Surface Finish N7
Scale: 1:2	0 50 100 All Dims in mm's UOS		
Approx. Mass: 35678.2kg	Instrument Design Technology Ltd Daresbury Innovation Centre Keckwick Lane, Daresbury Runcorn, WA4 4FS, UK		
Drn. PB Date. 12/7/05	Property of Instrument Design Technology Ltd. this document is to be published in confidence to the receiving party. It is not to be published nor its contents otherwise disseminated without the express written consent of Instrument Design Technology Ltd © Copyright Instrument Design Technology Ltd all rights reserved		
Chd. PB Date. 12/7/05	Title Power Limiting Assembly Left Bypass Hole		
Appd. PKM Date. 12/7/05	Size A0	Drawing No LSC-101-11-0012	Sheet1 of1
Used On LSC			Issue A

NOTE: THIS DRAWING REPRESENTS PART OF A MODIFICATION OF A DESIGN BY
B.RUSTHOVEN / ARGONNE NATIONAL LABORATORY REF DWG#31050403-400010





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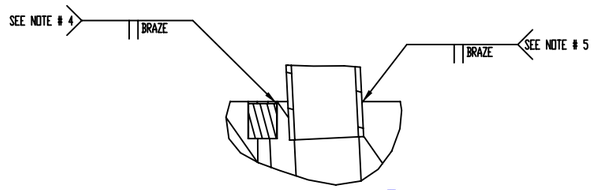
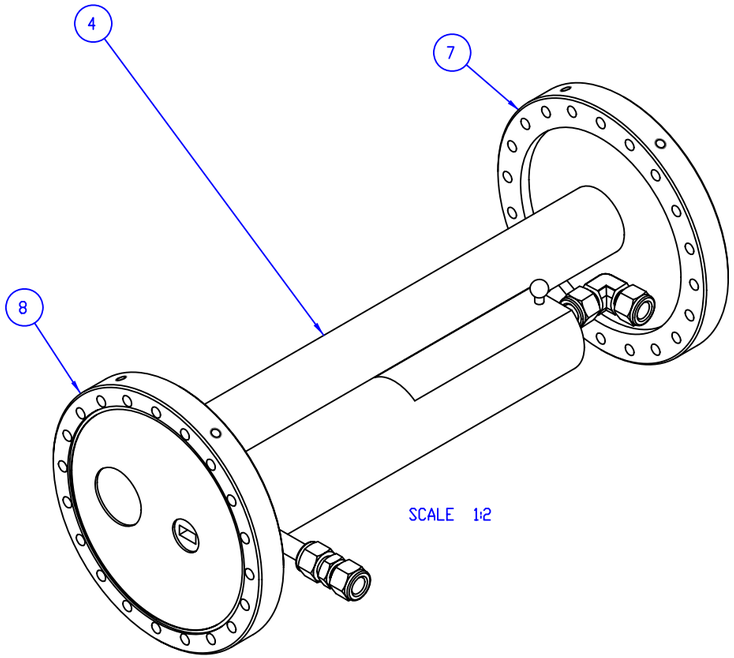
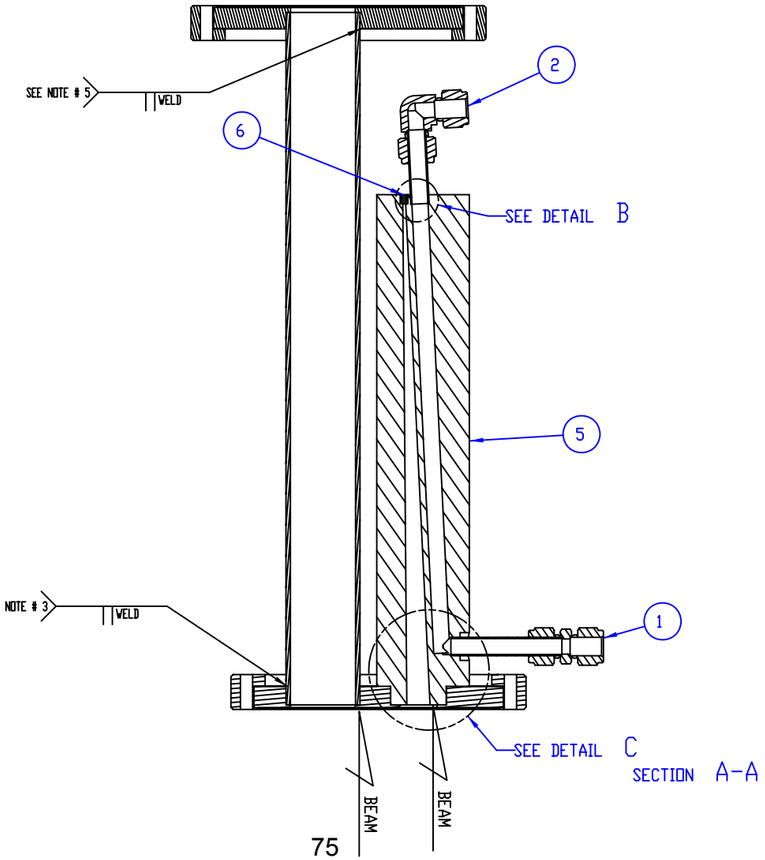


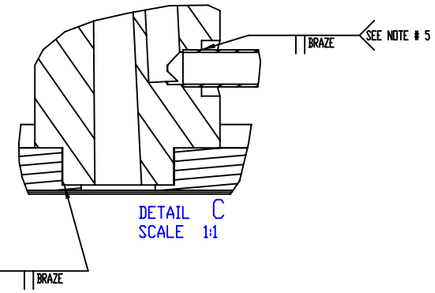
DETAIL B
SCALE 2:1



SCALE 1:2



SECTION A-A



DETAIL C
SCALE 1:1

NOTE:

1 CLEAN COMPONENTS FOR VACUUM SERVICE
(10 $^{-10}$Tor.). IF APPLICABLE PRIOR TO ASSEMBLY.

2 ALL WELDS TO BE INTERNAL EXCEPT AS NOTED, OR REQ'D, ALL EXT WELDS TO HAVE FULL PENETRATION

3 WELD JOINT MUST BE VACUUM TIGHT AND TESTED NO LEAK SHOULD BE DETECTABLE USING THE MOST SENSITIVE SCALE OF A HELIUM MASS SPECTROMETER WITH A MINIMUM SENSITIVITY OF 1.5x10 $^{-10}$ (Tor./Sec) OF HELIUM OR BETTER.

4 BRAZE JOINT MUST BE VACUUM TIGHT AND TESTED NO LEAK SHOULD BE DETECTABLE USING THE MOST SENSITIVE SCALE OF A HELIUM MASS SPECTROMETER WITH A MINIMUM SENSITIVITY OF 1.5x10 $^{-10}$ (Tor/Sec) OF HELIUM OR BETTER.

5 BRAZE TO BE WATER TIGHT & TESTED PER E ASTM 1003-84 @ 250 P.S.I..

NOTE:

TITLE BLOCK TOLERANCES TO BE SUPERSEDED AS FOLLOWS:

DECIMALS:
.XX .01
.XXX .005

SURFACE TEXTURE: 63

② SOURCE OR EQUIVALENT
McMASTER CARR
P.O. BOX 4355
CHICAGO, IL 60680-4355
(630) 833-0300
www.mcmaster.com

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
② 9	8484A13	TOOLING BALL - 1/4 SHANK	STL	2
8	21_IDB_BEAMSTOP_US_FLANGE	BEAM STOP FLANGE	SEE PARTS LIST	1
7	21_IDB_BEAMSTOP_DS_FLANGE	BEAM STOP DOWNSTREAM FLANGE	SEE PARTS LIST	1
6	21_IDB_BEAMSTOP_BRAZE_PLUG	BEAM STOP BRAZE PLUG	GLIDCOP AL-15	1
5	21_IDB_BEAMSTOP_CU_BLOCK	BEAM STOP	GLIDCOP AL-15	1
4	2" SST TUBING	2" 11 GA WALL SEAMLESS TUBING	304 SST TUBING	1
3	1/2" SST TUBING	1/2 20 GA SEAMLESS TUBING	304 SST TUBING	2
② 2	SS-810-9	ELBOW 1/2 X 1/2 TUBING 90	304 SST	1
② 1	SS-810-6	CONNECTOR BULKHEAD 1/2 TUBE	304 SST	1

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±1 XX ±.03 .XXX ±.010
ANGULAR ±.5°
SURFACE ROUGHNESS
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX. SURFACE TEXTURE IN ACCORDANCE WITH LATEST ASME B46.1-2002
DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH LATEST ASME Y14.5M-1994
MODEL NAME:
ELECTRONIC FILE NAME: 21_IDB_BEAMSTOP.DRW

DRAWN BY: B.STILLWELL — DATE: 18-Feb-05
CHECKED BY: — DATE:
DESIGNED BY: B.STILLWELL — DATE: 18-Feb-05
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:
MATERIAL: SEE PARTS LIST

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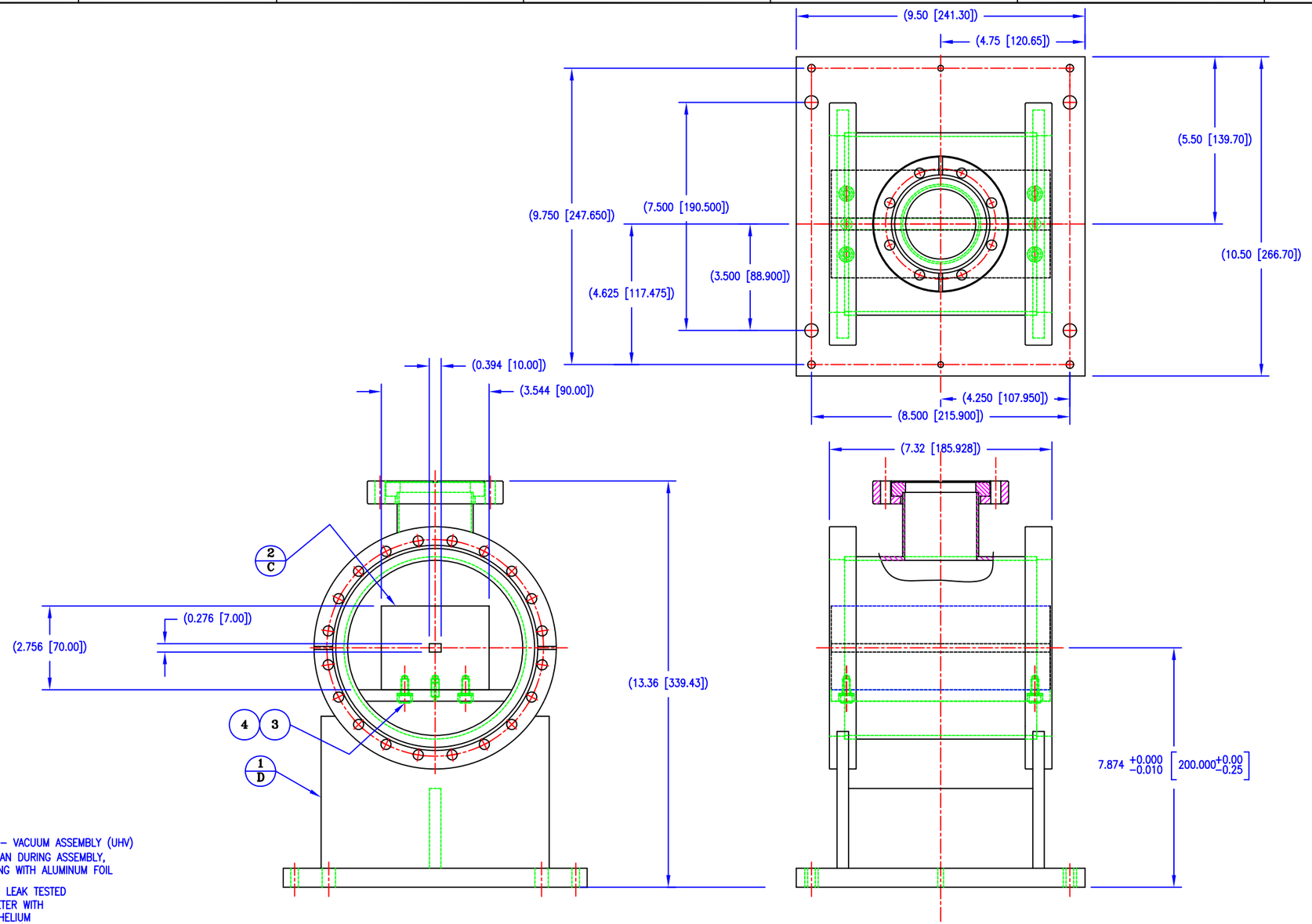
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE

TITLE:
21-IDB
BEAMSTOP
ASSEMBLY

SIZE D | FSCM No | DRAWING NUMBER: 21-1.8.1.18-100000-00 | REV: 0.0
SCALE: AS SHOWN | DO NOT SCALE DRAWING | SHEET 1 OF 1



NOTES:

1. THIS IS A ULTRA – HIGH – VACUUM ASSEMBLY (UHV) KEEP THE UHV PARTS CLEAN DURING ASSEMBLY, AND WRAP UP FOR PACKING WITH ALUMINUM FOIL
2. THE ASSEMBLY SHALL BE LEAK TESTED USING A MASS SPECTROMETER WITH MINIMUM SENSITIVITY FOR HELIUM OF 2X 10 −10 STANDARD CC/SEC PER LEAK METER DIVISION, SUCH AS:

   ALCATEL ASM−110TCL
   VARIAN NCR 925 OR 936
   VEECO MS−9, MS−90 OR MS−18
   Du PONT CEC 24−120B

   CALIBRATION OF THE LEAK DETECTOR SENSITIVITY SHALL BE PERFORMED JUST PRIOR TO TESTING.

   FINAL TEST WILL CONSIST OF SURROUNDING THE ASSEMBLY (BAGGING) WITH HELIUM. THE ASSEMBLY WILL BE REJECTED IF A 2% DEFLECTION IN THE MOST SENSITIVE RANGE OF THE LEAK DETECTOR IS SENSED WITHIN 1 MIN.
3. HANDLING: THE COMPONENTS SHALL BE HANDLED WITH WHITE SILK GLOVES AND WRAPPED WITH ALUMINUM FOIL TYPE DRY ANNEAL AND UNDER CLEAN ROOM CONDITIONS
4. DIMENSIONS IN [ ] ARE MILLIMETERS AND FOR REF. ONLY

ITEM	DWG/PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
4		1/4 ID STD. LOCK WASHER	SST	4
3		VENTED SKT. HD. SCR. 1/4−20 UNC X 0.50" LG.	SST/SILVER PLATED	4
2	K5−300002−00	K5 −30 COLLIMATOR	TUNGSTEN	1
1	K5−300001−00	K5 COLLIMATOR WELDMENT−ASSEMBLY		1

PARTS LIST

SYM	CHANGE DESCRIPTION	BY	CHKD	DATE
	REVISIONS			

UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±.1 [2.5]; .XX ±.01 [0.25]; .XXX ±.005 [0.13]
ANGULAR: ±0°30'
SURFACE ROUGHNESS 125
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX.
SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1
DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5
DO NOT SCALE DRAWING

LOG NUMBER: K5300000−00

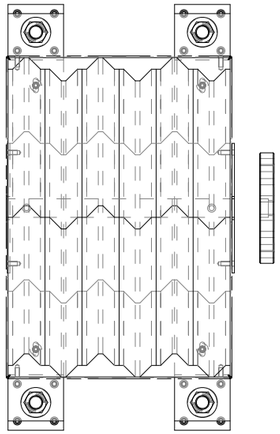
THIS DRAWING IS THE PROPERTY OF ARGONNE NATIONAL LABORATORY

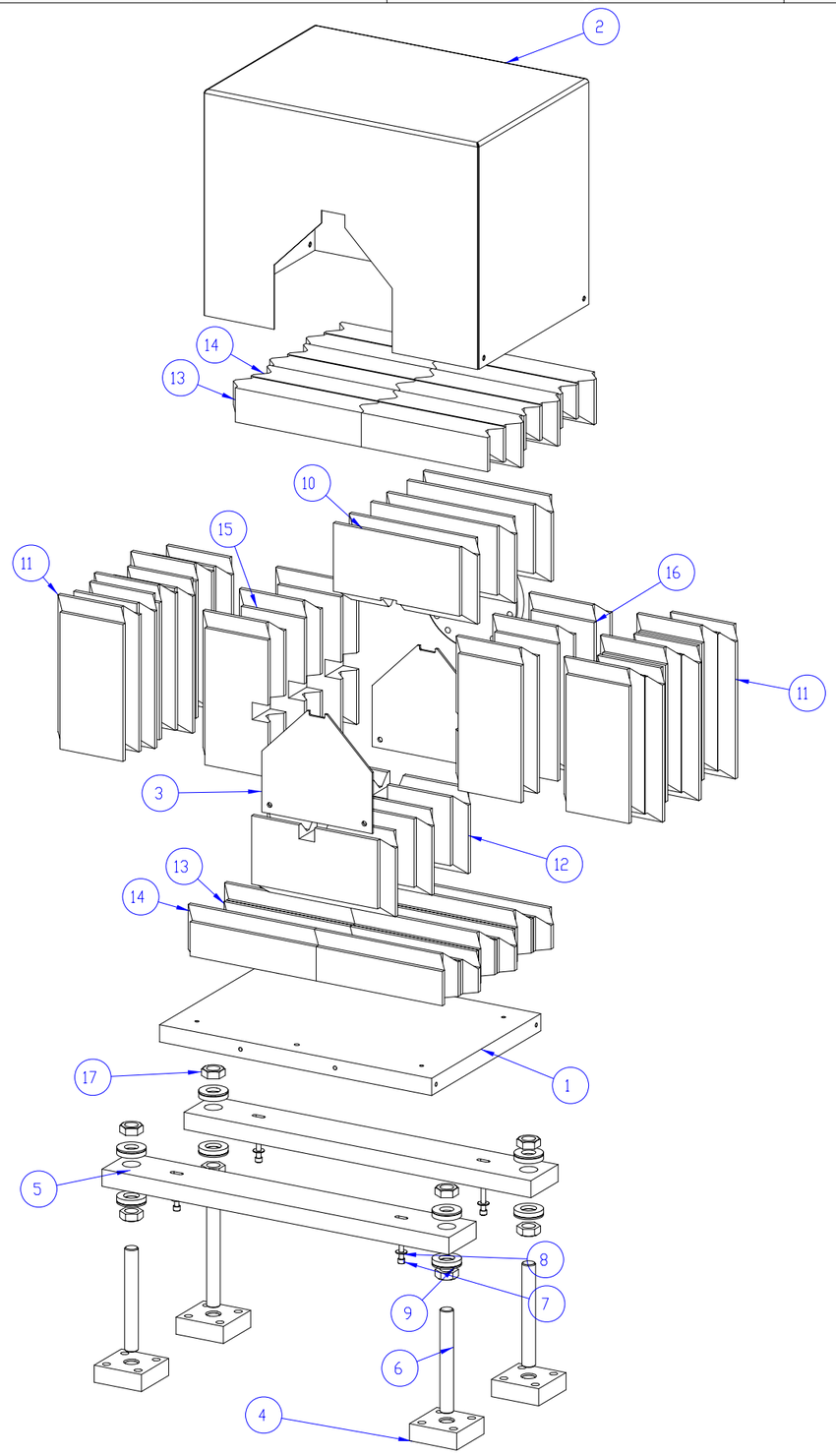
DRAWN BY: corcuera	DATE: 2/22/99	CHIEF DESIGN ENGINEER: D. SHU	DATE: 3/19/99
CHECKED BY: J. CHANG	DATE: 3/15/99	GP LEADER: T. KUZAY	DATE: 3/22/99
DESIGNER: CORCUERA/CHANG	DATE: 2/22/99	PROJECT MGR.	DATE
RESPONSIBLE ENGINEER: J. CHANG	DATE: 3/15/99	APPROVED/RELEASED	DATE

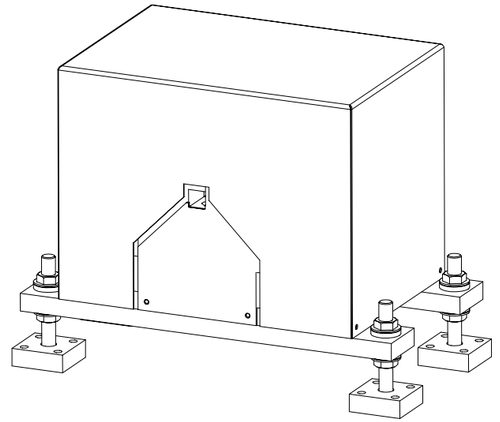
TITLE: ADVANCED PHOTON SOURCE
K5−30 4−ID−B COLLIMATOR ASSEMBLY

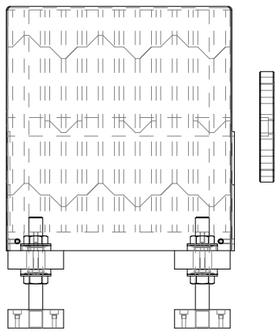
MATERIAL: SEE PARTS LIST
SCALE: 1:2
SHEET: 1 of 1
SIZE: D
DRAWING NUMBER: K5−300000−00

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE









SCALE 1:4

THIS DRAWING BASED ON A DESIGN BY D.SCHMIDT / ARGONNE NATIONAL LABORATORY
REF. DWG # 4102030110-100000

2 SOURCE OR EQUIVALENT
Mayco Industries
18 West Oxmoor Rd
Birmingham, AL 35209
800-749-6061

1 SOURCE OR EQUIVALENT
McMASTER-CARR
P.O. BOX 4355
CHICAGO, IL 60680-4355
630-833-0300

ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
17	91078A240	NUT HEX 3/4-16 27/64 THK UNF-2B	STEEL ZINC PLATED GRADE 2	8
16	21_IDC_BREM_COLL_100010	CHEVRON LEAD BRICK 2" X 4" X 8" - RIGHT CUT	SEE BOM	3
15	21_IDC_BREM_COLL_100008	CHEVRON LEAD BRICK 2" X 4" X 8" - LEFT CUT	SEE BOM	3
14	21_IDC_BREM_COLL_1000011	HALF CHEVRON BRICK - MALE SIDE	LEAD GRD "C" ASTM B29; QQ-L-201	12
13	21_IDC_BREM_COLL_1000012	HALF CHEVRON BRICK - FEMALE SIDE	LEAD GRD "C" ASTM B29; QQ-L-201	12
12	21_IDC_BREM_COLL_100009	CHEVRON LEAD BRICK 2" X 4" X 8" - BOTTOM CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
11	CHEVRON_LEAD_BRICK	CHEVRON LEAD BRICK 2" X 4" X 8"	LEAD GRD "C" ASTM B29; QQ-L-201	12
10	21_IDC_BREM_COLL_100007	CHEVRON LEAD BRICK - 2" X 4" X 8" - TOP CUT	LEAD GRD "C" ASTM B29; QQ-L-201	3
9	91131A090	WASH SPHR .75 ID X 1.63 OD X .38 THK	12L14 STEEL	8
8	92141A029	WASH FLT .25 NOM X .63 ODX .05 THK	18-8 SST	4
7	92196A546	SHCS 1/4-20 UNC-2A X 1.50 LG	18-8 SST	4
6	21_IDC_BREM_COLL_100006	THREADED STUD	STEEL, ZINC PLT	4
5	21_IDC_BREM_COLL_100005	SUPPORT BAR	6061-T6 ALUMINUM	2
4	21_IDC_BREM_COLL_100004	MOUNTING FOOT	6061-T6 ALUMINUM	4
3	21_IDC_BREM_COLL_101000	TUBE ASSEMBLY		1
2	21_IDC_BREM_COLL_100002	COVER	ALUM FLAT SHEET .063 THK	1
1	21_IDC_BREM_COLL_100001	BASE PLATE	ALUM ALLOY 6061-T6	1

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±.1, .XX ±.03, .XXX ±.010
ANGULAR ±.5°
SURFACE ROUGHNESS
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX.
SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1
DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5

MODEL NAME: 21_IDC_BREM_COLL_100000
ELECTRONIC FILE NAME: N/A

DRAWN BY: B.STILLWELL 08-24-05
CHECKED BY: DATE
DESIGNED BY: B.STILLWELL 08-24-05
RESPONSIBLE ENGINEER: DATE
GROUP LEADER: DATE
APPROVED BY: DATE
RELEASE LEVEL: WIP VERSION:
MATERIAL:

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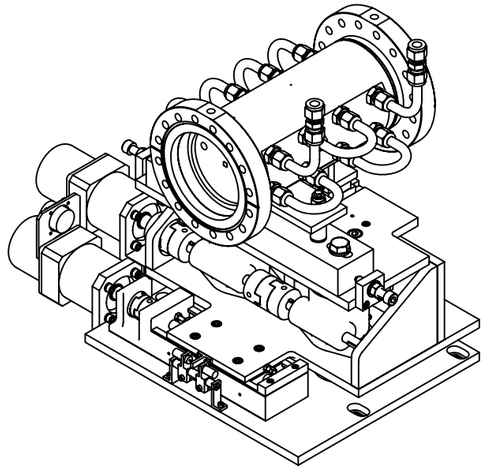
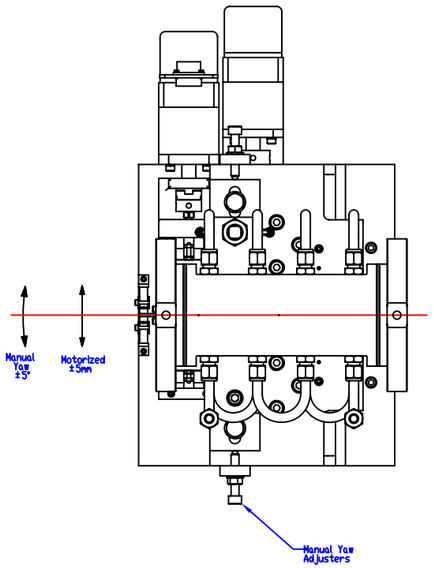
Bitmap Image
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE

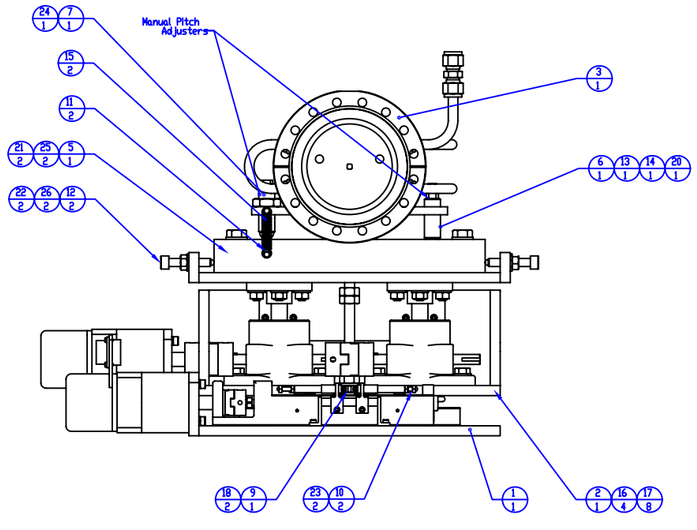
TITLE: (LS-CAT)
BREMSSTRAHLUNG COLLIMATOR #2
TOP LEVEL ASSEMBLY

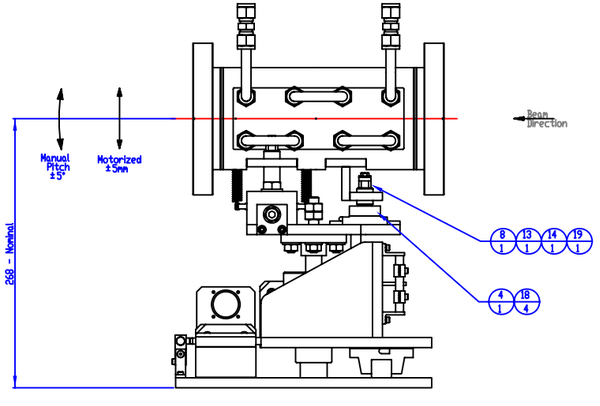
SIZE: D | FSCM No | DRAWING NUMBER: 21-1.8.1.21-100000 | REV: 0.0
SCALE: 1:2 | DO NOT SCALE DRAWING | SHEET 1 OF 1

Item No.	Qty	Part Number	Description	Remarks
1	1	NEC-100-11-0011	Linear Slide Assembly	
2	1	NEC-100-11-0009	Jack Assembly	
3	1	LSC-101-11-0010	Power Limiting Mask Assy	
4	1	NEC-100-11-1017	Vee Plate	
5	1	NEC-100-11-1020	Adjustable Block	
6	1	NEC-100-11-1021	Domed Adjustment Screw (Bored)	
7	1	NEC-100-11-1022	Domed Adjustment Screw	
8	1	NEC-100-11-1023	Domed Adjustment Screw (Short)	
9	1	NEC-100-11-1027	Limit Switch Striker	
10	2	NEC-100-11-1028	Mechanical Stop	
11	2	ID-30-1-0-0-09	Spring Pin - Short	
12	2	NEC-100-11-1029	Mtg Plate Adjustment Screw	
13	2	401-2004 F	M4 Spherical Washer - Female	WDS
14	2	401-2004 M	M4 Spherical Washer - Male	WDS
15	2	LEM-120D-3	Extension Spring	Lee
16	4	SHCS-050-010	M5 x 10 Lg Hex Skt Hd Cap Screw	
17	8	SHCS-060-012	M6 x 12 Lg Hex Skt Hd Cap Screw	
18	6	SHCS-030-010	M3 x 10 Lg Hex Skt Hd Cap Screw	
19	1	HHS-040-0375	M4 x 37.5 Lg Hex Hd Screw	
20	1	HHS-040-075	M4 x 75 Lg Hex Hd Screw	
21	2	HHS-100-045	M10 x 45 Lg Hex Hd Screw	
22	2	NUT-080	M8 Hexagon Nut	
23	2	NUT-040	M4 Hexagon Nut	
24	1	FTN-NUT-160	M16 (Fine Series) Hex Thin Nut	
25	2	WAS-100	M10 Plain Washer	
26	2	WAS-080	M8 Plain Washer	



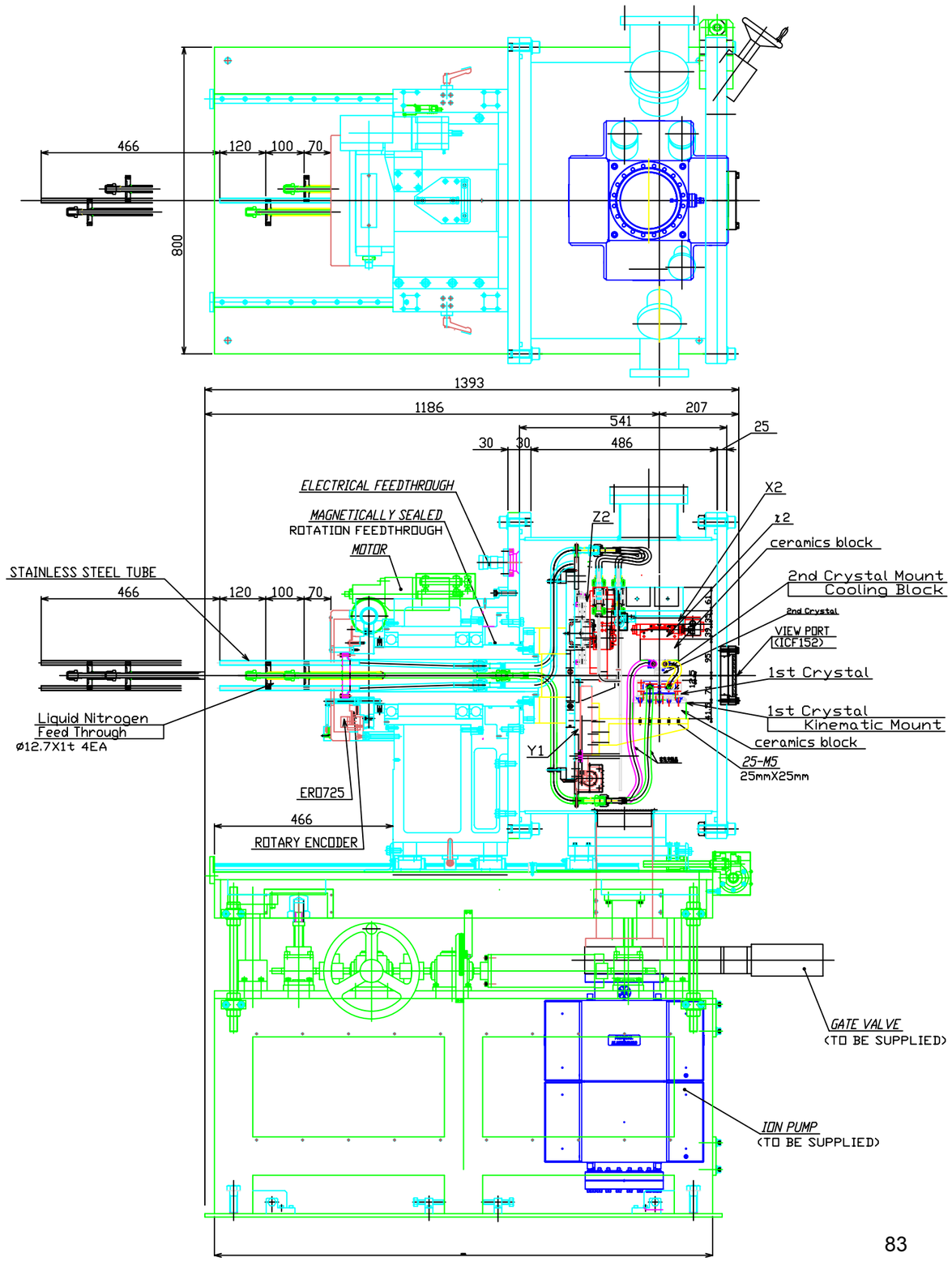


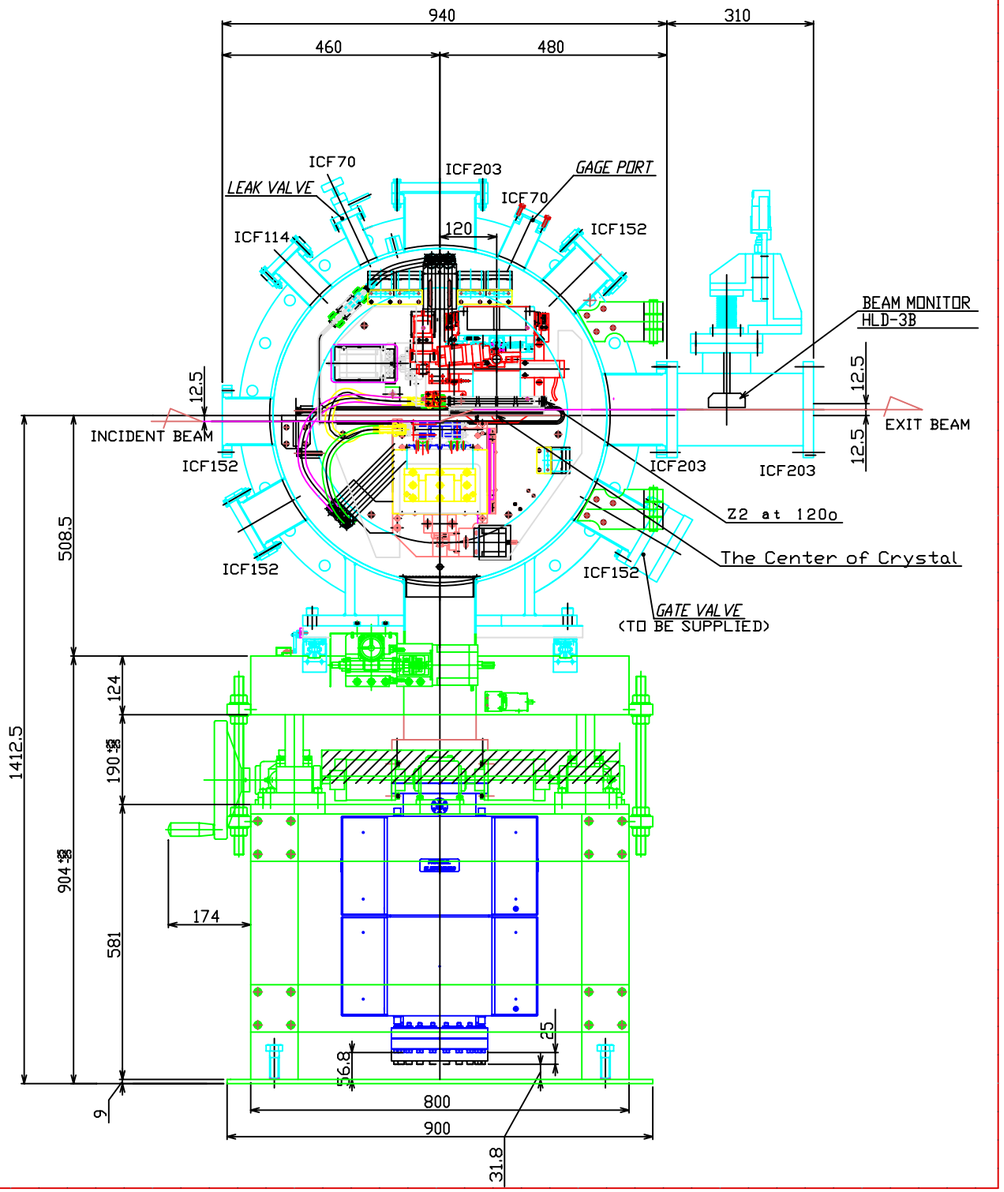




Material: See Notes			
Tolerances: UOS Linear Dimensions: Upto 120 ± 0.25, Over 120 ± 0.5	Angular Dimensions: ± 0.25°	Surface Finish: N7	
Scale: 1:2	All Dims in mm's UOS		
Approx. Mass: 3332514g	Instrument Design Technology Ltd Daresbury Innovation Centre Keckwick Lane, Daresbury Runcorn, WA4 4FS, UK		
Drn. PB, Date. 12/7/05	Property of Instrument Design Technology Ltd. This document is to be published in confidence to the receiving party. It is not to be published nor its contents otherwise disseminated without the express written consent of Instrument Design Technology Ltd © Copyright Instrument Design Technology Ltd All rights reserved		
Chd. PB, Date. 12/7/05	Title: Power Limiting Assembly Central Hole		
Appd. PKM, Date. 12/7/05	Size: A0	Drawing No: LSC-101-11-0013	Sheet1 of1
Used On: LSC			Issue: A

組立図:ASSEMBLE DRAWING			
HLD-8NE		110550-0001	
2003.08.08		1/6	UEHARA
Double Crystal Monochromator			UEHARA
神津精機株式会社 KOHZU Precision Co., Ltd.	A1		
FILE No,	110550-0001		





NOTES:

1. THIS IS A ULTRA – HIGH – VACUUM ASSEMBLY (UHV) KEEP THE UHV PARTS CLEAN DURING ASSEMBLY, AND WRAP UP FOR PACKING WITH ALUMINUM FOIL
2. THE ASSEMBLY SHALL BE LEAK TESTED USING A MASS SPECTROMETER WITH MINIMUM SENSITIVITY FOR HELIUM OF 2X 10 –10 STANDARD CC/SEC PER LEAK METER DIVISION, SUCH AS:

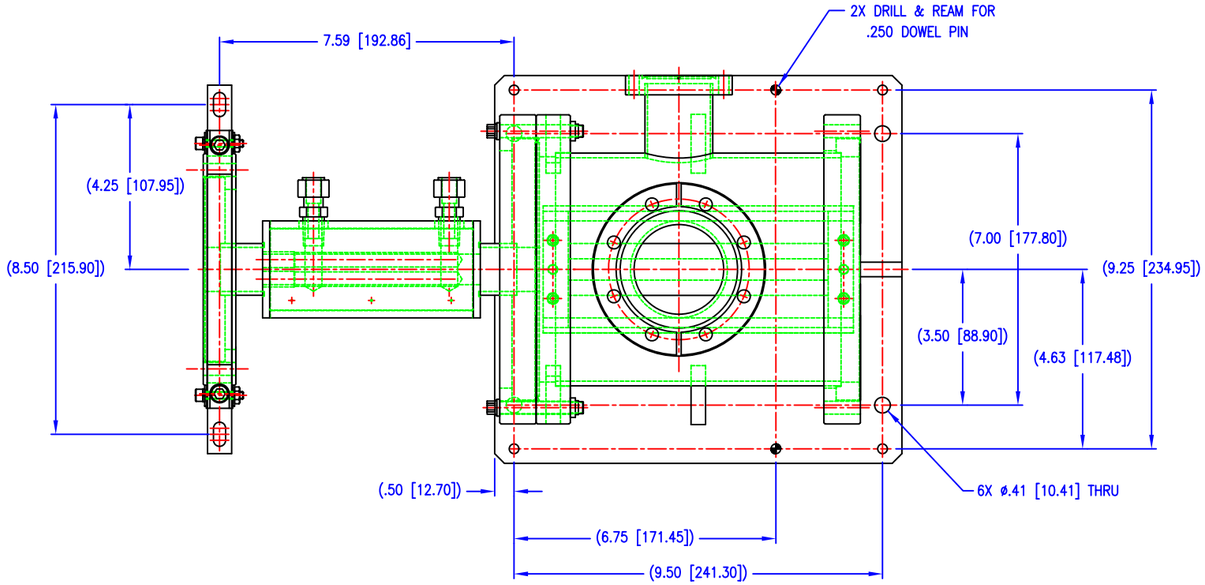
   ALCATEL ASM–110TCL
   VARIAN NCR 925 OR 936
   VEECO MS–9, MS–90 OR MS–18
   Du PONT CEC 24–120B

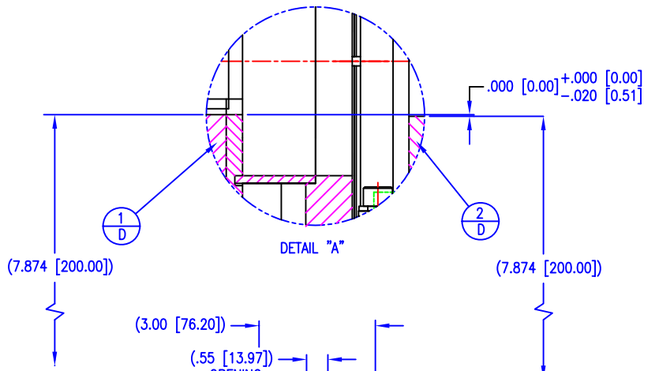
   CALIBRATION OF THE LEAK DETECTOR SENSITIVITY SHALL BE PERFORMED JUST PRIOR TO TESTING.

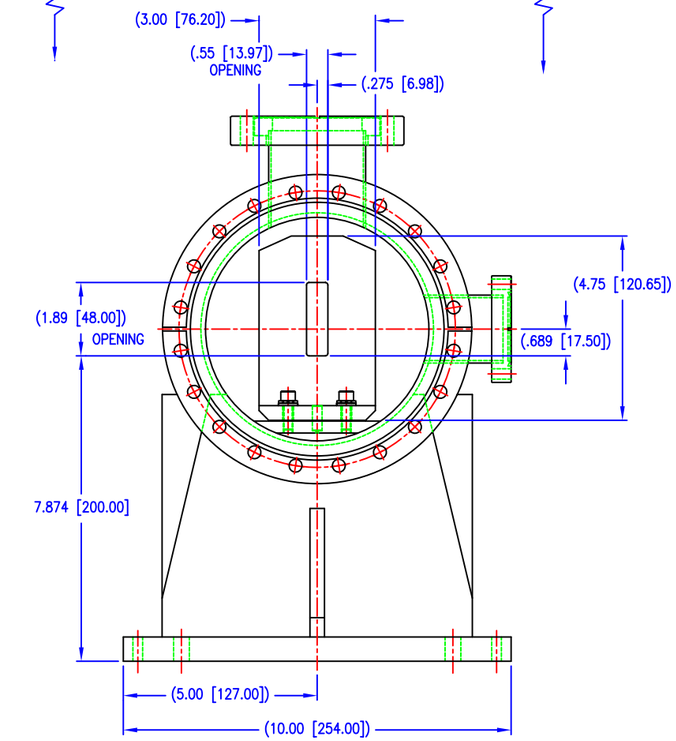
   FINAL TEST WILL CONSIST OF SURROUNDING THE ASSEMBLY (BAGGING) WITH HELIUM. THE ASSEMBLY WILL BE REJECTED IF A 2% DEFLECTION IN THE MOST SENSITIVE RANGE OF THE LEAK DETECTOR IS SENSED WITHIN 1 MIN.
3. DIMENSIONS IN [ ] ARE MILLIMETERS AND FOR REF. ONLY
4. SHIM AS NEEDED

REF. SOURCE

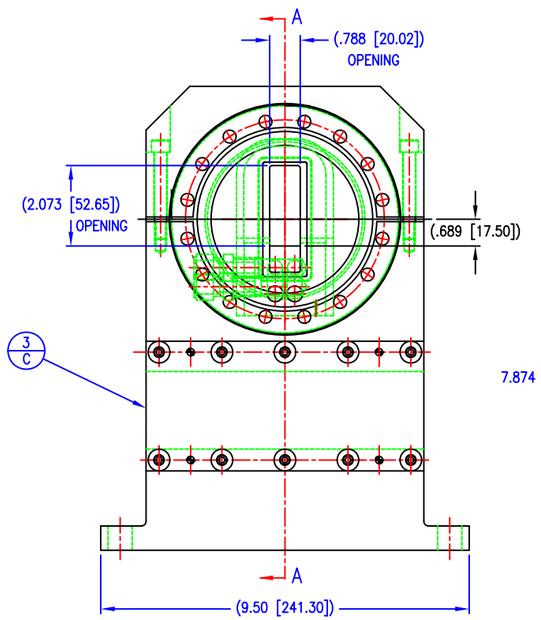
1 MDC VACUUM PRODUCTS CORP.
23842 CABOT BLVD.
HAYWARD, CA 94545–1651
TEL: 1–800–443–8817
FAX: 1–510–887–0626

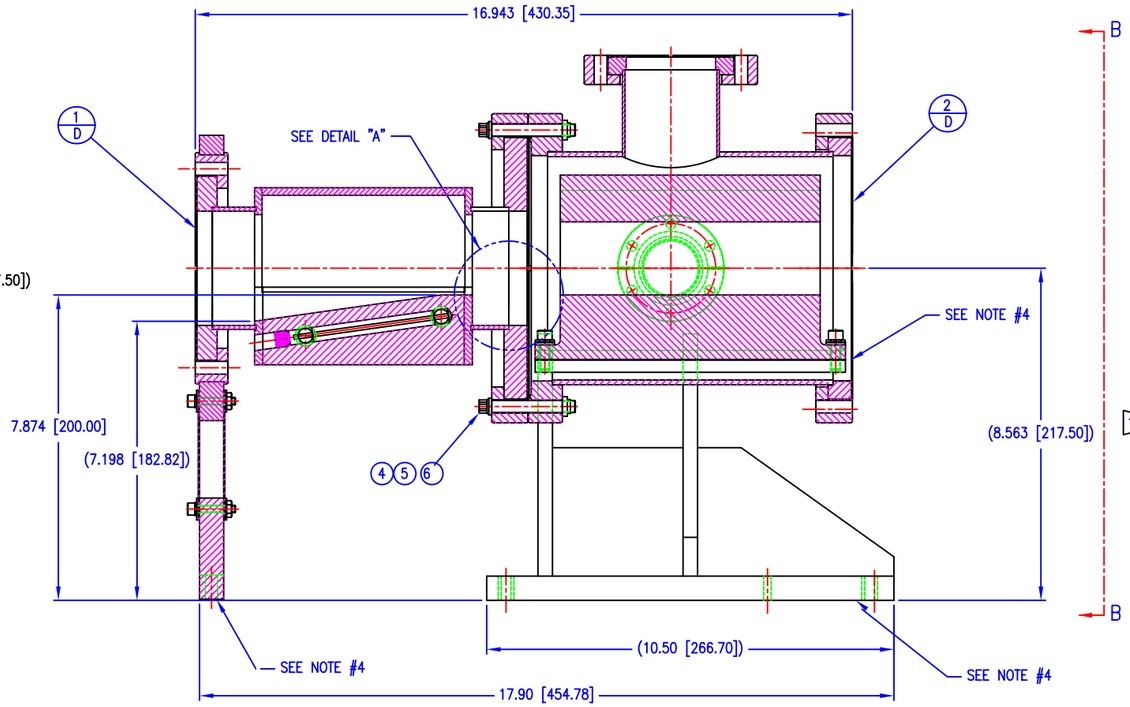






VIEW B – B





SECTION A – A

ITEM	DWG/PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
6	#191017	COPPER GASKET	OFHC COPPER	1
5		5/16 – 24 PLATE NUT	SST	10
4		5/16 – 24 X 2 1/4" LG. 12–PT BOLT	SST SILVER PLATE	20
3	P4–630000–00	P4–60 SUPPORT PLATFORM ASSEMBLY	SST	1
2	P4–620000–00	P4–60 WHITE BEAM COLLIMATOR ASSEMBLY	SST/TUNGSTEN	1
1	P4–610000–00	P4–60 WHITE BEAM STOP ASSEMBLY	SST/GLID–COP	1

PARTS LIST

SYM	CHANGE DESCRIPTION	BY	CHKD	DATE

REVISIONS

UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±.1 [2.5], .XX ±.01 [0.25], .XXX ±.005 [0.13]
ANGULAR: ±0°30'
SURFACE ROUGHNESS 125
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX.
SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1
DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5
DO NOT SCALE DRAWING

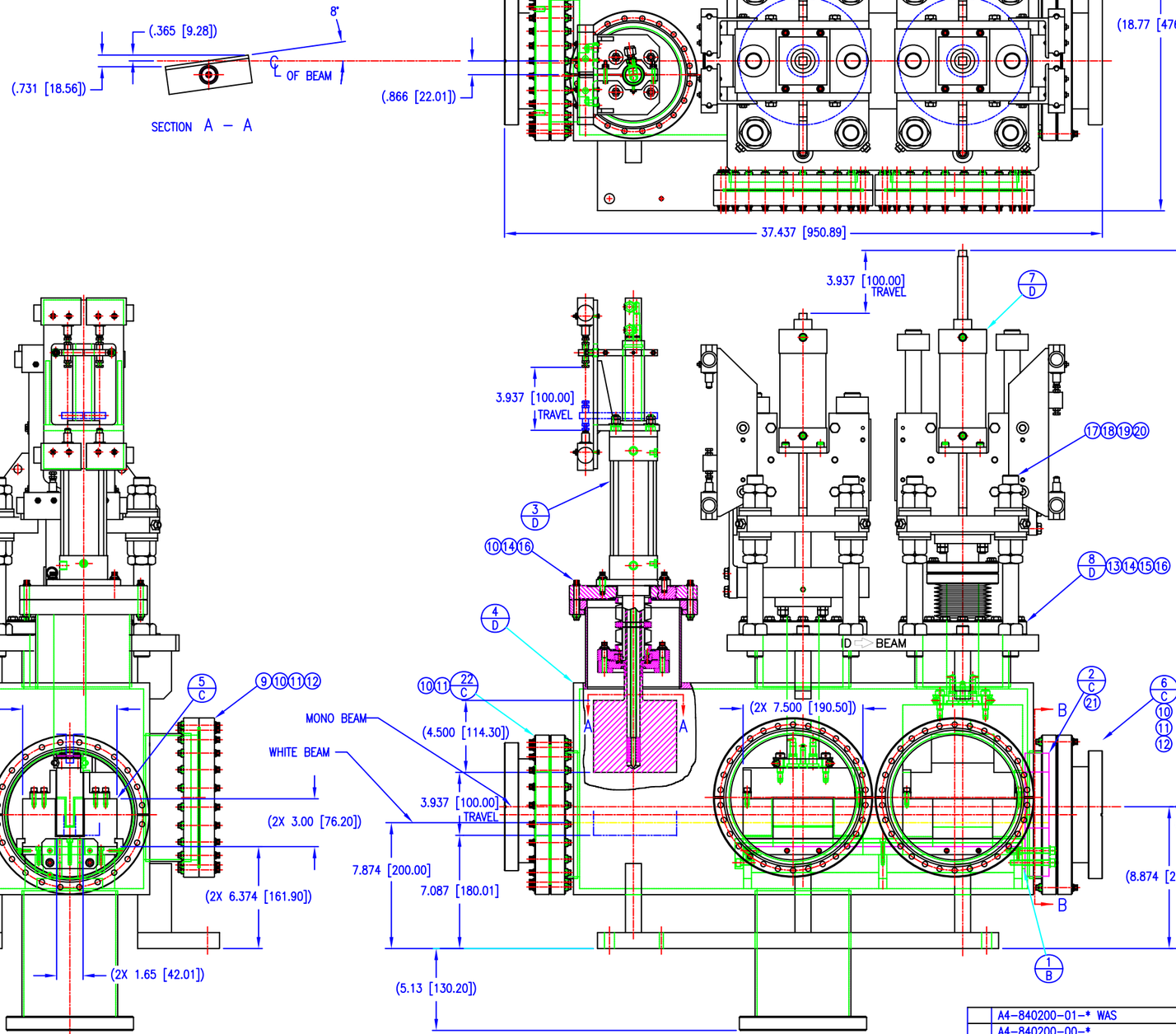
LOG NUMBER: P4600000–00

THIS DRAWING IS THE PROPERTY OF ARGONNE NATIONAL LABORATORY

DRAWN BY: MUSCIA	DATE: 9/13/99	CHIEF DESIGN ENGINEER: D. SHU	DATE: 10/6/99
CHECKED BY: J. CHANG	DATE: 9/24/99	GP LEADER: T.M. KUZAY	DATE: 10/11/99
DESIGNER: CHANG/MUSCIA	DATE: 9/13/99	PROJECT MGR.	DATE
RESPONSIBLE ENGINEER: J. CHANG	DATE: 9/24/99	APPROVED/RELEASED	DATE

TITLE: ADVANCED PHOTON SOURCE
P4–60 INTEGRAL WHITE BEAM STOP & COLLIMATOR FOR 4–ID–B ASSEMBLY

MATERIAL: SEE PARTS LIST
SCALE: HALF
SIZE: D
SHEET 1 OF 1
DRAWING NUMBER: P4–600000–00



NOTES:

1. THIS IS A ULTRA – HIGH – VACUUM ASSEMBLY (UHV) KEEP THE UHV PARTS CLEAN DURING ASSEMBLY, AND WRAP UP FOR PACKING WITH ALUMINUM FOIL
2. THE ASSEMBLY SHALL BE LEAK TESTED USING A MASS SPECTROMETER WITH MINIMUM SENSITIVITY FOR HELIUM OF 2X 10 –10 STANDARD CC/SEC PER LEAK METER DIVISION, SUCH AS:

   ALCATEL ASM–11OTCL
   VARIAN NCR 925 OR 936
   VEECO MS–9, MS–90 OR MS–18
   Du PONT CEC 24–120B

   CALIBRATION OF THE LEAK DETECTOR SENSITIVITY SHALL BE PERFORMED JUST PRIOR TO TESTING.

   FINAL TEST WILL CONSIST OF SURROUNDING THE ASSEMBLY (BAGGING) WITH HELIUM. THE ASSEMBLY WILL BE REJECTED IF A 2% DEFLECTION IN THE MOST SENSITIVE RANGE OF THE LEAK DETECTOR IS SENSED WITHIN 1 MIN.
3. DIMENSIONS IN [ ] ARE MILLIMETERS AND FOR REF. ONLY
4. ALL DRILLED HEAD SCREWS SHALL BE SECURED WITH SAFETY LOCK WIRE

THIS IS A MODIFICATION OF A DESIGN BY ARGONNE NATIONAL LABORATORY / ADVANCED PHOTON SOURCE. REF DWG # P4400000–01–2

87

SYM	CHANGE DESCRIPTION	BY	CHKD	DATE
	REVISIONS			
	A4–840200–01–* WAS			
	A4–840200–00–*			
	A4–840100–01 WAS A4–840100–00			
	P4–430000–01 WAS P4–430000–00			
	P4–420000–01–* WAS			
	P4–420000–00–*			
	P4–410000–01 WAS P4–410000–00			
	DRAWING BROUGHT UP TO DATE	M.J.M.	J.C.	1/31/00
1	FLANGE ADAPTER DOWNSTREAM	B.K.S.		1/16/06
1	CORRECTED FLANGE ADAPTERS	B.K.S.		3/16/06

ITEM	DWG/PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
		PARTS LIST		

UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±1DECTOL; .XX ±2DECTOL; .XXX ±3DECTOL
ANGULAR: ±ANGTOL
SURFACE ROUGHNESS
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX.
SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1
DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5
DO NOT SCALE DRAWING

LOG NUMBER

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
SYNCHROTRON RADIAITION CENTER

DRAWN BY: B. STILLWELL	DATE: 3/16/06	CHIEF DESIGN ENGINEER	DATE
CHECKED BY	DATE	GP LEADER	DATE
DESIGNER	DATE	PROJECT MGR.	DATE
RESPONSIBLE ENGINEER	DATE	APPROVED/RELEASED	DATE

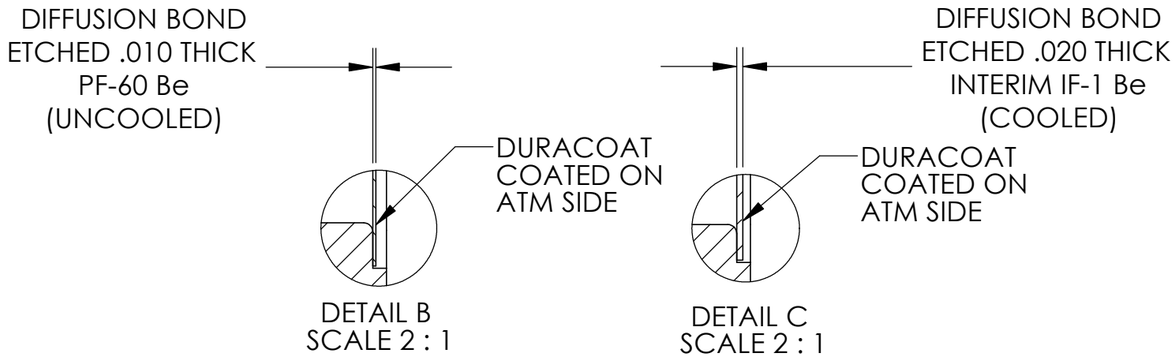
TITLE: P4–40 INTEGRAL SHUTTER FOR 21–ID ASSEMBLY

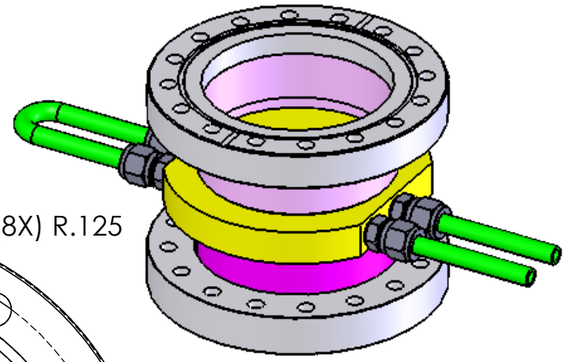
MATERIAL

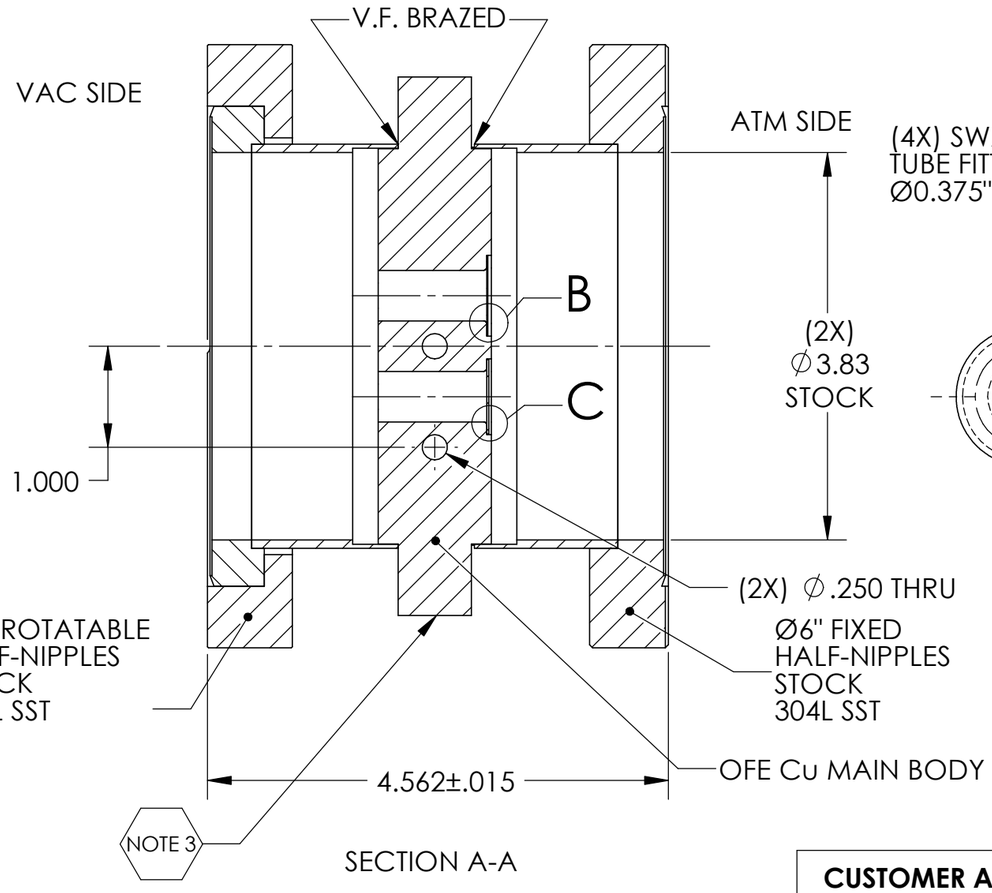
SCALE | SIZE | DRAWING NUMBER: D | 21–1.8.1.32–100000–02–2

SHEET 2 OF 3

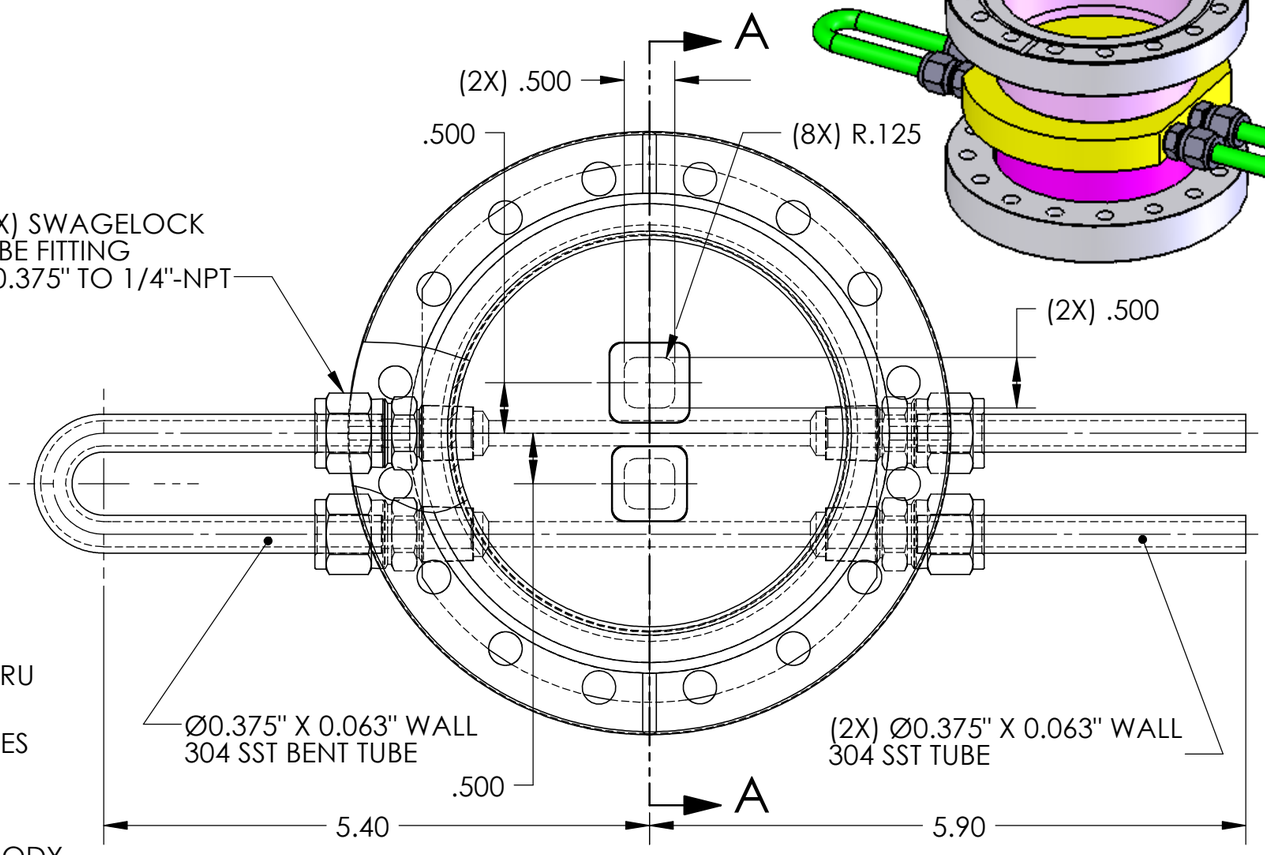
REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
5C&8C	A	ADD NOTES TO INDICATE ATMOSPHERE SIDE AND VACUUM SIDE	01/12/06	HC
5C&8C	B	REVERSE THE ATMOSPHERE SIDE VS. VACUUM SIDE FOR THE HALF-NIPPLES	01/12/06	HC
8D	C	REVISE THE FOIL THICKNESS FOR PF-60 Be WDW FROM 0.025" TO 0.010"	02/23/06	HC
2C	D	REVISE FOIL CORNER RADIUS AND REDUCE THE THICKNESS OF Cu MAIN BODY	03/01/06	HC
3C/5D/7D	E	REDUCE APERTURE SIZE, INCREASE FOIL THICKNESS & ADD COATING ON ATM SIDE	03/13/06	HC
5D&8D	F	MODIFY FOIL THICKNESS FOR PF-60 Be & ADD NOTES FOR ETCHED SURFACE CONDITION	03/14/06	HC







SECTION A-A



NOTES:
1) DIMENSIONS SHOWN ARE TO BE INSPECTED AGAINST SEPARATE DETAIL DRAWINGS.
2) PARTS TO BE MASS SPECTROMETER LEAK TIGHT WITH NO LEAKS GREATER THAN 2X10-10 STD CC/SEC OF He.
3) SCRIBE BEAM DIRECTION ON THE SIDE OF THE Cu MAIN BODY.

CUSTOMER APPROVAL DRAWING

APPROVED BY _____

DATE _____

BW-ELECTROFUSION PROPRIETARY
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	NAME	DATE
DRAWN	H. Chen	01/10/06
ENG APPR.	.	.

Interpret Drawing per ANSI Y14.5 M-1994.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
ANGLES ±0.5
.X ±0.03
.XX ±0.005

FINISH (Ra): 63

CUSTOMER: Northwestern Univ.

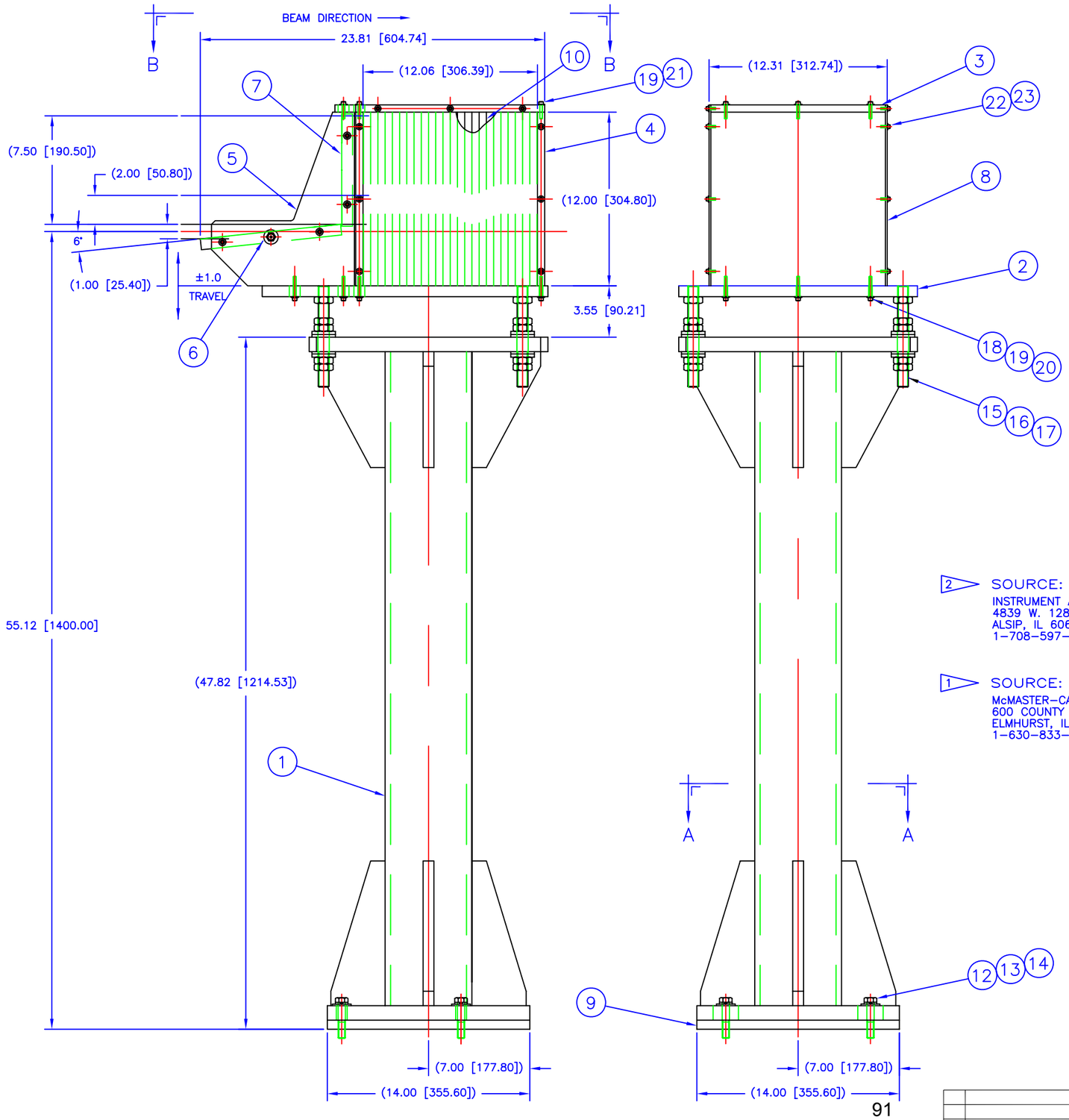
MATERIAL: As Shown

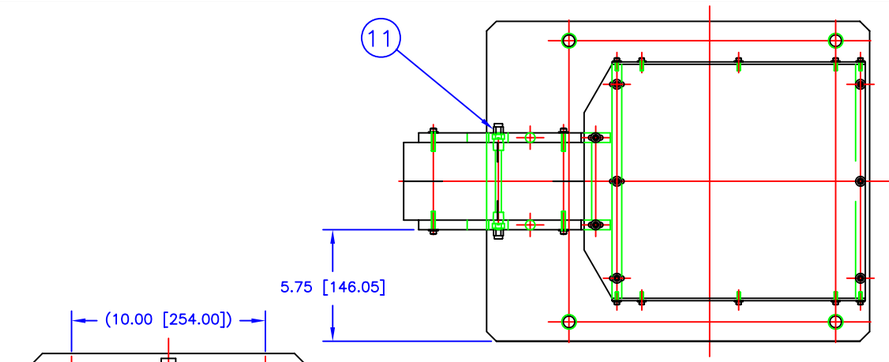
BRUSH WELLMAN
ELECTROFUSION PRODUCTS

TITLE:
INTERIM IF-1/PF-60 Be ACWA
DIFF. BOND 0.010" THK IF-1 Be TO 6" AC WDW ASSY

SIZE	DWG. NO.	REV
B	B-2510146-200	F

SCALE: 2:3 | DO NOT SCALE DRAWING | SHEET 1 OF 1





APRROXIMATE WEIGHT 1100 LBS.

2 SOURCE:
INSTRUMENT ASSOCIATES
4839 W. 128TH PLACE
ALSIP, IL 60658
1–708–597–9880

1 SOURCE:
McMASTER–CARR COMP.
600 COUNTY LINE RD.
ELMHURST, IL 60126–2081
1–630–833–0300

ITEM	DWG/PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
23		.19 ID FLAT WASHER	STEEL	18
22		#8–32 X .50 SOC HD CPSC	STEEL	18
21		#10–24 X 1.00 SOC HD CPSC	STEEL	16
20		.20 ID LOCK WASHER	STEEL	10
19		.25 ID FLAT WASHER	STEEL	26
18		#10–24 X 1.50 SOC HD CPSC	STEEL	10
1 17	91131A090	SPHERICAL WASHER	STEEL BLK OXIDE	8
16		.75–10 JAM NUT	STEEL	20
15		THREADED ROD–.75–10 X 7.0	STEEL	4
14		.5 ID LOCK WASHER	STEEL	4
13		.5 ID FLAT WASHER	STEEL	4
12		.5–13 X 2.5 LG HEX HD CPSC	STEEL	4
2 11	2-4 FBZ-SS-C	PARKER 1/4 NPT x 1/8 T CONN.	STAINLESS STEEL	2
1 10	9032K117	SHEET LEAD	12 x 12 x 1/2	24
9	K5–500009	K5–50 SHIM	SEE DETAIL	1
8	K5–500008	K5–50 COVER PLATE	SEE DETAIL	2
7	K5–500007	K5–50 COOLING PLATE	SEE DETAIL	1
6	K5–500006	K5–50 COOLING PLATE	SEE DETAIL	1
5	K5–500005	K5–50 SUPPORT PLATE	SEE DETAIL	2
4	K5–500004	K5–50 SIDE PLATE	SEE DETAIL	2
3	K5–500003	K5–50 TOP PLATE	SEE DETAIL	1
2	K5–500002	K5–50 SUPPORT PLATE	SEE DETAIL	1
1	K5–500001	K5–50 WELDMENT SUPPORT	SEE DETAIL	1

PARTS LIST

SYM	CHANGE DESCRIPTION	BY	CHKD	DATE

REVISIONS

UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±.03 [.76] .XX ±.01 [0.25] .XXX ±.005 [0.13]
ANGULAR: ±.25°
SURFACE ROUGHNESS 125
REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX.
SURFACE TEXTURE TO BE IN ACCORDANCE WITH LATEST ANSI B46.1
DIMENSIONING & TOLERANCING IN ACCORDANCE WITH LATEST ANSI Y14.5
DO NOT SCALE DRAWING

LOG NUMBER K5500000–00

THIS DRAWING IS THE PROPERTY OF
ARGONNE NATIONAL LABORATORY

DRAWN BY G.CHERRY	DATE 2/7/00	CHIEF DESIGN ENGINEER D. SHU	DATE 3/21/00
CHECKED BY J.BARRAZA	DATE 3/14/00	GP LEADER T.M. KUZAY	DATE
DESIGNER J.BARRAZA	DATE	PROJECT MGR.	DATE
RESPONSIBLE ENGINEER J.BARRAZA	DATE 3/14/00	APPROVED/RELEASED	DATE

TITLE
ADVANCED PHOTON SOURCE
K5–50
4–ID–D
END STOP ASSEMBLY

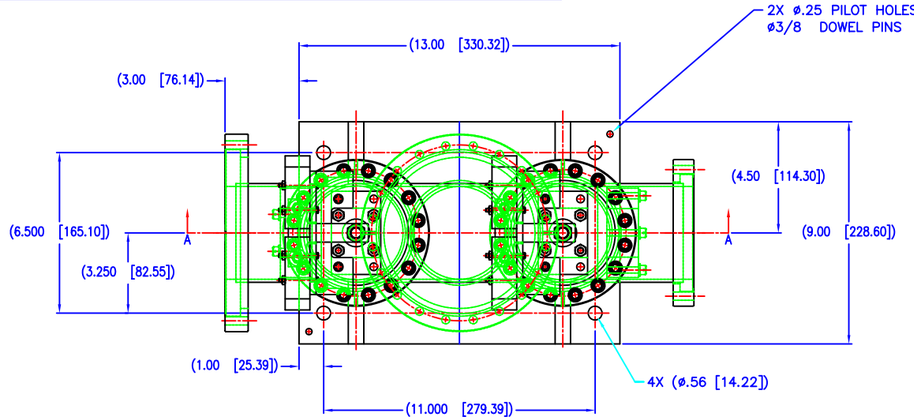
MATERIAL SEE PARTS LIST

SCALE 1:4 SHEET 1 OF 1

SIZE D

DRAWING NUMBER K5–500000–00

THIS IS A MODIFICATION OF A DESIGN BY ARGONNE NATIONAL LABORATORY / ADVANCED PHOTON SOURCE. REF DWG # P8-500000-02

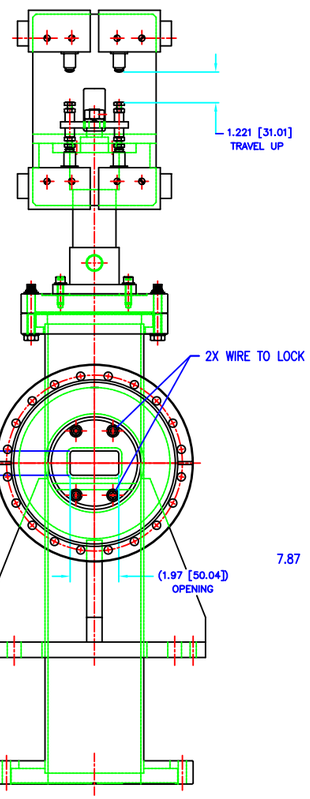


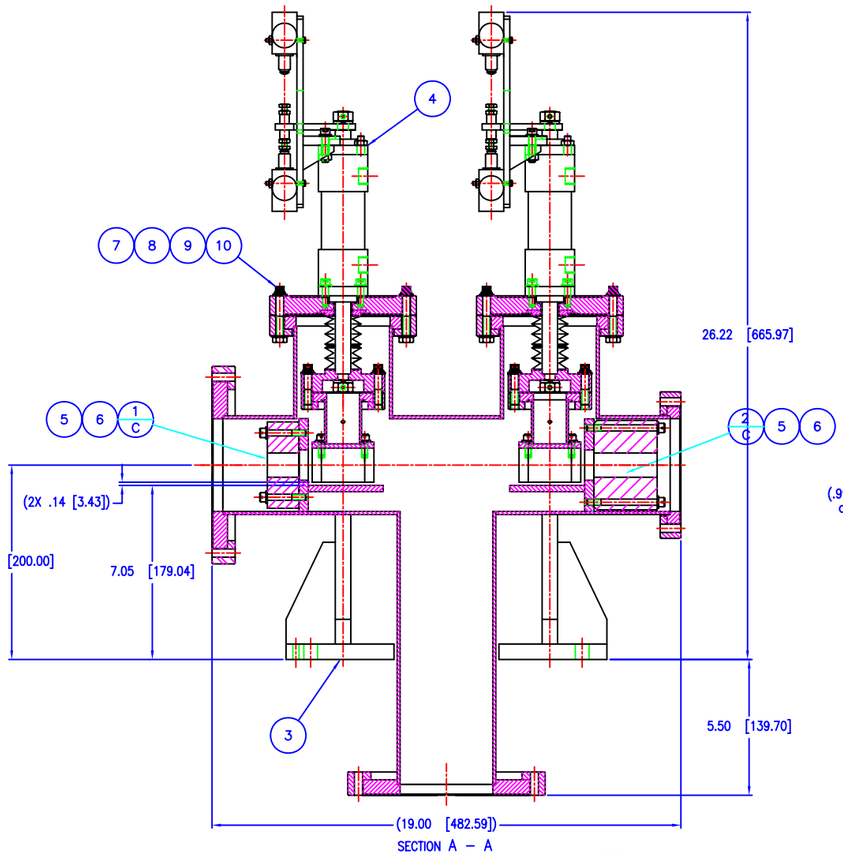
NOTES:

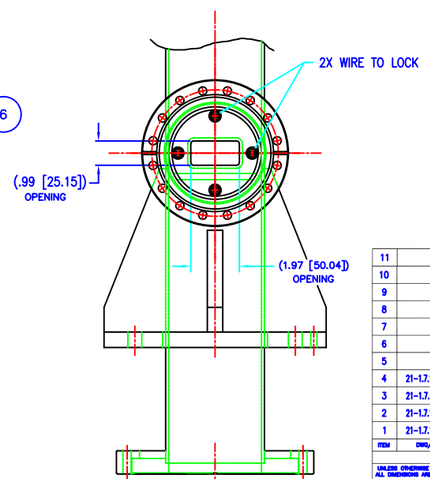
1. THIS IS A UHV ASSEMBLY, KEEP THE PARTS CLEAN DURING ASSEMBLY AND WRAP UP FOR UHV PACKING WITH ALUMINUM FOIL
2. SHUTTER ASSEMBLY SHALL BE LEAK TESTED USING A MASS SPECTROMETER WITH MINIMUM SENSITIVITY FOR HELIUM OF 2X 10 −10 STANDARD CC/SEC PER LEAK METER DIVISION, SUCH AS:
   ALCATEL ASM−110TCL
   VARIAN NCR 925 OR 936
   VECCO MS−9, MS−90 OR MS−18
   Du PONT CEC 24−120B
   CALIBRATION OF THE LEAK DETECTOR SENSITIVITY SHALL BE PERFORMED JUST PRIOR TO TESTING.
   FINAL TEST WILL CONSIST OF SURROUNDING THE SHUTTER ASSEMBLY (BAGGING) WITH HELIUM. THE CHAMBER WILL BE REJECTED IF A 2% DEFLECTION IN THE MOST SENSITIVE RANGE OF THE LEAK DETECTOR IS SENSED WITHIN 1 MIN.
3. ALL DRILLED HEAD SCREWS SHALL BE WIRED TO LOCK IN PLACE
4. DIMENSIONS IN [ ] ARE MILLIMETERS AND FOR REF. ONLY

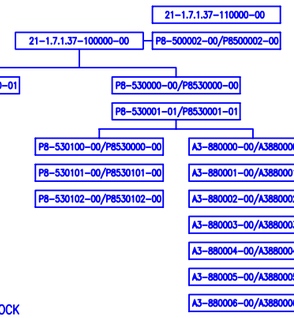
DESIGN SPECIFICATIONS FOR
P8 ID MONOCHROMATIC MONO SHUTTER 1

1. LOCATION: 36.7M FROM APS ID STRAIGHT SECTION CENTER
2. OPTICAL APERATURE AREA: 50mm (H) X 25mm (V)
3. HORIZONTAL MAXIMUM ACCEPTANCE: 1.36 mrd
4. VERTICAL MAXIMUM ACCEPTANCE: 0.68 mrd
5. CLOSE TIME: 1 SEC.
6. BASE FLANGE O.D.: 8 INCHES
7. FEETHROUGH FLANGE O.D.: 3.38 INCHES
8. MASK INCIDENCE ANGLE: 90 DEGREE
9. MASK MATERIAL: HEAVY METAL (TUNGSTEN)
10. COOLING: RADIATION
11. ACTUATOR: PNEUMATIC
    END POINT REPRODUCIBILITY: .05mm
12. DAMPER: PNEUMATIC
13. HEAVY METAL SIZE: 76.2mm (H) X 35.0mm (V) X 63.5mm (L)
14. VACUUM: UHV COMPATIBLE
15. MAXIMUM TOTAL POWER: 0.1 W
16. DEVICE FLANGE TO FLANGE LENGTH = 482.6mm









ITEM	DWG/PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
11		1/4 − 20 X 2 7/8" LG. DR. HD. SOC. SCREW	SST/SILVERPLATED	4
10		6" O.D. VACUUM FLANGE COPPER GASKET	OFHC COPPER	2
9		5/16 − 24 BOLT FLANGE NUT	SST	16
8		.344 I.D. X .065 THK PLAIN WASHER	SST	16
7		5/16 − 24 X 2 1/4" LG. 12−PT HD. BOLT	SST/SILVERPLATED	16
6		.262 I.D. X .062 THK. LOCK WASHER	SST	8
5		1/4 − 20 X 1 5/8" LG. DR. HD. SOC. SCREW	SST/SILVERPLATED	4
4	21-1.7.1.37-120000-00	P8-50 SUB-ASSEMBLY		2
3	21-1.7.1.37-110000-01	P8-50 CHAMBER WELDMENT ASSEMBLY	304L SST	1
2	21-1.7.1.37-100002-00	P8-50 TUNGSTEN OPENING	TUNGSTEN	1
1	21-1.7.1.37-100001-00	P8-50 TUNGSTEN OPENING	TUNGSTEN	1

PARTS LIST

SYM	CHANGE DESCRIPTION	BY	CHKD	DATE
2	REPLACED 2ND W BLOCK PER PT DWG	B.K.S		9/14/05
1	ADDED 8" CF NIPPLE FOR ION PUMP	B.K.S		7/19/05

REVISIONS

UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES TOLERANCES: DECIMALS .X ±(NEXT) .XX ±(NEXT) .XXX ±(NEXT) ANGULAR ±(NEXT) SURFACE ROUGHNESS ALL OTHERS ALL SURFS AND EDGES BREAK SHARP LIGHT AT 90DEG. BREAK EDGES TO .01 R. ACCEPT EDGES TO .03 R. DIMENSIONS & TOLERANCES IN ACCORDANCE W/ ASME Y14.5 DO NOT SCALE DRAWING

LOG NUMBER

DRAWN BY: D. STILLWELL DATE: 7/6/05
CHECKED BY — DATE
DESIGNER — DATE
RESPONSIBLE ENGINEER — DATE
MATERIAL

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
SYNCHROTRON RADIATION CENTER

CHIEF DESIGN ENGINEER — DATE
SP LEADER — DATE
PROJECT MGR. — DATE
APPROVED/RELEASED — DATE

TITLE: P8-50 MONOCHROMATIC SHUTTER FOR 21-ID ASSEMBLY

SCALE: 1 SHEET 1 of 1
SIZE: C
DRAWING NUMBER: 21-1.7.1.37-100000-00

Appendix F: IDT Masks – Heat Load Calculation

LSC-101 Heat load calculation

This programme calculates the Gaussian distribution

Constants & Units

$GeV := 1.60217646 \cdot 10^{-10} \cdot joule$ Gigaelectron volt (1eV is equal to the energy of 1.602 x 10^{-19} Joules)

$mrad := 10^{-3} \cdot rad$

$Tesla := 10000 \cdot Oe$

$m_{electron} := 9.11 \cdot 10^{-28} \cdot gm$ Mass of electron

$c := 2.998 \cdot 10^{10} \cdot cm \cdot sec^{-1}$ Speed of light

1. Source parameters

$E := 7 \cdot GeV$ Electron energy

$I := 130 \cdot mA$ Maximum current

2. Wiggler parameters

$B_0 := 0.72 \cdot Tesla$ Peak magnetic field

$\lambda_u := 3.3 \cdot cm$ The period length of the undulator (cm)

$N := 60$ N is the number of periods

3. Beam & optic parameters

$T_p := 46 \cdot m$ Tangent point distance from ID to optic

$h := \left(\frac{3}{25.5}\right) \cdot mrad$ Horizontal beam size

$v := \left(\frac{2}{25.5}\right) \cdot mrad$ Vertical beam size

3. Calculate K (deflection parameter) of undulator

$K := 0.934 \cdot \lambda_u \cdot B_0 \cdot \left(\frac{1}{cm \cdot Tesla}\right)$ Divide by units to make non-dimensional

$K = 2.219$

4. Calculate γ of synchrotron source

$$\gamma := \frac{E}{m_{electron} \cdot c^2}$$

$\gamma = 13697$

5. Distribution functions

The normalising factor is given in the following equation

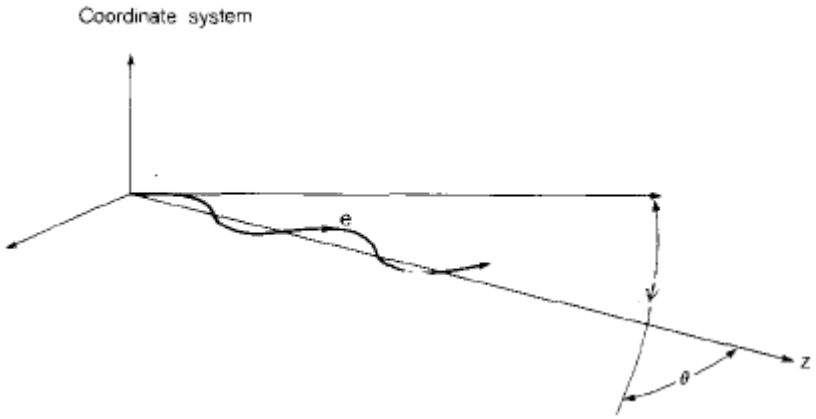
$$G(K) := K \cdot \frac{\left(K^6 + \frac{24}{7} \cdot K^4 + 4 \cdot K^2 + \frac{16}{7}\right)}{\left(1 + K^2\right)^{\frac{7}{2}}}$$

$G(K) = 0.985$

The factor $f_K(\theta, \psi)$ gives the angular dependence and is defined as

$$f_K(\theta, \psi) := \frac{16 \cdot K}{7 \cdot \pi \cdot G(K)} \cdot \int_{-\pi}^{\pi} \left[\frac{1}{\left[1 + \gamma^2 \cdot \psi^2 + (\gamma \cdot \theta - K \cdot \cos(\alpha))^2\right]^3} - 4 \cdot \frac{(\gamma \cdot \theta - K \cdot \cos(\alpha))^2}{\left[1 + \gamma^2 \cdot \psi^2 + (\gamma \cdot \theta - K \cdot \cos(\alpha))^2\right]^5} \right] \cdot (\sin(\alpha))^2 \, d\alpha$$

where θ is the horizontal angular displacement (mrad) and ψ is the vertical angular displacement (mrad), as shown below.



The electron moves on a sinusoidal trajectory in the horizontal plane x-z. θ and ψ are the angles of observation in the horizontal and vertical directions, respectively.

The function f_K is normalised so that

$$f_K(0,0) = 1$$

Power density (W/mrad^2) in the forward direction is given by

$$P_u(\theta,\psi) := 10.84 \cdot E^4 \cdot B_0 \cdot I \cdot N \cdot G(K) \cdot f_K(\theta,\psi) \cdot \left(\frac{1}{GeV^4 \cdot Tesla \cdot A}\right) \cdot \left(\frac{W}{mrad^2}\right)$$

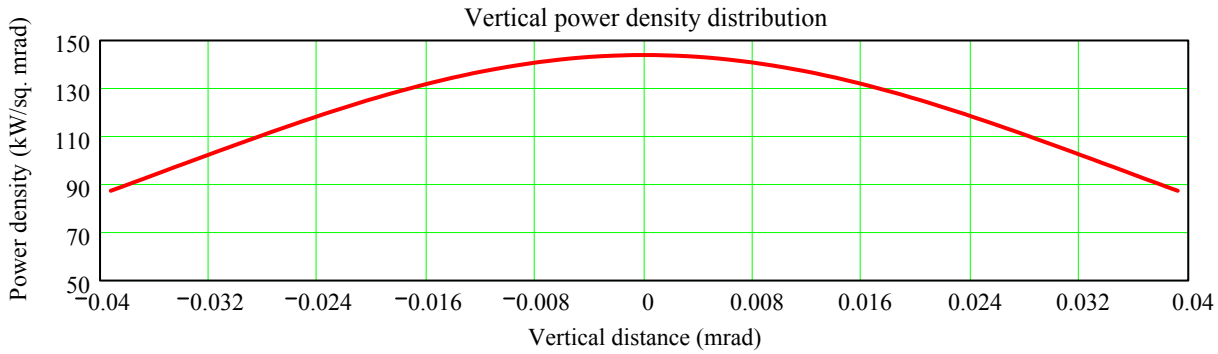
Peak power density at the centre of the cone (i.e. θ = 0 and ψ = 0)

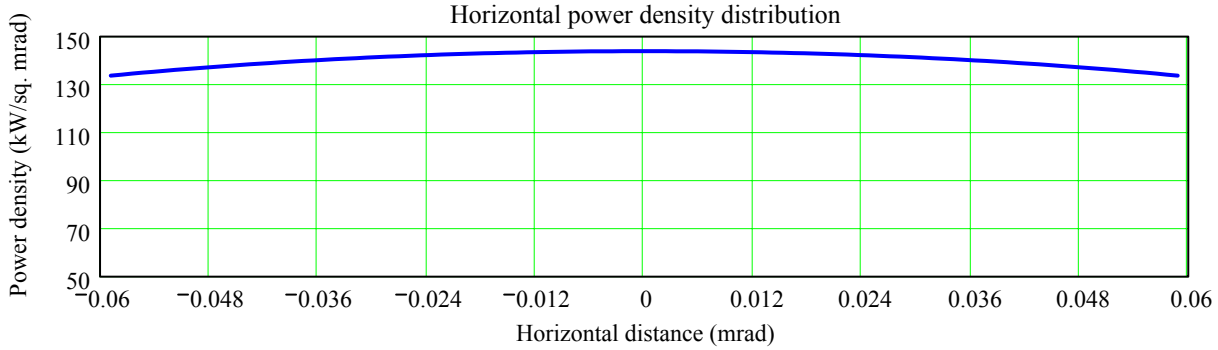
$$P_{maximum} := P_u(0,0)$$

$$P_{maximum} = 143.9\ kW \cdot mrad^{-2}$$

Define horizontal and vertical range as that of the beam size

$$\theta := \frac{-h}{2}, \left(\frac{-h}{2} + \frac{h}{100}\right) .. \frac{h}{2} \qquad \psi := \frac{-v}{2}, \left(\frac{-v}{2} + \frac{v}{100}\right) .. \frac{v}{2}$$





Calculate the total power incident on optic

The total power (integrated to beam dimenions) is given by

$$P := \int_{\frac{-v}{2}}^{\frac{v}{2}} \left(\int_{\frac{-h}{2}}^{\frac{h}{2}} P_u(\theta, \psi)\, d\theta \right) d\psi$$

$$P = 1.1\,\text{kW}$$

Calculate Vertical and Horizontal Full Width Half Maximum values

To calculate the Full Width Half Maximum value the power distribution has to be corrected so that half power occurs at 'zero' power. This is undertaken so that MathCad can automatically give the values of angular displacement (using the root function) where zero power density occurs.

Half Maximum value is

$$P_{halfmaximum} := \frac{P_{maximum}}{2}$$

$$P_{halfmaximum} = 71950.7\,\text{W}\,\text{mrad}^{-2}$$

Define function to give half maximum power at 'zero' power.

$$P_{hm}(\theta, \psi) := P_u(\theta, \psi) - P_{halfmaximum}$$

Use the root function to calculate angular displacement at zero power density.
Vertical Half Maximum,

$$\psi_{halfmaximum} := \text{root}\left[\left(P_u(0, \psi) - P_{halfmaximum}\right), \psi, 0\cdot\text{mrad}, 0.5\cdot\text{mrad}\right]$$

$$\psi_{halfmaximum} = 0.047\,\text{mrad}$$

Horizontal Half Maximum,

$$\theta_{halfmaximum} := \text{root}\left[\left(P_u(\theta, 0) - P_{halfmaximum}\right), \theta, 0\cdot\text{mrad}, 0.5\cdot\text{mrad}\right]$$

$$\theta_{halfmaximum} = 0.136\,\text{mrad}$$

Calculate Full Width Half Maximum.
Vertically,

$$fwhm_{vert} := 2\cdot\sqrt{\left(\psi_{halfmaximum}\cdot T_p\right)^2}$$

$$fwhm_{vert} = 4.3\,\text{mm}$$

Horizontally,

$$fwhm_{horz} := 2\cdot\sqrt{\left(\theta_{halfmaximum}\cdot T_p\right)^2}$$

$$fwhm_{horz} = 12.5\,\text{mm}$$

Calculate 1st sigma value.
Vertically,

$$\sigma_{vert} := \frac{fwhm_{vert}}{2.36}$$

$$\sigma_{vert} = 1.8\,\text{mm}$$

Horizontally,

$$\sigma_{horz} := \frac{fwhm_{horz}}{2.36}$$

$$\sigma_{horz} = 5.3\,\text{mm}$$

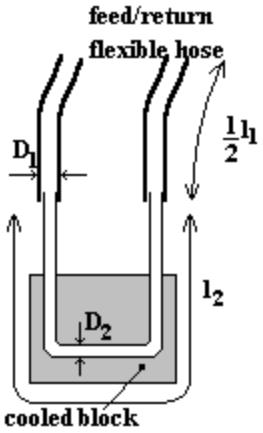
Appendix G: IDT Masks – Coolant Flow Calculation

LSC-101 Power limiting aperture - Coolant flow calculation

This file calculates minimum flow rates for Beamline components and then checks to see if minimum flow can be achieved given normal demin mains pressure and water circuit friction losses ..etc.

System Parameters

Consider a water circuit with heat loaded section of diameter D_2, length l_2 and feed and return pipes of diameter D_1 and total length of l_1, as shown.



Power input(W) $Q := 1142$

Feed/Return pipe ID(m) $D_1 := 8 \cdot 10^{-3}$

Feed/Ret pipe length(m) $l_1 := 8$

Cooled pipe ID(m) $D_2 := 8 \cdot 10^{-3}$

Cooled pipe total length(m) $l_2 := 0.09 \cdot 8 + 0.06 \cdot 8 + 0.08 \cdot 2$

Pipe length in heated zone (m) $l := 0.08 \cdot 8$

Note: l, Assumes only 2 x 4m hose to water panel and larger bore pipe, but boundary layer rise and in/out temp difference relatively insensitive to this length change

$l_2 = 1.36$

$l = 0.64$

Cooled wetted area (m^2)
(assumes full pipe bore through block) $A := \pi \cdot D_2 \cdot l$

$A = 0.01608$

Minimum Flow Rates

First find the minimum allowable flow rate Fa from consideration of maximum permissible boundary layer temperature difference

Max. boundary layer ΔT $\Delta Ta := 10$

Minimum flow rate F_a is then given by the following expression [1];

$$F_a := 3.76 \cdot D_2^{2.25} \cdot \left[\frac{Q}{(A \cdot \Delta Ta)} \right]^{1.25}$$

Minimum boundary rate

$F_a = 4.69$ llitres/min

Next find minimum flow rate F_b due to bulk temperature rise

Max. bulk temperature rise ΔT $\Delta Tb := 5$

Minimum flow rate F_b is then given by the following expression ;

$$F_b := 60 \cdot \frac{Q}{(4190 \cdot \Delta Tb)}$$

Minimum bulk rise rate

$F_b = 3.271$ llitres/min

Take the higher of either F_a or F_b as the minimum flow rate then use the information below to determine whether such flow is possible given the pressure losses in the system defined

Turbulent flow pressure loss in smooth bore pipe with differing cross sections

$p := 300000$	Set Demin mains pressure in N/m^2	1 bar = 100,000 N/m^2
$\rho := 1000$	Density of water kg/m^3	
$f := .008$	Conservative Fanning friction factor	
$\mu := .001002$	Dynamic viscosity Ns/m^2	
$N_1 := 4$	Number of 90 degree bends in each channel	
$N_2 := 16$		
$CL_1 := 0.5$	Special coefficient factors (see opposite)	
$CL_2 := 0.5$		

CL - coefficients of loss

Close return bend (180)	- 2.2
Standard Tee	- 1.8
Standard 90o bend	- 0.9
Medium sweep 90 bend	- 0.75
Long sweep 90 elbow	- 0.60
45o bend	- 0.42
Rounded inlet	- 0.04
Re-entrant inlet	- 0.8
Sharp edged inlet	- 0.5

Contraction

A2/A1	CL
0.1	0.37
0.2	0.35
0.4	0.27
0.6	0.17
0.8	0.06
0.9	0.02

$$p = \rho\cdot\frac{u_1^2}{2}\cdot\left[\left(4\cdot f\cdot\frac{l_1}{D_1} + 0.9\cdot N_1 + CL_1\right) + \rho\cdot\frac{u_2^2}{2}\cdot\left(\left(4\cdot f\cdot\frac{l_2}{D_2} + 0.9\cdot N_2 + CL_2\right)\right)\right]$$

Now $u_2 = u_1\cdot\left(\frac{D_1}{D_2}\right)^2$ Therefore,

$$p = \rho\cdot\frac{u_1^2}{2}\cdot\left[\left(4\cdot f\cdot\frac{l_1}{D_1} + 0.9\cdot N_1 + CL_1\right) + \left(\frac{D_1}{D_2}\right)^4\cdot\left(4\cdot f\cdot\frac{l_2}{D_2} + 0.9\cdot N_2 + CL_2\right)\right]$$

Rearranging to find mean stream velocities u_1 and u_2;

$$u_1 := \sqrt{\frac{(2\cdot p)}{\left[\rho\cdot\left[\left(4\cdot f\cdot\frac{l_1}{D_1} + 0.9\cdot N_1 + CL_1\right) + \left(\frac{D_1}{D_2}\right)^4\cdot\left(4\cdot f\cdot\frac{l_2}{D_2} + 0.9\cdot N_2 + CL_2\right)\right]\right]}} \qquad u_2 := u_1\cdot\left(\frac{D_1}{D_2}\right)^2 \quad \text{m/sec.}$$

Now find flow rates F1 & F2 in litres/min

$$F_1 := 60000\cdot\pi\cdot\left(\frac{D_1}{2}\right)^2\cdot u_1 \qquad F_2 := 60000\cdot\pi\cdot\left(\frac{D_2}{2}\right)^2\cdot u_2$$

And Reynolds number Re1, Re2 in each part of circuit (should be >2300 for turbulent)

$$Re_1 := \rho\cdot u_1\cdot\frac{D_1}{\mu} \qquad Re_2 := \rho\cdot u_2\cdot\frac{D_2}{\mu}$$

Mean stream velocity	$u_1 = 3.26$	$u_2 = 3.26$	metres/sec
Quantity flow	$F_1 = 9.833$	$F_2 = 9.833$	litres/min
Reynolds Number	$Re_1 = 26031.811$	$Re_2 = 26031.811$	

Now calculate heat transfer coefficient in required area

Density (kg/m^3) $\rho := 1000$

Dynamic viscosity $\mu := .001002$

Specific Heat (J/kg K) $Cp := 4180$

Conductivity (W/m K) $K := 0.618$

Calculate;

Reynolds number $Re := Re_2$ $Re = 26031.811$

Prandtl Number $Pr := \mu \cdot \frac{Cp}{K}$ $Pr = 6.777$

Nusselt Number $Nu := 0.023 \cdot Re^{.8} \cdot Pr^{0.4}$ $Nu = 168.482$

Note: Have used the Dittus-Boelter empirical correlation to calculate the Nusselt number
(Ref Heat Transfer, J P Holman, 6th Ed)

Heat Transfer Coefficient $h := \frac{K}{D_2} \cdot (Nu)$ $h = 13015.236$ $W/m^2/K$

Calculate total temperature rise across boundary layer

Boundary layer temperature rise is given by;

$$\Delta Ta := \frac{Q}{(h \cdot A)} \qquad \Delta Ta = 5.455 \quad \text{degrees C}$$

Calculate total bulk temperature rise

Determine the bulk rise in temperature, i.e difference between the inlet and exit water temperatures

$$\Delta Tb := \frac{Q}{\left[\rho \cdot \pi \cdot \left(\frac{D_2}{2}\right)^2 \cdot u_2 \cdot Cp\right]} \qquad \Delta Tb = 1.667 \quad \text{degrees C}$$

$\Delta T := \Delta Ta + \Delta Tb$

$\Delta T = 7.122$

Appendix H: IDT Masks – Finite Element Analysis

fea report

Project/ESR Ref:	LSC-101 Beam line	**Analysis by:**	Chris Rycroft
Component Ref:	Fixed aperture mask	**Date:**	24-Feb-06

Finite Element Analysis of Aperture mask

Related documents refer to;

1. General Description

This analysis is to confirm the thermal and structural integrity for the fixed aperture mask design chosen for the LSC-101 project.

Based on an front end aperture of 3mm (horizontal) x 2mm (vertical) at a distance of 25.5m from the insertion device and the fixed apertures positioned 46m from the centre of the insertion device, this results in beam width of 5.4mm (3mm / 25.5m x 46m) and a beam height of 3.6mm (2mm / 25.5m x 46m).

Based on a maximum beam miss-steer of +/- 8.4mm horizontal and +/- 8.4 mm vertical[1], the further most corner of the beam footprint at the maximum miss-steer from the beam centreline will be 11.1mm in the horizontal and 10.2mm in the vertical. The entrance to the mask is therefore set at 24mm wide by 22mm high to allow full acceptance of the beam on one mask (see Figure 1). This is considered the worst case scenario with the maximum miss-steer used with the beam focused into the corner of the tapered surface.

2. Analysis Features

The models have been made approximate and mostly conservative with full 3D elements used throughout. Element sizing can be characterized as fine to very fine, especially around a heat loaded area and regions considered susceptible to high stress concentrations. The main approximations in idealizing the model are in application of boundary conditions. These are described in this section.

2.1. FEA Thermal Model

The beam footprint is set at 5.4mm wide by 3.6mm high, as defined by an upstream aperture. Based on the undulator parameters and beam size, the power through the front end aperture has been calculated to be 1.1kW[2].

[1] Ref. Email from John Quintana to Chris Rycroft, dated 01/04/2005.
[2] Ref. MathCAD document LSC-101-60-01 iss1

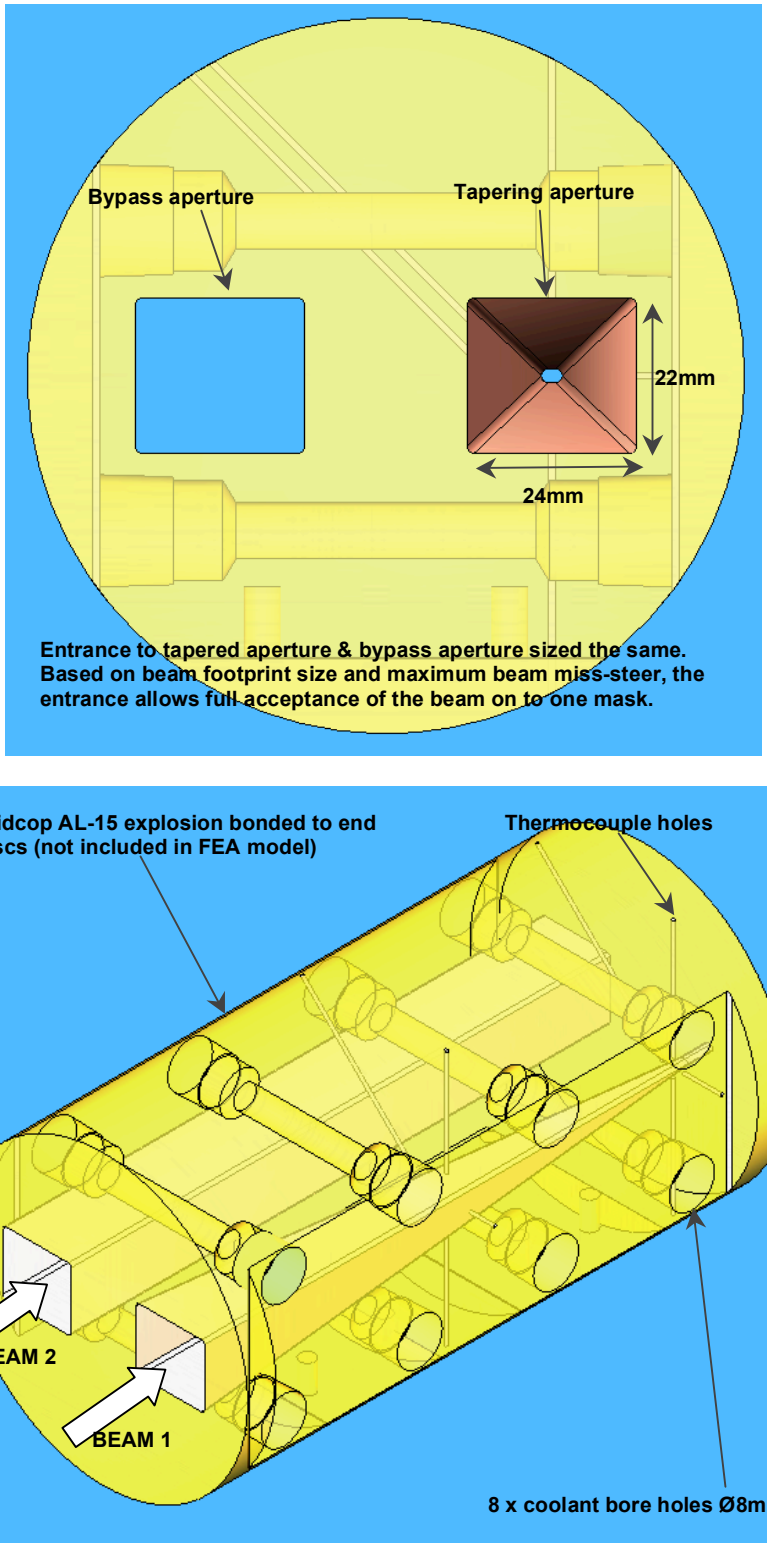


Figure 1; Front & iso-view of the mask assembly

Since the beam footprint is relatively small, there will be a limit to the number of elements that can be used vertically and horizontally in the area directly affected by the beam footprint (i.e. the optically active area). Introducing high numbers of elements concentrated into a small area will tend towards singularity errors and erroneous data from the FEA model. Based on this, the decision was made to limit the element concentration to give a more accurate solution and apply a uniform heat load in the volume rather than use power profiles in the horizontal and vertical directions.

Convective cooling (Heat Transfer Coefficient of 1.3W/cm^2 K @ 293K[3], achieved at flow rates typically 10 L/min with 3 bar pressure) has been applied to the external surface of the bore holes made for the coolant.

2.2. FEA Stress Model

The results from the thermal model is then loaded into a structural model; with restraints applied to allow free expansion

2.3. Material Properties

Material	Glidcop AL-15[4]
E Youngs Modulus (N/mm^2)	130,000
ν Poisson Ratio	0.25
ρ Density (x10^{-6}kg/mm^3)	8.9
α Coefficient of Expansion (/ºC x 10^{-6})	16.6
K Thermal Conductivity (W/mm/ºC)	0.365
σ Design yield stress (N/mm^2)	300
Melting Point (°C)	1083

Table 1; Glidcop AL-15 material properties.

3. Model Information

Analysis was undertaken using COSMOSWORKS 2005[5].

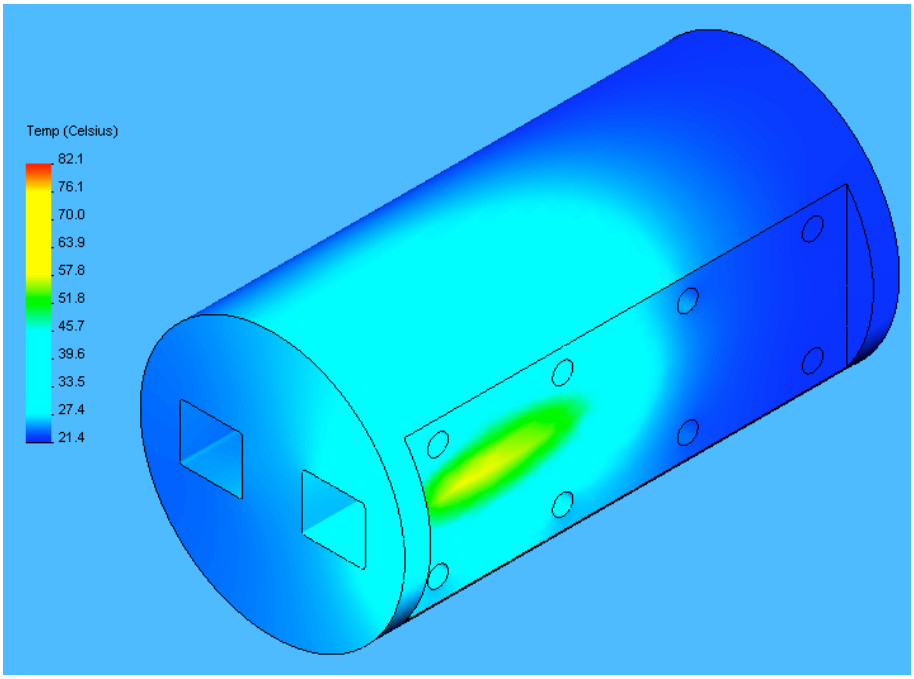
4. Analysis Validation

At applicable points in the Results section, comments have been included to give comparison between FEA model results and referenced calculations.

[3] Ref. MathCAD document LSC-101-60-02 iss2
[4] Properties taken from MatWeb – Online Materials Information Resource.
[5] Previous modelling of similar mask designs using both the ANSYS and COSMOSWorks FEA design tool have shown very good agreement.

5. Results

5.1. Thermal Model



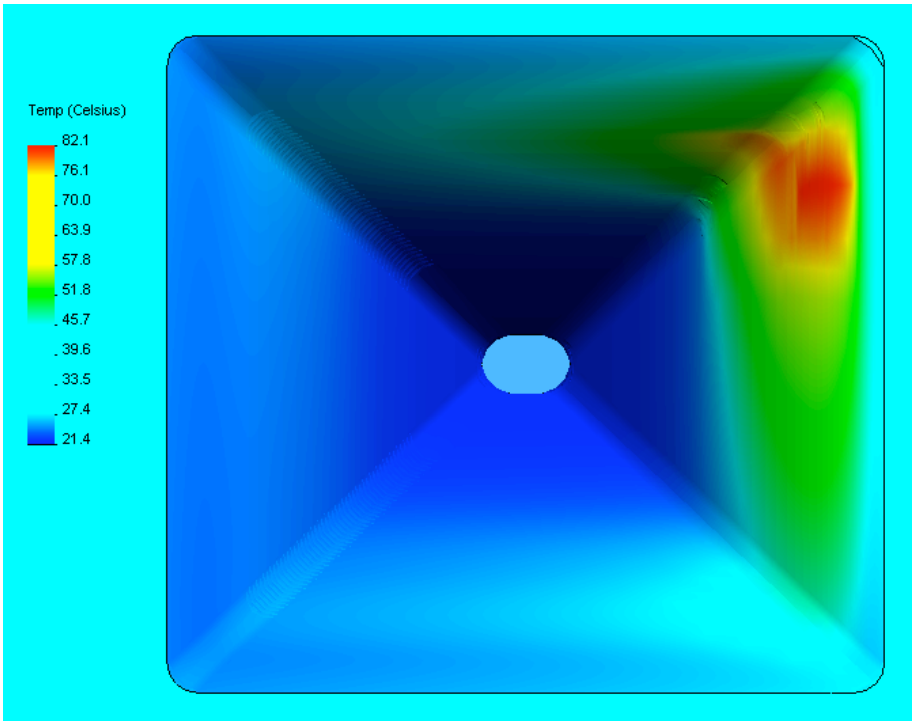


Figure 2; Temperature distribution (˚C).

Note: The heatload distribution is visible as temperature gradient along the internal surfaces of the mask.

5.2. Stress Model

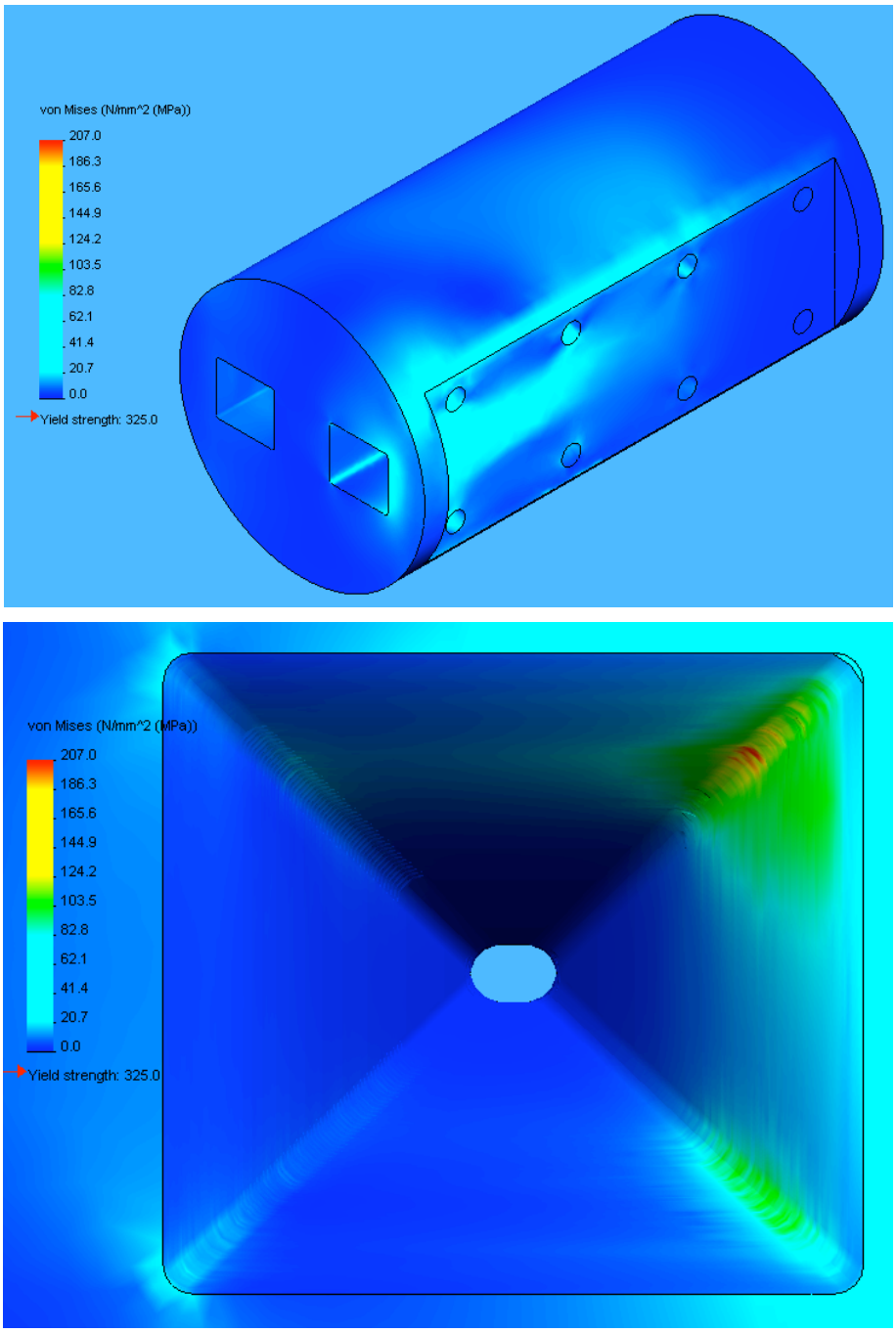


Figure 3; Stress distribution (MPa).

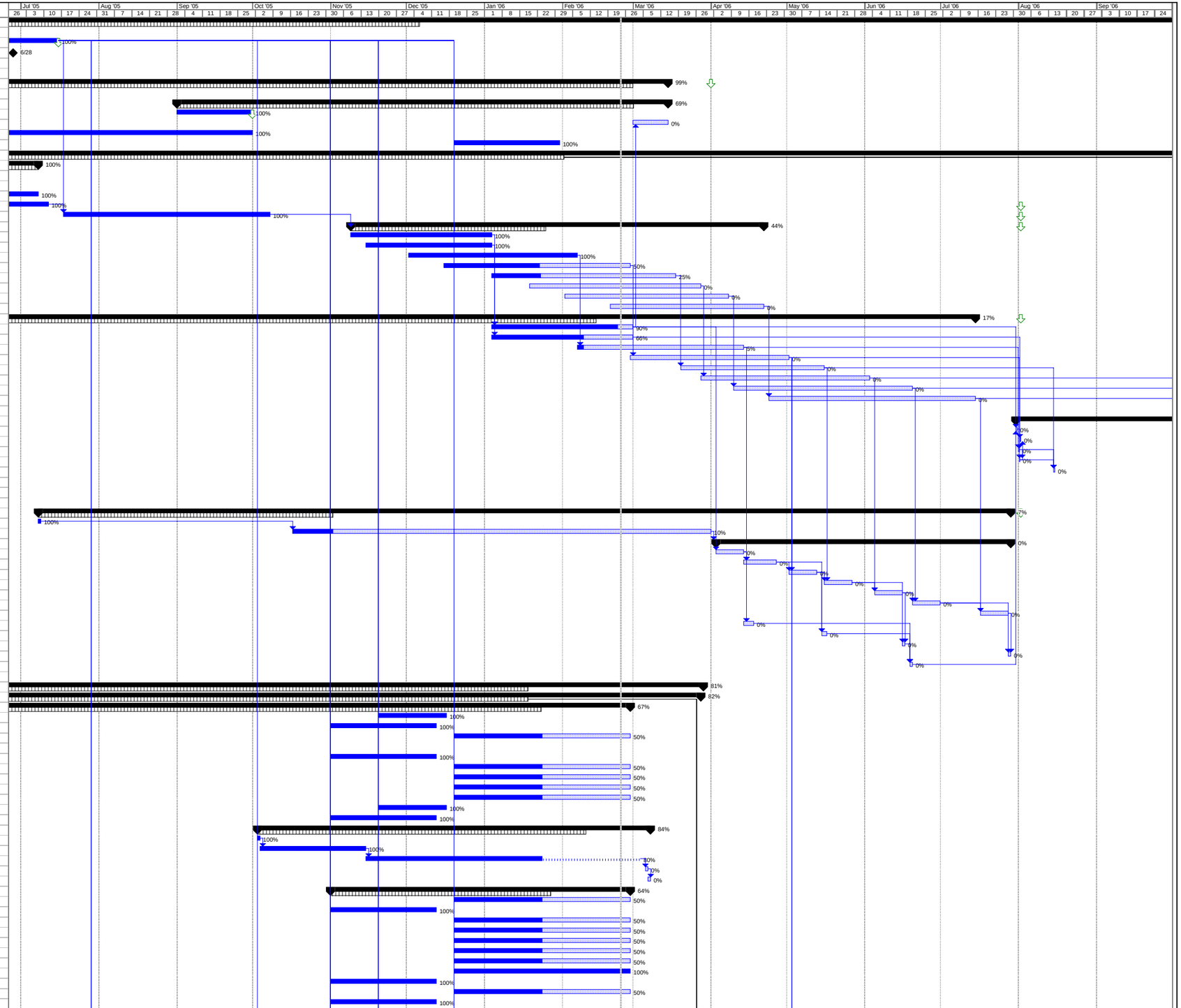
Note: Maximum stress in the Glidcop mask = 207MPa. Design fatigue stress for Glidcop AL-15 is quoted at 300MPa. Therefore the mask block design is not stress critical.

6. Conclusion

FEA analysis has shown that the proposed design for the LSC-101 fixed aperture masks to satisfy operational and material constraints.

Appendix I: Work Breakdown Schedule

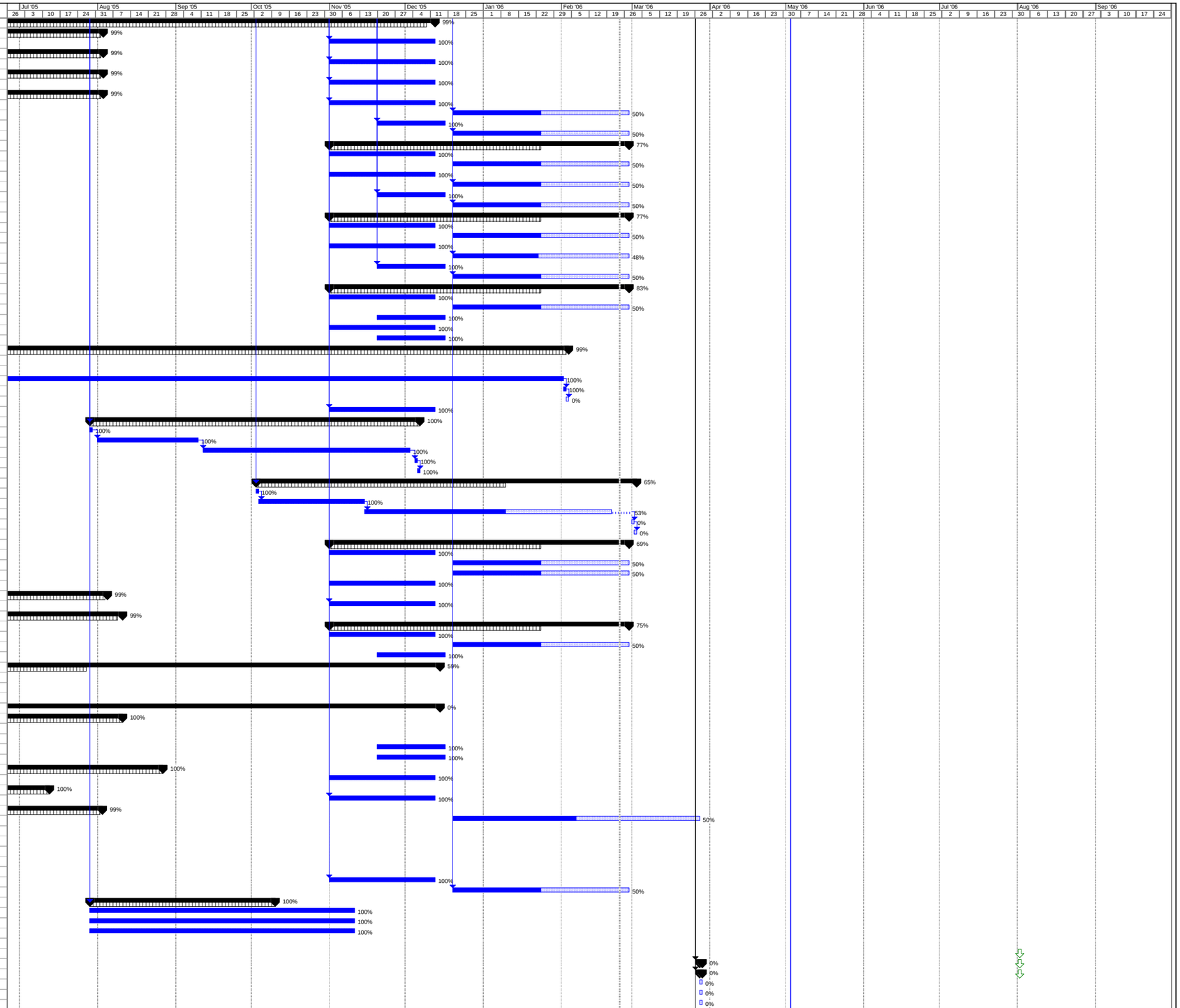
ID	WBS	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**1**	**LS-CAT Construction**	**1149.5 days?**	**Wed 10/23/02**	**Tue 3/20/07**		
2	1.1	CDR	1 day?	Tue 4/8/03	Tue 4/8/03		
3	1.2	PDR	55 days?	Mon 5/2/05	Fri 7/15/05		
4	1.3	Tape Floor	1.6 wks	Tue 6/28/05	Thu 7/7/05		
5	**1.4**	**FDR**	**1 day?**	**Tue 4/8/03**	**Tue 4/8/03**		
6	1.4.1	Create Ray Tracings	1 day?	Tue 4/8/03	Tue 4/8/03		
7	**1.5**	**Front End and Undulators**	**766 days?**	**Tue 4/8/03**	**Tue 3/14/06**		
8	1.5.1	Fund APS Account	1 day?	Tue 4/8/03	Tue 4/8/03		
9	**1.5.2**	**Install Front End**	**139 days?**	**Thu 9/1/05**	**Tue 3/14/06**		
10	1.5.2.1	Upstream of Ratchet Wall	22 days?	Thu 9/1/05	Fri 9/30/05		
11	1.5.2.2	Downstream of Ratchet Wall	10 days	Wed 3/1/06	Tue 3/14/06	31	
12	1.5.3	Install First Undulator	649 days?	Tue 4/8/03	Fri 9/30/05		
13	1.5.4	Install Final Undulator	30 days?	Tue 12/20/05	Mon 1/30/06		
14	**1.6**	**Radiation Enclosures and Utilities**	**1139.5 days?**	**Wed 10/23/02**	**Tue 3/6/07**		
15	**1.6.1**	**Drawings**	**707 days?**	**Wed 10/23/02**	**Thu 7/7/05**		
16	1.6.1.1	Initial	440 days?	Wed 10/23/02	Tue 6/29/04		LR Designs
17	1.6.1.2	Preliminary AutoCAD	120 days?	Mon 1/3/05	Fri 6/17/05	16	BS
18	1.6.1.3	ProE	14 days	Mon 6/20/05	Thu 7/7/05	16,17	APS
19	1.6.2	Specify	51 days?	Mon 5/2/05	Mon 7/11/05		BS,KB,RS,JQ
20	1.6.3	Procure	60 days	Mon 7/18/05	Fri 10/7/05	19,3	APS
21	**1.6.4**	**Manufacture**	**118 days?**	**Wed 11/9/05**	**Fri 4/21/06**	**20**	**TeckNit**
22	1.6.4.1	21-ID-A	40 days?	Wed 11/9/05	Tue 1/3/06		
23	1.6.4.2	21-ID-WBT	36 days?	Tue 11/15/05	Tue 1/3/06		
24	1.6.4.3	21-ID-B	47 days?	Fri 12/2/05	Mon 2/6/06		
25	1.6.4.4	21-ID-C	52 days?	Fri 12/16/05	Mon 2/27/06		
26	1.6.4.5	21-ID-D	53 days?	Wed 1/4/06	Fri 3/17/06		
27	1.6.4.6	21-ID-E	48 days?	Thu 1/19/06	Mon 3/27/06		
28	1.6.4.7	21-ID-F	47 days?	Thu 2/2/06	Fri 4/7/06		
29	1.6.4.8	21-ID-G	45 days?	Mon 2/20/06	Fri 4/21/06		
30	**1.6.5**	**Install**	**785 days?**	**Mon 7/14/03**	**Fri 7/14/06**		**Tech Staff**
31	1.6.5.1	21-ID-A	40 days?	Wed 1/4/06	Tue 2/28/06	22	
32	1.6.5.2	21-ID-WBT	40 days?	Wed 1/4/06	Tue 2/28/06	23	
33	1.6.5.3	21-ID-B	48 days?	Tue 2/7/06	Thu 4/13/06	24	
34	1.6.5.4	21-ID-C	45 days?	Tue 2/28/06	Mon 5/1/06	25	
35	1.6.5.5	21-ID-D	41 days?	Mon 3/20/06	Mon 5/15/06	26	
36	1.6.5.6	21-ID-E	49 days?	Tue 3/28/06	Fri 6/2/06	27	
37	1.6.5.7	21-ID-F	51 days?	Mon 4/10/06	Mon 6/19/06	28	
38	1.6.5.8	21-ID-G	60 days?	Mon 4/24/06	Fri 7/14/06	29	
39	1.6.5.9	Utility Hookup	1 day?	Mon 7/14/03	Mon 7/14/03		
40	**1.6.6**	**Shielding Verification**	**156.5 days**	**Mon 7/31/06**	**Tue 3/6/07**		
41	1.6.6.1	21-ID-A	4 hrs	Mon 7/31/06	Mon 7/31/06	31,64	
42	1.6.6.2	21-ID-WBT	2 hrs	Tue 8/1/06	Tue 8/1/06	32,44	
43	1.6.6.3	21-ID-B	3 hrs	Tue 8/1/06	Tue 8/1/06	33,41	
44	1.6.6.4	21-ID-C	3 hrs	Tue 8/1/06	Tue 8/1/06	34,43	
45	1.6.6.5	21-ID-D	4 hrs	Tue 8/15/06	Tue 8/15/06	35,43,44	
46	1.6.6.6	21-ID-E	4 hrs	Tue 3/6/07	Tue 3/6/07	36	
47	1.6.6.7	21-ID-F	4 hrs	Tue 2/6/07	Tue 2/6/07	37	
48	1.6.6.8	21-ID-G	4 hrs	Tue 2/6/07	Tue 2/6/07	38	
49	**1.7**	**PSS**	**276 days?**	**Fri 7/8/05**	**Fri 7/28/06**		**APS**
50	1.7.1	Specify	1 day?	Fri 7/8/05	Fri 7/8/05		KB,APS
51	1.7.2	Off-line construction/testing	120 days?	Mon 10/17/05	Fri 3/31/06	50	APS
52	**1.7.3**	**Installation**	**85 days?**	**Mon 4/3/06**	**Fri 7/28/06**	**51**	**APS**
53	1.7.3.1	Station Wiring 21-ID-A	9 days	Mon 4/3/06	Thu 4/13/06	31	
54	1.7.3.2	Station Wiring 21-ID-B	9 days	Fri 4/14/06	Wed 4/26/06	33,53	
55	1.7.3.3	Station Wiring 21-ID-C	9 days	Tue 5/2/06	Fri 5/12/06	34,54	
56	1.7.3.4	Station Wiring 21-ID-D	9 days	Tue 5/16/06	Fri 5/26/06	35,55	
57	1.7.3.5	Station Wiring 21-ID-E	9 days	Mon 6/5/06	Thu 6/15/06	36,56	
58	1.7.3.6	Station Wiring 21-ID-F	9 days	Tue 6/20/06	Fri 6/30/06	37,57	
59	1.7.3.7	Station Wiring 21-ID-G	9 days	Mon 7/17/06	Thu 7/27/06	38,58	
60	1.7.3.8	Interconnect 21-ID-A	2 days	Fri 4/14/06	Mon 4/17/06	53	
61	1.7.3.9	Interconnect 21-ID-B	2 days	Mon 5/15/06	Tue 5/16/06	54,55	
62	1.7.3.10	Interconnect 21-ID-D	1 day?	Fri 6/16/06	Fri 6/16/06	56,57	
63	1.7.3.11	Interconnect 21-ID-F	1 day?	Fri 7/28/06	Fri 7/28/06	58,59	
64	1.7.3.12	Interlock checkout, Phase I	1 day?	Mon 6/19/06	Mon 6/19/06	60,61,62	
65							
66	**1.8**	**Assemble Beamline Components**	**776 days?**	**Tue 4/8/03**	**Tue 3/28/06**		
67	**1.8.1**	**Phase I**	**775 days?**	**Tue 4/8/03**	**Mon 3/27/06**		
68	**1.8.1.1**	**Minor Components**	**755 days?**	**Tue 4/8/03**	**Mon 2/27/06**		
69	1.8.1.1.1	Adapter: 4.5-8	4 wks	Sun 11/20/05	Fri 12/16/05		
70	1.8.1.1.2	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
71	1.8.1.1.3	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
72	1.8.1.1.4	Cross	1 day?	Tue 4/8/03	Tue 4/8/03		
73	1.8.1.1.5	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
74	1.8.1.1.6	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
75	1.8.1.1.7	Cross	10 wks	Tue 12/20/05	Mon 2/27/06		
76	1.8.1.1.8	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
77	1.8.1.1.9	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
78	1.8.1.1.10	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05		
79	1.8.1.1.11	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
80	**1.8.1.2**	**Brehmsstrahlung Collimator**	**112 days?**	**Mon 10/3/05**	**Tue 3/7/06**		
81	1.8.1.2.1	Specify	1 day?	Mon 10/3/05	Mon 10/3/05		BS,JQ
82	1.8.1.2.2	Procure	30 days?	Tue 10/4/05	Mon 11/14/05	81	
83	1.8.1.2.3	Manufacture	15.8 wks?	Tue 11/15/05	Fri 3/3/06	82	
84	1.8.1.2.4	Accept	1 day	Mon 3/6/06	Mon 3/6/06	83	BS
85	1.8.1.2.5	Scribe Fiducial Marks	1 day?	Tue 3/7/06	Tue 3/7/06	84	APS
86	**1.8.1.3**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
87	1.8.1.3.1	Cross	10 wks	Tue 12/20/05	Mon 2/27/06		
88	1.8.1.3.2	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
89	1.8.1.3.3	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
90	1.8.1.3.4	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
91	1.8.1.3.5	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
92	1.8.1.3.6	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
93	1.8.1.3.7	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
94	1.8.1.3.8	Valve	10 wks	Tue 12/20/05	Mon 2/27/06		
95	1.8.1.3.9	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
96	1.8.1.3.10	Cross	10 wks	Tue 12/20/05	Mon 2/27/06		
97	1.8.1.3.11	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		



Project: LS-CAT-KB
Date: Fri 2/24/06

Critical
Critical Split
Critical Progress
Task
Split
Task Progress
Baseline
Baseline Split
Baseline Milestone
Milestone
Summary Progress
Summary
Project Summary
External Tasks
External Milestone
Deadline

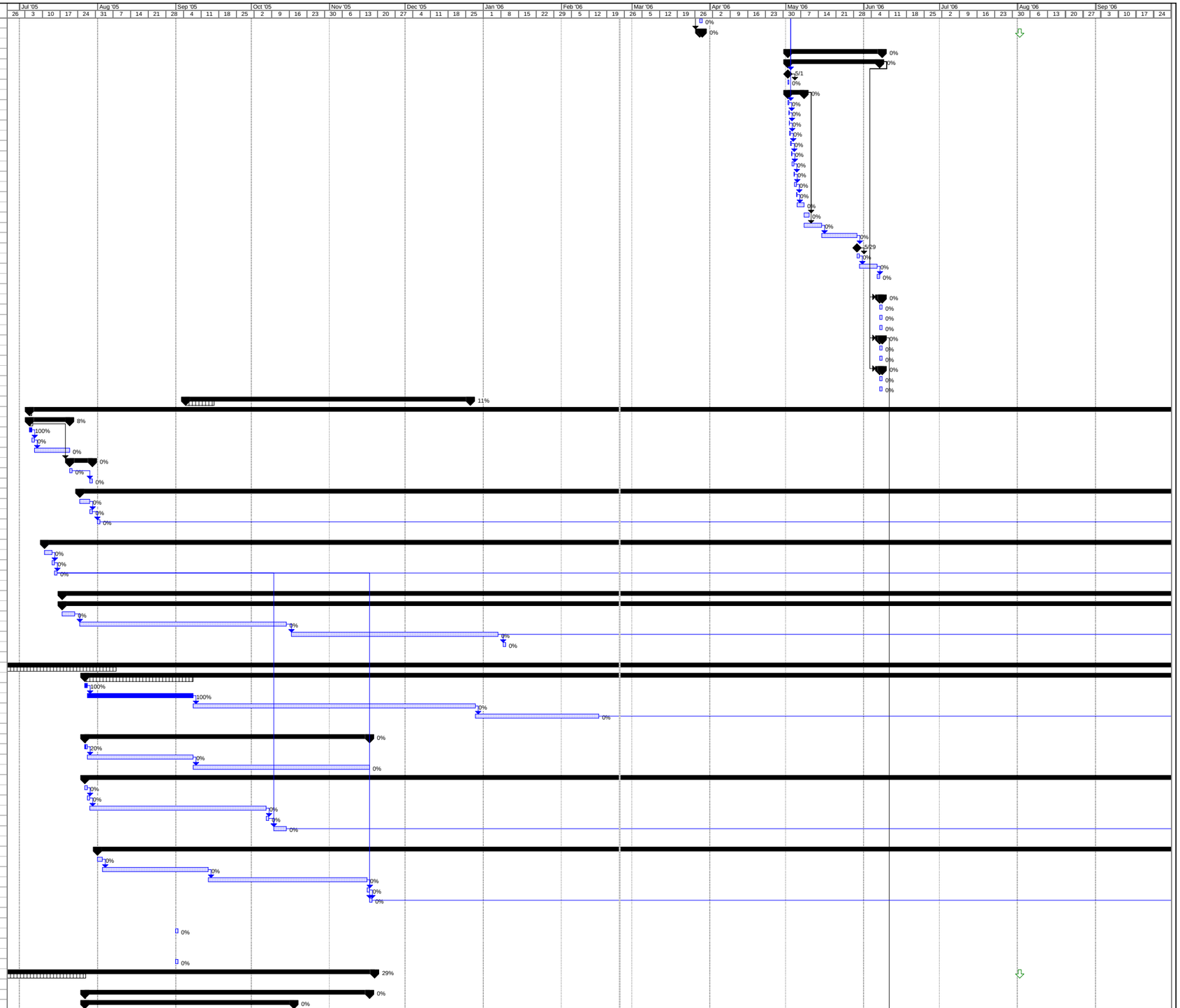
ID	WBS	Task Name	Duration	Start	Finish	Predecessors	Resource Names
98	**1.8.1.4**	**Mask Assembly**	**243 days?**	**Thu 1/6/05**	**Mon 12/12/05**		
99	**1.8.1.4.1**	**Fixed Mask 1**	**149.1 days?**	**Thu 1/6/05**	**Wed 8/3/05**		
105	1.8.1.4.2	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
106	**1.8.1.4.3**	**Fixed Mask 2**	**149.1 days?**	**Thu 1/6/05**	**Wed 8/3/05**		
112	1.8.1.4.4	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
113	**1.8.1.4.5**	**Fixed Mask 3**	**149.1 days?**	**Thu 1/6/05**	**Wed 8/3/05**		
119	1.8.1.4.6	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
120	**1.8.1.4.7**	**Fixed Mask 4**	**149.1 days?**	**Thu 1/6/05**	**Wed 8/3/05**		
126	1.8.1.5	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
127	1.8.1.6	Cross	10 wks	Tue 12/20/05	Mon 2/27/06	3	
128	1.8.1.7	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05	3	
129	1.8.1.8	Mono D3 Placeholder	10 wks	Tue 12/20/05	Mon 2/27/06	3	
130	**1.8.1.9**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
131	1.8.1.9.1	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
132	1.8.1.9.2	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
133	1.8.1.9.3	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
134	1.8.1.10	Cross	10 wks	Tue 12/20/05	Mon 2/27/06	3	
135	1.8.1.11	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05	3	
136	1.8.1.12	Mono D2 Placeholder	10 wks	Tue 12/20/05	Mon 2/27/06	3	
137	**1.8.1.13**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
138	1.8.1.13.1	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
139	1.8.1.13.2	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
140	1.8.1.13.3	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
141	1.8.1.14	Cross	10 wks	Tue 12/20/05	Mon 2/27/06	3	
142	1.8.1.15	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05	3	
143	1.8.1.16	Mono D1 Placeholder	10 wks	Tue 12/20/05	Mon 2/27/06	3	
144	**1.8.1.17**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
145	1.8.1.17.1	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
146	1.8.1.17.2	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
147	1.8.1.17.3	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05		
148	1.8.1.17.4	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
149	1.8.1.17.5	Adapter	4 wks	Sun 11/20/05	Fri 12/16/05		
150	**1.8.1.18**	**Sharma Beamstop**	**244 days?**	**Tue 3/1/05**	**Fri 2/3/06**		
151	1.8.1.18.1	Specify	1 day?	Tue 3/1/05	Tue 3/1/05		JQ
152	1.8.1.18.2	Procure	1 day?	Thu 3/3/05	Thu 3/3/05	151	
153	1.8.1.18.3	Manufacture	47.8 wks?	Fri 3/4/05	Wed 2/1/06	152	ANL
154	1.8.1.18.4	Accept	1 day	Thu 2/2/06	Thu 2/2/06	153	BS
155	1.8.1.18.5	Feducialize	1 day?	Fri 2/3/06	Fri 2/3/06	154	APS
156	1.8.1.19	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
157	**1.8.1.20**	**K5-30 Collimator**	**93 days?**	**Fri 7/29/05**	**Tue 12/6/05**	**3**	
158	1.8.1.20.1	Specify	1 day?	Fri 7/29/05	Fri 7/29/05		BS
159	1.8.1.20.2	Procure	30 days?	Mon 8/1/05	Fri 9/9/05	158	
160	1.8.1.20.3	Manufacture	12 wks	Mon 9/12/05	Fri 12/2/05	159	
161	1.8.1.20.4	Accept	1 day	Mon 12/5/05	Mon 12/5/05	160	BS
162	1.8.1.20.5	Feducialize	1 day?	Tue 12/6/05	Tue 12/6/05	161	APS
163	**1.8.1.21**	**Brehmsstrahlung Collimator**	**109 days?**	**Mon 10/3/05**	**Thu 3/2/06**	**3**	
164	1.8.1.21.1	Specify	1 day?	Mon 10/3/05	Mon 10/3/05		BS,JQ
165	1.8.1.21.2	Procure	30 days?	Tue 10/4/05	Mon 11/14/05	164	
166	1.8.1.21.3	Manufacture	15.2 wks?	Tue 11/15/05	Tue 2/28/06	165	
167	1.8.1.21.4	Accept	1 day	Wed 3/1/06	Wed 3/1/06	166	BS
168	1.8.1.21.5	Feducialize	1 day?	Thu 3/2/06	Thu 3/2/06	167	APS
169	**1.8.1.22**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
170	1.8.1.22.1	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
171	1.8.1.22.2	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
172	1.8.1.22.3	Cross	10 wks	Tue 12/20/05	Mon 2/27/06		
173	1.8.1.22.4	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
174	**1.8.1.23**	**L5-92**	**151 days?**	**Thu 1/6/05**	**Thu 8/4/05**		
180	1.8.1.24	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
181	**1.8.1.25**	**L5-92**	**142 days?**	**Tue 1/25/05**	**Wed 8/10/05**		
188	**1.8.1.26**	**Minor Components**	**85 days**	**Tue 11/1/05**	**Mon 2/27/06**		
189	1.8.1.26.1	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
190	1.8.1.26.2	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06		
191	1.8.1.26.3	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05		
192	**1.8.1.27**	**Monochromator Assembly**	**702 days?**	**Tue 4/8/03**	**Wed 12/14/05**		
193	1.8.1.27.1	Kohzu Monochromator	1 day?	Mon 4/14/03	Mon 4/14/03		
194	1.8.1.27.2	Kohzu Upgrade	1 day?	Tue 4/8/03	Tue 4/8/03		
195	1.8.1.27.3	CryoCooler	1 day?	Tue 4/8/03	Tue 4/8/03		
196	**1.8.1.27.4**	**Crystal Holders**	**142 days?**	**Tue 5/31/05**	**Wed 12/14/05**		
202	**1.8.1.27.5**	**Crystals**	**50 days?**	**Thu 6/2/05**	**Wed 8/10/05**		
207	1.8.1.27.6	Scribe Fiducial Marks	1 day?	Tue 4/15/03	Tue 4/15/03	193	
208	1.8.1.28	Fluorescence Screen	1 day?	Tue 4/8/03	Tue 4/8/03		
209	1.8.1.29	UHV Valve	4 wks	Sun 11/20/05	Fri 12/16/05		
210	1.8.1.30	Adapter	4 wks	Sun 11/20/05	Fri 12/16/05		
211	**1.8.1.31**	**Quad Diode BPM**	**113 days?**	**Wed 3/23/05**	**Fri 8/26/05**		
217	1.8.1.32	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05		
218	**1.8.1.33**	**P4-60**	**88 days?**	**Fri 3/11/05**	**Tue 7/12/05**		
224	1.8.1.34	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
225	**1.8.1.35**	**P4-40**	**103 days?**	**Fri 3/11/05**	**Tue 8/2/05**		
231	1.8.1.36	Shielded Nipple	14 wks	Tue 12/20/05	Mon 3/27/06		
232	1.8.1.37	Be Window	1 day?	Tue 4/8/03	Tue 4/8/03		
233	1.8.1.38	He Flow Tube	1 day?	Tue 4/8/03	Tue 4/8/03		
234	1.8.1.39	Be Window	1 day?	Tue 4/8/03	Tue 4/8/03		
235	**1.8.1.40**	**Mirror System**	**1 day?**	**Tue 4/8/03**	**Tue 4/8/03**		
238	1.8.1.41	Be Window	1 day?	Tue 4/8/03	Tue 4/8/03		
239	1.8.1.42	Bellows	6 wks	Tue 11/1/05	Mon 12/12/05	3	
240	1.8.1.43	Nipple	10 wks	Tue 12/20/05	Mon 2/27/06	3	
241	**1.8.1.44**	**K5-50 End Stop**	**51.6 days?**	**Fri 7/29/05**	**Mon 10/10/05**	**3**	
247	1.8.1.45	P8-50 (21-ID-E)	75 days?	Fri 7/29/05	Thu 11/10/05		
248	1.8.1.46	P8-50 (21-ID-F)	75 days?	Fri 7/29/05	Thu 11/10/05		
249	1.8.1.47	P8-50 (21-ID-G)	75 days?	Fri 7/29/05	Thu 11/10/05		
250	1.8.1.48	Quad Table BPM	44 days?	Thu 5/13/04	Tue 7/13/04		
251	1.8.1.49	Pump Cart	6 days?	Thu 6/10/04	Thu 6/17/04		
252	**1.8.2**	**Phase II: 21-ID-E**	**1 day?**	**Tue 3/28/06**	**Tue 3/28/06**	**67**	
257	**1.8.3**	**Phase II: 21-ID-F**	**1 day?**	**Tue 3/28/06**	**Tue 3/28/06**	**67**	
258	1.8.3.1	Mirror System	1 day?	Tue 3/28/06	Tue 3/28/06		
259	1.8.3.2	Mono Cyrstals	1 day?	Tue 3/28/06	Tue 3/28/06		
260	1.8.3.3	Mono	1 day?	Tue 3/28/06	Tue 3/28/06		



Project: LS-CAT-KB
Date: Fri 2/24/06

Critical
Critical Split
Critical Progress
Task
Split
Task Progress
Baseline
Baseline Split
Baseline Milestone
Milestone
Summary Progress
Summary
Project Summary
External Tasks
External Milestone
Deadline

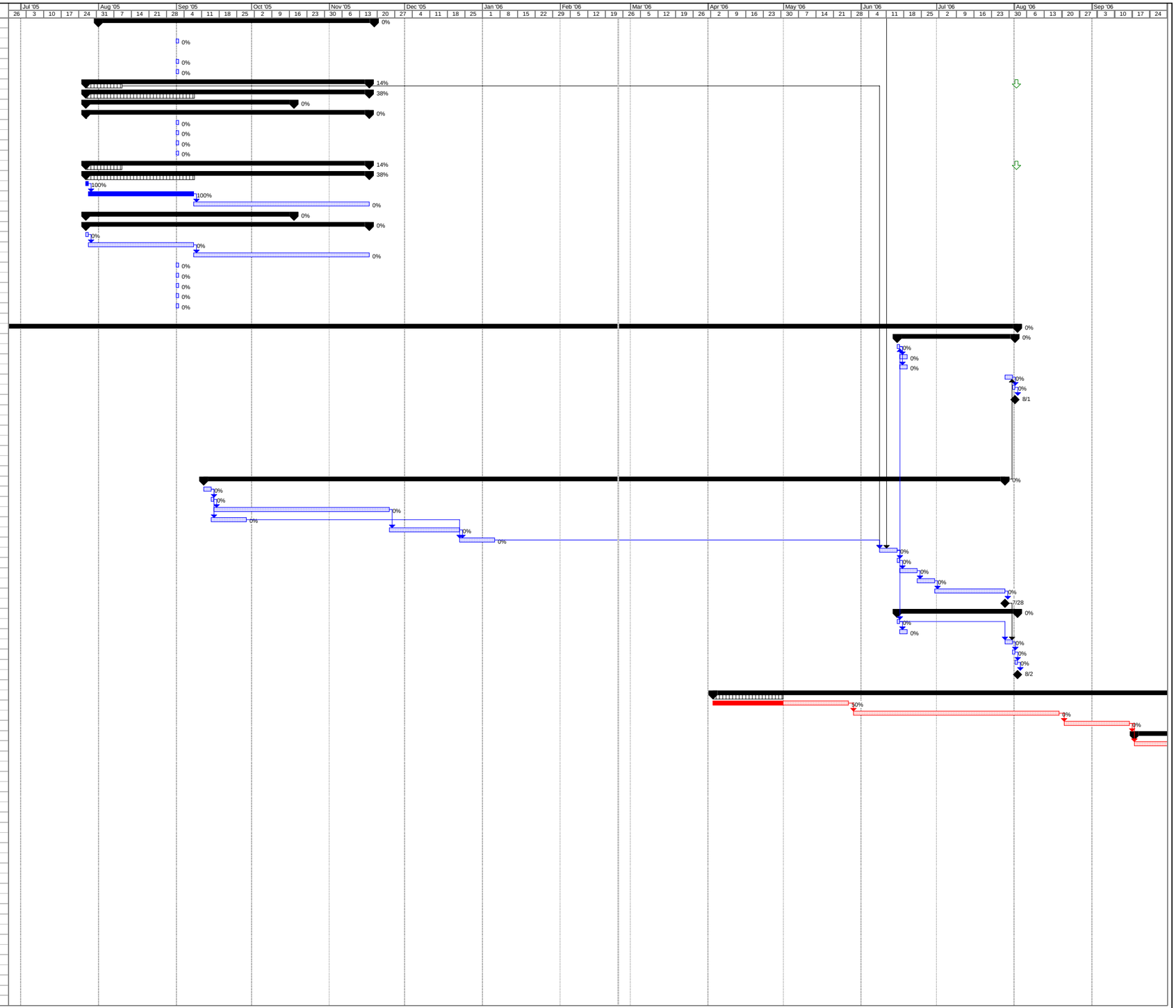
ID	WBS	Task Name	Duration	Start	Finish	Predecessors	Resource Names
261	1.8.3.4	Vacuum Components	1 day?	Tue 3/28/06	Tue 3/28/06		
262	**1.8.4**	**Phase II: 21-ID-G**	**1 day?**	**Tue 3/28/06**	**Tue 3/28/06**	**67**	
266							
267	**1.9**	**Install Beamline Components**	**27.25 days?**	**Mon 5/1/06**	**Thu 6/8/06**		
268	**1.9.1**	**Phase 1**	**26.25 days?**	**Mon 5/1/06**	**Wed 6/7/06**		
269	1.9.1.1	Survey New Landmarks	0 days	Mon 5/1/06	Mon 5/1/06	34	
270	1.9.1.2	21-ID-A Collimator Surveyed	2 hrs	Tue 5/2/06	Tue 5/2/06	269	
271	**1.9.1.3**	**Install/Survey SOE/TOE Critical Components**	**4.25 days**	**Tue 5/2/06**	**Mon 5/8/06**		
272	1.9.1.3.1	K5-50	0.25 days	Tue 5/2/06	Tue 5/2/06	34	APS,LS-Techs
273	1.9.1.3.2	P4-40	0.25 days	Tue 5/2/06	Tue 5/2/06	272	APS,LS-Techs
274	1.9.1.3.3	P4-60	0.25 days	Tue 5/2/06	Tue 5/2/06	273	APS,LS-Techs
275	1.9.1.3.4	L5-92	0.25 days	Tue 5/2/06	Tue 5/2/06	274	APS,LS-Techs
276	1.9.1.3.5	Brehmsstrahlung Collimator 2	0.25 days	Wed 5/3/06	Wed 5/3/06	275	APS,LS-Techs
277	1.9.1.3.6	K5-30	0.5 days	Wed 5/3/06	Wed 5/3/06	276	APS,LS-Techs
278	1.9.1.3.7	Sharma Beamstop	0.5 days	Wed 5/3/06	Thu 5/4/06	277	APS,LS-Techs
279	1.9.1.3.8	IDT Masks	0.5 days	Thu 5/4/06	Thu 5/4/06	278	APS,LS-Techs
280	1.9.1.3.9	P8-50, ID-E	0.5 days	Thu 5/4/06	Fri 5/5/06	279	APS,LS-Techs
281	1.9.1.3.10	P8-50, ID-G	0.5 days	Fri 5/5/06	Fri 5/5/06	280	APS,LS-Techs
282	1.9.1.3.11	P8-50, ID-F	0.5 days	Fri 5/5/06	Mon 5/8/06	281	APS,LS-Techs
283	1.9.1.4	Install/Survey Monochromator	2 days	Mon 5/8/06	Wed 5/10/06	271	LS-Techs
284	1.9.1.5	Install Remaining Beamline Components IDB/C	1 wk?	Mon 5/8/06	Mon 5/15/06	271	LS-Techs
285	1.9.1.6	Pump Down	2 wks	Mon 5/15/06	Mon 5/29/06	284	
286	1.9.1.7	First Light, ID-D	0 days	Mon 5/29/06	Mon 5/29/06	285	
287	1.9.1.8	Shielding Verification, ID-B/G	1 day?	Mon 5/29/06	Tue 5/30/06	286	
288	1.9.1.9	Mitigation	1 wk?	Tue 5/30/06	Tue 6/6/06	287	
289	1.9.1.10	Shielding Verification, ID-B/G	1 day?	Tue 6/6/06	Wed 6/7/06	288	
290							
291	**1.9.2**	**Phase II: 21-ID-E**	**1 day?**	**Wed 6/7/06**	**Thu 6/8/06**	**268**	
292	1.9.2.1	First Crystal and Holder	1 day?	Wed 6/7/06	Thu 6/8/06		
293	1.9.2.2	Second Crystal and Holder	1 day?	Wed 6/7/06	Thu 6/8/06		
294	1.9.2.3	Mirror System	1 day?	Wed 6/7/06	Thu 6/8/06		
295	**1.9.3**	**Phase II: 21-ID-F**	**1 day?**	**Wed 6/7/06**	**Thu 6/8/06**	**268**	
296	1.9.3.1	Mirror System	1 day?	Wed 6/7/06	Thu 6/8/06		
297	1.9.3.2	Diamond Mono	1 day?	Wed 6/7/06	Thu 6/8/06		
298	**1.9.4**	**Phase II: 21-ID-G**	**1 day?**	**Wed 6/7/06**	**Thu 6/8/06**	**268**	
299	1.9.4.1	Mirror System	1 day?	Wed 6/7/06	Thu 6/8/06		
300	1.9.4.2	Diamond Mono	1 day?	Wed 6/7/06	Thu 6/8/06		
301	**1.10**	**EPS**	**81 days?**	**Mon 9/5/05**	**Mon 12/26/05**		
307	**1.11**	**Beamline Controls**	**445.5 days?**	**Tue 7/5/05**	**Tue 3/20/07**		
308	**1.11.1**	**Alignment Procedures**	**12 days?**	**Tue 7/5/05**	**Wed 7/20/05**		
309	1.11.1.1	Enumerate Degrees of Freedom	1 day?	Tue 7/5/05	Tue 7/5/05		KB
310	1.11.1.2	Algorithm to Eliminate Degrees of Freedom	1 day?	Wed 7/6/05	Wed 7/6/05	309	KB
311	1.11.1.3	Add Appropriate Diagnostics to Beamline Components	2 wks?	Thu 7/7/05	Wed 7/20/05	310	KB,BS
312	**1.11.2**	**Motors and Motor Drivers**	**7 days?**	**Thu 7/21/05**	**Fri 7/29/05**	**308**	
313	1.11.2.1	Check torque/precision/feedback Specs	1 day?	Thu 7/21/05	Thu 7/21/05		KB
314	1.11.2.2	Select Driver/Motor Combinations Each Motor	1 day?	Fri 7/29/05	Fri 7/29/05	313	KB
315	**1.11.3**	**Cabling**	**431.5 days?**	**Mon 7/25/05**	**Tue 3/20/07**		
316	1.11.3.1	Route Cables	4 days?	Mon 7/25/05	Thu 7/28/05		BS,KB
317	1.11.3.2	Procure Components	1 day?	Fri 7/29/05	Fri 7/29/05	316	BS
318	1.11.3.3	Assemble/Test Wiring	1 day?	Mon 8/1/05	Mon 8/1/05	317	Tech Staff
319	1.11.3.4	Install Cabling	2 wks?	Tue 3/6/07	Tue 3/20/07	14,318	Tech Staff
320	**1.11.4**	**Control System Infrastructure**	**441.5 days?**	**Mon 7/11/05**	**Tue 3/20/07**		
321	1.11.4.1	Determine Type/Distribution of IOCs	3 days?	Mon 7/11/05	Wed 7/13/05		
322	1.11.4.2	Procure Hardware	1 day?	Thu 7/14/05	Thu 7/14/05	321	
323	1.11.4.3	Install/Test in Lab	1 day?	Fri 7/15/05	Fri 7/15/05	322	
324	1.11.4.4	Install/Test in Final Location	2 wks?	Tue 3/6/07	Tue 3/20/07	323,14	
325	**1.11.5**	**Control System Software**	**427.5 days?**	**Mon 7/18/05**	**Wed 3/7/07**		
326	**1.11.5.1**	**Drivers and IOC Software**	**427.5 days?**	**Mon 7/18/05**	**Wed 3/7/07**		
327	1.11.5.1.1	Identify needed Software	1 wk	Mon 7/18/05	Fri 7/22/05		
328	1.11.5.1.2	Obtain and Install Software on Test IOC	60 days?	Mon 7/25/05	Fri 10/14/05	327	
329	1.11.5.1.3	Procure IOCs	60 days?	Mon 10/17/05	Fri 1/6/06	328	
330	1.11.5.1.4	Install Software and Test in Lab	1 day?	Mon 1/9/06	Mon 1/9/06	329	
331	1.11.5.1.5	Install on Beamline	1 day?	Tue 3/6/07	Wed 3/7/07	14,329	
332	**1.12**	**21-ID-D End Station Equipment**	**1021.5 days?**	**Tue 4/8/03**	**Wed 3/7/07**		
333	**1.12.1**	**Detector System**	**420.5 days?**	**Wed 7/27/05**	**Wed 3/7/07**		
334	1.12.1.1	Select Detector	1 day?	Wed 7/27/05	Wed 7/27/05		LS-CAT ESEC
335	1.12.1.2	Procure	30 days?	Thu 7/28/05	Wed 9/7/05	334	
336	1.12.1.3	Manufacture	4 mons?	Thu 9/8/05	Wed 12/28/05	335	
337	1.12.1.4	Accept	7 wks?	Thu 12/29/05	Wed 2/15/06	336	
338	1.12.1.5	Install	1 day?	Tue 3/6/07	Wed 3/7/07	337,14	
339	**1.12.2**	**Diffractometer**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
340	1.12.2.1	Select System	1 day?	Wed 7/27/05	Wed 7/27/05		LS-CAT ESEC
341	1.12.2.2	Procure	30 days?	Thu 7/28/05	Wed 9/7/05	340	
342	1.12.2.3	Manufacture	10 wks?	Thu 9/8/05	Wed 11/16/05	341	
343	**1.12.3**	**Positioning System**	**420.5 days?**	**Wed 7/27/05**	**Wed 3/7/07**		
344	1.12.3.1	Select System	1 day?	Wed 7/27/05	Wed 7/27/05		LS-CAT ESEC
345	1.12.3.2	Procure	1 day?	Thu 7/28/05	Thu 7/28/05	344	
346	1.12.3.3	Manufacture	10 wks?	Fri 7/29/05	Thu 10/6/05	345	
347	1.12.3.4	Accept	1 day?	Fri 10/7/05	Fri 10/7/05	346	
348	1.12.3.5	Integrate Into Beamline Controls	1 wk?	Mon 10/10/05	Fri 10/14/05	323,347	
349	1.12.3.6	Install on Beamline	1 day?	Tue 3/6/07	Wed 3/7/07	14,348	
350	**1.12.4**	**Fluorescence Detector**	**417.5 days?**	**Mon 8/1/05**	**Wed 3/7/07**		
351	1.12.4.1	Select	2 days?	Mon 8/1/05	Tue 8/2/05		KB
352	1.12.4.2	Procure	30 days?	Wed 8/3/05	Tue 9/13/05	351	
353	1.12.4.3	Manufacture	45 days?	Wed 9/14/05	Tue 11/15/05	352	
354	1.12.4.4	Accept	1 day?	Wed 11/16/05	Wed 11/16/05	353	
355	1.12.4.5	Integrate Into Beamline Controls	1 day?	Thu 11/17/05	Thu 11/17/05	323,354	
356	1.12.4.6	Install on Beamline	1 day?	Tue 3/6/07	Wed 3/7/07	14,355	
357	1.12.5	Table BPM	42 days?	Thu 5/13/04	Fri 7/9/04		
358	1.12.6	Current Preamplifiers	1 day?	Thu 9/1/05	Thu 9/1/05		
359	1.12.7	Coldstream	10 wks?	Wed 2/16/05	Tue 4/26/05		
360	1.12.8	Slits	1 day?	Tue 4/8/03	Tue 4/8/03		
361	1.12.9	Ion Chambers	1 day?	Thu 9/1/05	Thu 9/1/05		
362	**1.13**	**21-ID-E End Station Equipment**	**490 days?**	**Mon 1/5/04**	**Fri 11/18/05**		
363	1.13.1	Detector: MAR 225	1 day?	Mon 1/5/04	Mon 1/5/04		
364	**1.13.2**	**Diffractometer**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
368	**1.13.3**	**Positioning System**	**59 days?**	**Wed 7/27/05**	**Mon 10/17/05**		



Project: LS-CAT-KB
Date: Fri 2/24/06

Critical | Critical Progress | Split | Baseline | Baseline Milestone | Summary Progress | Project Summary | External Milestone
Critical Split | Task | Task Progress | Baseline Split | Milestone | Summary | External Tasks | Deadline

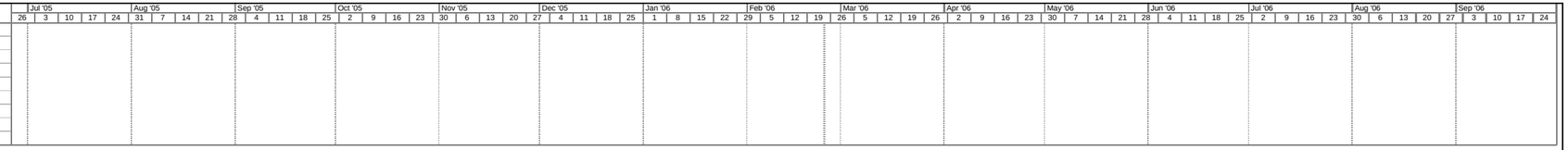
ID	WBS	Task Name	Duration	Start	Finish	Predecessors	Resource Names
375	**1.13.4**	**Fluorescence Detector**	**80 days?**	**Mon 8/1/05**	**Fri 11/18/05**		
382	1.13.5	Table BPM	42 days?	Thu 5/13/04	Fri 7/9/04		
383	1.13.6	Current Preamplifiers	1 day?	Thu 9/1/05	Thu 9/1/05		
384	1.13.7	Coldstream	10 wks?	Wed 2/16/05	Tue 4/26/05		
385	1.13.8	Slits	1 day?	Thu 9/1/05	Thu 9/1/05		
386	1.13.9	Ion Chambers	1 day?	Thu 9/1/05	Thu 9/1/05		
387	**1.14**	**21-ID-F End Station Equipment**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
388	**1.14.1**	**Detector**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
392	**1.14.2**	**Positioning System**	**59 days?**	**Wed 7/27/05**	**Mon 10/17/05**		
399	**1.14.3**	**Diffractomater**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
404	1.14.4	Table BPM	1 day?	Thu 9/1/05	Thu 9/1/05		
405	1.14.5	Current Preamplifiers	1 day?	Thu 9/1/05	Thu 9/1/05		
406	1.14.6	Slits	1 day?	Thu 9/1/05	Thu 9/1/05		
407	1.14.7	Ion Chambers	1 day?	Thu 9/1/05	Thu 9/1/05		
408	**1.15**	**21-ID-G End Station Equipment**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
409	**1.15.1**	**Detector**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
410	1.15.1.1	Select System	1 day?	Wed 7/27/05	Wed 7/27/05		LS-CAT ESEC
411	1.15.1.2	Procure	30 days?	Thu 7/28/05	Wed 9/7/05	410	
412	1.15.1.3	Manufacture	10 wks?	Thu 9/8/05	Wed 11/16/05	411	
413	**1.15.2**	**Positioning System**	**59 days?**	**Wed 7/27/05**	**Mon 10/17/05**		
420	**1.15.3**	**Diffractomater**	**81 days?**	**Wed 7/27/05**	**Wed 11/16/05**		
421	1.15.3.1	Select System	1 day?	Wed 7/27/05	Wed 7/27/05		LS-CAT ESEC
422	1.15.3.2	Procure	30 days?	Thu 7/28/05	Wed 9/7/05	421	
423	1.15.3.3	Manufacture	10 wks?	Thu 9/8/05	Wed 11/16/05	422	
424	1.15.4	Coldstream	1 day?	Thu 9/1/05	Thu 9/1/05		
425	1.15.5	Table BPM	1 day?	Thu 9/1/05	Thu 9/1/05		
426	1.15.6	Current Preamplifiers	1 day?	Thu 9/1/05	Thu 9/1/05		
427	1.15.7	Slits	1 day?	Thu 9/1/05	Thu 9/1/05		
428	1.15.8	Ion Chambers	1 day?	Thu 9/1/05	Thu 9/1/05		
429							
430	**2**	**Robotics**	**866.25 days?**	**Tue 4/8/03**	**Wed 8/2/06**		
431	**2.1**	**21-ID-D**	**33 days?**	**Thu 6/15/06**	**Tue 8/1/06**		
432	2.1.1	Obtain Components	1 day?	Thu 6/15/06	Fri 6/16/06	452	
433	2.1.2	Assemble Components	1 day?	Fri 6/16/06	Mon 6/19/06	432	
434	2.1.3	Bench Test Hardare/Software	1 day?	Fri 6/16/06	Mon 6/19/06	432	
435	2.1.4	Dry Run on Beamline	1 day?	Fri 7/28/06	Mon 7/31/06	445	
436	2.1.5	Friendly User Run	1 day?	Mon 7/31/06	Tue 8/1/06	435	
437	2.1.6	Add to User Program	0 days	Tue 8/1/06	Tue 8/1/06	436	
438	**2.2**	**21-ID-E**	**2 days?**	**Tue 4/8/03**	**Wed 4/9/03**		
439	2.2.1	Obtain Components	1 day?	Tue 4/8/03	Tue 4/8/03		
440	2.2.2	Assemble Components	1 day?	Wed 4/9/03	Wed 4/9/03	439	
441	2.2.3	Bench Test Hardare/Software	1 day?	Wed 4/9/03	Wed 4/9/03	439	
442	2.2.4	Dry Run on Beamline	1 day?	Tue 4/8/03	Tue 4/8/03		
443	2.2.5	Friendly User Run	1 day?	Wed 4/9/03	Wed 4/9/03	442	
444	2.2.6	Add to User Program	0 days	Wed 4/9/03	Wed 4/9/03	443	
445	**2.3**	**21-ID-F**	**229.25 days?**	**Mon 9/12/05**	**Fri 7/28/06**		
446	2.3.1	Specify	3 days?	Mon 9/12/05	Wed 9/14/05		
447	2.3.2	Plan GUI	1 day?	Thu 9/15/05	Thu 9/15/05	446	
448	2.3.3	Obtain Components	10 wks?	Fri 9/16/05	Thu 11/24/05	447	
449	2.3.4	Obain Base Software	2 wks?	Thu 9/15/05	Wed 9/28/05	446	
450	2.3.5	Assemble Components	4 wks	Fri 11/25/05	Thu 12/22/05	448	
451	2.3.6	Bench Test Hardare/Software	2 wks?	Fri 12/23/05	Thu 1/5/06	449,450	
452	2.3.7	1st Dry Run on Beamline	1 wk?	Thu 6/8/06	Thu 6/15/06	451,387,295	
453	2.3.8	Fix Problems	1 day?	Thu 6/15/06	Fri 6/16/06	452	
454	2.3.9	2nd Dry Run on Beamline	1 wk?	Fri 6/16/06	Fri 6/23/06	453	
455	2.3.10	Friendly User Run	1 wk?	Fri 6/23/06	Fri 6/30/06	454	
456	2.3.11	Fix Problems	4 wks?	Fri 6/30/06	Fri 7/28/06	455	
457	2.3.12	Add to User Program	0 days	Fri 7/28/06	Fri 7/28/06	456	
458	**2.4**	**21-ID-G**	**34 days?**	**Thu 6/15/06**	**Wed 8/2/06**		
459	2.4.1	Obtain Components	1 day?	Thu 6/15/06	Fri 6/16/06	452	
460	2.4.2	Assemble Components	1 day?	Fri 6/16/06	Mon 6/19/06	459	
461	2.4.3	Bench Test Hardare/Software	1 day?	Fri 7/28/06	Mon 7/31/06	457,459	
462	2.4.4	Dry Run on Beamline	1 day?	Mon 7/31/06	Tue 8/1/06	461	
463	2.4.5	Friendly User Run	1 day?	Tue 8/1/06	Wed 8/2/06	462	
464	2.4.6	Add to User Program	0 days	Wed 8/2/06	Wed 8/2/06	463	
465							
466	**3**	**Micro Focusing**	**788 days?**	**Mon 4/3/06**	**Wed 4/8/09**		
467	3.1	Review Options	8 wks	Mon 4/3/06	Fri 5/26/06		
468	3.2	Provide Detailed Plans	60 days	Mon 5/29/06	Fri 8/18/06	467	
469	3.3	Review Detailed Plans	4 wks	Mon 8/21/06	Fri 9/15/06	468	
470	**3.4**	**21-ID-D**	**284 days?**	**Mon 9/18/06**	**Thu 10/18/07**	**469**	
471	3.4.1	Specify Components	4 wks?	Mon 9/18/06	Fri 10/13/06	469	
472	3.4.2	Procure	30 days?	Mon 10/16/06	Fri 11/24/06	471	
473	3.4.3	Manufacture	9 mons?	Mon 11/27/06	Fri 8/3/07	472	
474	3.4.4	Accept	4 days?	Mon 8/6/07	Thu 8/9/07	473	
475	3.4.5	Install	2 wks?	Fri 8/10/07	Thu 8/23/07	474	
476	3.4.6	Staff Testing	4 wks?	Fri 8/24/07	Thu 9/20/07	475	
477	3.4.7	Friendly User Testing	4 wks?	Fri 9/21/07	Thu 10/18/07	476	
478	3.4.8	Add to User Program	0 days	Thu 10/18/07	Thu 10/18/07	477	
479	**3.5**	**21-ID-E**	**284 days?**	**Fri 10/19/07**	**Wed 11/19/08**	**470**	
480	3.5.1	Specify Components	4 wks	Fri 10/19/07	Thu 11/15/07		
481	3.5.2	Procure	30 days	Fri 11/16/07	Thu 12/27/07	480	
482	3.5.3	Manufacture	9 mons?	Fri 12/28/07	Thu 9/4/08	481	
483	3.5.4	Accept	4 days?	Fri 9/5/08	Wed 9/10/08	482	
484	3.5.5	Install	2 wks?	Thu 9/11/08	Wed 9/24/08	483	
485	3.5.6	Staff Testing	4 wks?	Thu 9/25/08	Wed 10/22/08	484	
486	3.5.7	Friendly User Testing	4 wks?	Thu 10/23/08	Wed 11/19/08	485	
487	3.5.8	Add to User Program	0 days	Wed 11/19/08	Wed 11/19/08	486	
488	**3.6**	**21-ID-F**	**334 days?**	**Fri 10/19/07**	**Wed 1/28/09**	**470**	
489	3.6.1	Specify Components	4 wks?	Fri 10/19/07	Thu 11/15/07		
490	3.6.2	Procure	30 days?	Fri 11/16/07	Thu 12/27/07	489	
491	3.6.3	Manufacture	9 mons?	Fri 12/28/07	Thu 9/4/08	490	
492	3.6.4	Accept	4 days?	Fri 9/5/08	Wed 9/10/08	491	
493	3.6.5	Install	2 wks?	Thu 11/20/08	Wed 12/3/08	492,479	
494	3.6.6	Staff Testing	4 wks?	Thu 12/4/08	Wed 12/31/08	493	
495	3.6.7	Friendly User Testing	4 wks?	Thu 1/1/09	Wed 1/28/09	494	
496	3.6.8	Add to User Program	0 days	Wed 1/28/09	Wed 1/28/09	495	



Project: LS-CAT-KB
Date: Fri 2/24/06

Critical | Critical Progress | Split | Baseline | Baseline Milestone | Summary Progress | Project Summary | External Milestone
Critical Split | Task | Task Progress | Baseline Split | Milestone | Summary | External Tasks | Deadline

ID	WBS	Task Name	Duration	Start	Finish	Predecessors	Resource Names
497	**3.7**	**21-ID-G**	**384 days?**	**Fri 10/19/07**	**Wed 4/8/09**	**470**	
498	3.7.1	Specify Components	4 wks?	Fri 10/19/07	Thu 11/15/07		
499	3.7.2	Procure	30 days?	Fri 11/16/07	Thu 12/27/07	498	
500	3.7.3	Manufacture	9 mons?	Fri 12/28/07	Thu 9/4/08	499	
501	3.7.4	Accept	4 days?	Fri 9/5/08	Wed 9/10/08	500	
502	3.7.5	Install	2 wks?	Thu 1/29/09	Wed 2/11/09	501,488	
503	3.7.6	Staff Testing	4 wks?	Thu 2/12/09	Wed 3/11/09	502	
504	3.7.7	Friendly User Testing	4 wks?	Thu 3/12/09	Wed 4/8/09	503	
505	3.7.8	Add to User Program	0 days	Wed 4/8/09	Wed 4/8/09	504	



Appendix J: BLEPS Requirements Document

LS-CAT 21-ID
Equipment Protection System: Requirements Document

Keith Brister
LS-CAT Operations Manager

051006

Introduction

This document lays out the requirements for the LS-CAT Sector 21 Equipment Protection System (EPS). This system includes the Beamline EPS (BLEPS) as well as the signal to be used in series with the Front End EPS (FEEPS) to generate the FE Shutter Permit. The intent of this system is to specify engineering controls to protect equipment from synchrotron radiation damage due to the following exceptional conditions:

1) High pressure present in any vacuum zone
2) Vacuum system gate valve closed
3) Low coolant flow rate in a non-PSS monitored component
4) Cryogenic System(s) Failure
    a) Low or absent LN2 flow
    b) High pressure in cryocooler closed system

The EPS will control 5 relay contacts to be interpreted as shutter permits:
1) 21-ID-FE Shutter Permit
2) 21-ID-D Shutter Permit
3) 21-ID-E Shutter Permit
4) 21-ID-F Shutter Permit
5) 21-ID-G Shutter Permit

In each case a closed contact grants said shutter permit. Wiring of the 21-ID-D/E/F/G shutter permits will follow "**Beamline Equipment Protection (BE-EPS) [sic] Interface Control Document for the Generation-3 Personnel Safety System (PSS)...**" document 4104-913053-00. Wiring of the 21-ID-FE shutter permit is assumed similar but had yet to be defined as of this writing.

Note that LS-CAT Sector 21-ID beamlines and station layouts place most of the EPS protected equipment before the first so-called integral shutter and hence this document mainly talks to the control of the FE Shutter Permit through the FEEPS.

General Requirements:

The system shall consist of two distinct parts:
1) PLC based control with the following properties:
    a) Logic states visible through EPICS Process Variables (PV's)
    b) Resetting of latched exceptions through EPICS PV's
2) EPICS based monitoring of analog signals used in this EPS
    a) Programmable alerts for various conditions may be generated here
    b) Not part of the logic chain that results in a shutter permit being granted or revoked

That is, all logic required to generate shutter permits and to control vacuum gate valves shall use PLC logic components while logging of analog signals may be done using EPICS based Input/Output Controllers (IOC's). LS-CAT requires that resets are possible from a remote location, preferably through one or more EPICS PV's.

Input Overrides

The ability to override inputs is considered essential. This may be accomplished through either authenticated engineering controls (such as password protected PLC logic) or specific administrative

controls (such as tagged toggle switches). Of the two, authenticated engineering controls are preferred. In either case, the LS-CAT Operations Manager shall approve conditions where input overriding may occur.

Code

The system shall include all components necessary to perform routine maintenance of the PLC firmware and EPICS software. In the case of the PLC code, complete documentation, hardware, software, and licenses shall be in the possession of the LS-CAT Operations Manager before the system is deemed operational. All software tools shall be installed on a computer system owned and controlled by LS-CAT and this installation shall be used to generate the final version of the code used in systems installed at LS-CAT.

Requirements and Rollout

The entire system does not need simultaneous implementation. Rollout may be accomplished in four steps starting with the main beamline. The following outline specifies the requirements and rollout:

1) Main Line, Top Priority
    a) Vacuum gate valves
        i) Any gate valve is not open revokes FE Shutter Permit
        ii) Gate valve commanded to close revokes FE Shutter Permit. When the FE Shutter Permit is in a granted state, the FE Shutters must be verified closed before the gate valve is actuated.
    b) Vacuum trip points
        i) Any gauge controller trip point activated for a set period of time (initially 1 second) revokes the appropriate shutter permit.
        ii) Trips are latched or logged with an indication of the cause
    c) Low flow to non-PSS monitored components
        i) Any low flow indicated for a set period of time (initial 1 second) revokes the appropriate shutter permit.
        ii) Trips are latched or logged with an indication of the cause
    d) Cryocooler system for 21-ID-D beam
        i) May require secondary system to generate relay contacts
        ii) Trips latched revoking FE Shutter Permit without delay
2) Main Line, Lower Priority
    a) Monitor Vacuum Levels using EPICS PV's
    b) Log vacuum levels
        i) Suppress noise to reduce log size
        ii) Record transients with 100 mSec precision and acuracy
        iii) Software settable alarms resulting in email alerts
    c) Log system state
3) Branch Lines, Top Priority
    a) Vacuum and flow as for the main line
    b) Possible LN2 monitoring of branch E monochromator. To be decided later.
4) Branch Lines, Lower Priority
    a) Monitoring as for the main line
    b) Logging as for the main line

Water Flows

Water flow through the following devices shall be monitored and appropriate trip points set:

1) Downstream Insertion Device Heatload Mask A at 46.75 m (MID46)
2) Downstream Insertion Device Heatload Mask B at 47.10 m (MID47)
3) Upstream Insertion Device Heatload Mask A at 58.32 m (MID58)
4) Upstream Insertion Device Heatload Mask B at 58.68 m (MID59)
5) Kohzu monochromator Compton Shield near 60.10 m (KCSID60)
6) Beryllium window located at 63.30 m (WID63)
7) Monochromator for E line (Future capability)
8) Monochromator for F line (Future capability)
9) Monochromator for G line (Future capability)
10) Beryllium windows (2-4) for white beam mirror tank in 21-ID-D (Future capability, controls 21-ID-D Shutter Permit)
11) White Beam Mirror(s) in 21-ID-D (Future capability, controls 21-ID-D Shutter Permit)

Drawings

LS-CAT will provide elevation drawings of the layout for use on panel and/or EPICS displays.

Vacuum Zones:

Table 1 lists the vacuum zones that we need to monitor and control. The chart includes the approximate positions of components in meters from the source point along with the name to be used. Branch lines include the designator E, F, or G. For example, CCGID27 refers to the Cold Cathode Gauge on the ID line approximately 27 meters from the source while IPID55G refers to the Ion Pump on the ID line approximately 55 meters from the source along the G branch. Items that include a name but no precise location, such as TCID27, are anticipated but have not yet been included in the beamline design.

Allowance must be made for the future inclusion of additional equipment and additional vacuum zones. In particular, additional cold cathode gauges will likely be added near the valves furthest from the ion pumps; we a near term expansion by a factor of 2 may be anticipated.

Trips in any of these zones shall remove the 21-ID-FE shutter permit with the exception of 21-ID-D mirror zones (10 and 11) that shall only remove the 21-ID-D shutter permit.

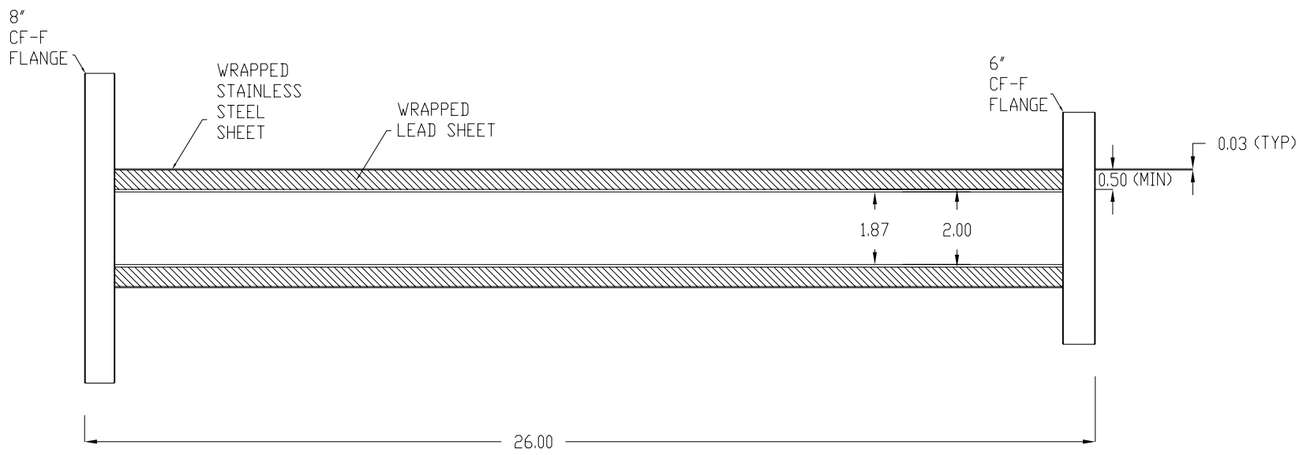
Table 1: Vacuum Zones

Zone	Isolates	Location	US Valve	DS Valve	DS2 Valve	US Be Window	DS Be Window	Ion Pump	Ion Gauge	Cold Cathode Gauge	Thermal Couple Gauge
1	--	21-ID-A	26.45	34.2				26.87	26.87		
			VID26	VID34				IPID27	IGID27	CCGID27	TCGID27
2	WBT	White Beam Transport	34.2	45.2				34.98	34.98	34.98	34.98
			VID34	VID45				IPID35	IGID35	CCGID35	TCGID35
3	Masks/DS-ID Aperture	21-ID-B	45.2	47.64				45.58	45.58		
			VID45	VID48				IPID46	IGID46	CCGID46	TCGID46

Zone	Isolates	Location	US Valve	DS Valve	DS2 Valve	US Be Window	DS Be Window	Ion Pump	Ion Gauge	Cold Cathode Gauge	Thermal Couple Gauge
4	Mono D3	21-ID-B	47.64	49.95	48.82 F			48	48		
			VID48	VID50	VID49F			IPID48	IGID48	CCGID48	TCGID48
5	Mono D2	21-ID-B	49.95	53.56	51.08 G			50.31	50.31		
			VID50	VID54	VID51G			IPID50	IGID50	CCGID50	TCGID50
6	Mono D1-1	21-ID-B/C	53.56	55.8	54.46 E			53.92	53.92		
			VID54	VID56	VID54E			IPID54	IGID54	CCGID54	TCGID54
7	US-ID Aperture	21-ID-C	55.8	59.46				57.87	57.87		
			VID56	VID59				IPID58	IGID58	CCGID58	TCGID58
8	Kohzu Mono	21-ID-C	59.46	60.83				59.96	59.96		
			VID59	VID61				IPID60	IGID60	CCGID60	TCGID60
9	Shutters	21-ID-C/D	60.83				63.26	61.83	61.83		
			VID61				WID63	IPID62	IGID62	CCGID62	TCGID62
10	21-ID-D Mirror (Vertical)	21-ID-D				64.01	65.66	64.81	64.81		
						WID64	WID66	IPID65	IGID65	CCGID65	TCGID65
11	21-ID-D Mirror (Horizontal)	21-ID-D				66	68	67	67		
						WID66	WID68	IPID67	IGID67	CCGID67	TCGID67
12	21-ID-F Mirror	21-ID-B	48.82 F	50.5 F				49.63 F	49.63 F		
			VID49F	VID51F				IPID50F	IGID50F	CCGID50F	TCGID50F
13	21-ID-F Shutters	21-ID-B/F	50.5 F				55.38 F	54.35 F	54.35 F		
			VID51F				WID55F	IPID54F	IGID54F	CCGID54F	TCGID54F
14	21-ID-G Mirror	21-ID-B	51.08 G	52.81 G				51.94 G	51.94 G		
			VID51G	VID53G				IPID52G	IGID52G	CCGID52G	TCGID52G
15	21-ID-G Shutters	21-ID-B/F/G	52.81 G				55.64 G	54.61 G	54.61 G		
			VID53G				WID56G	IPID55G	IGID55G	CCGID55G	TCGID55G
16	Mono D1-2	21-ID-C	54.46 E	60.10 E				59.25 E	59.25 E		
			VID54E	VID60E				IPID59E	IGID59E	CCGID59E	TCGID59E
17	21-ID-E Mirror	21-ID-C	60.10 E				61.91 E	61.06 E	61.06 E		
			VID60E				WID62E	IPID61E	IGID61E	CCGID61E	TCGID61E
18	21-ID-E Shutters	21-ID-C				61.91 E	63.88 E	62.85 E	62.85 E		
						WID62E	WID64E	IPID63E	IGID63E	CCGID63E	TCGID63E

Appendix K: Shielded Beampipe Drawings

REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY.

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED
ALL DIMENSIONS ARE IN INCHES
TOLERANCES
DECIMALS: .X ±.1, .XX ±.03, .XXX ±.010
ANGULAR ±.5°

DRAWN BY: B.STILLWELL — DATE: 05-15-06
CHECKED BY: — DATE:
DESIGNED BY: — DATE:
RESPONSIBLE ENGINEER: — DATE:
GROUP LEADER: — DATE:
APPROVED BY: — DATE:
RELEASE LEVEL: — MODEL VER: — DRWG VER:

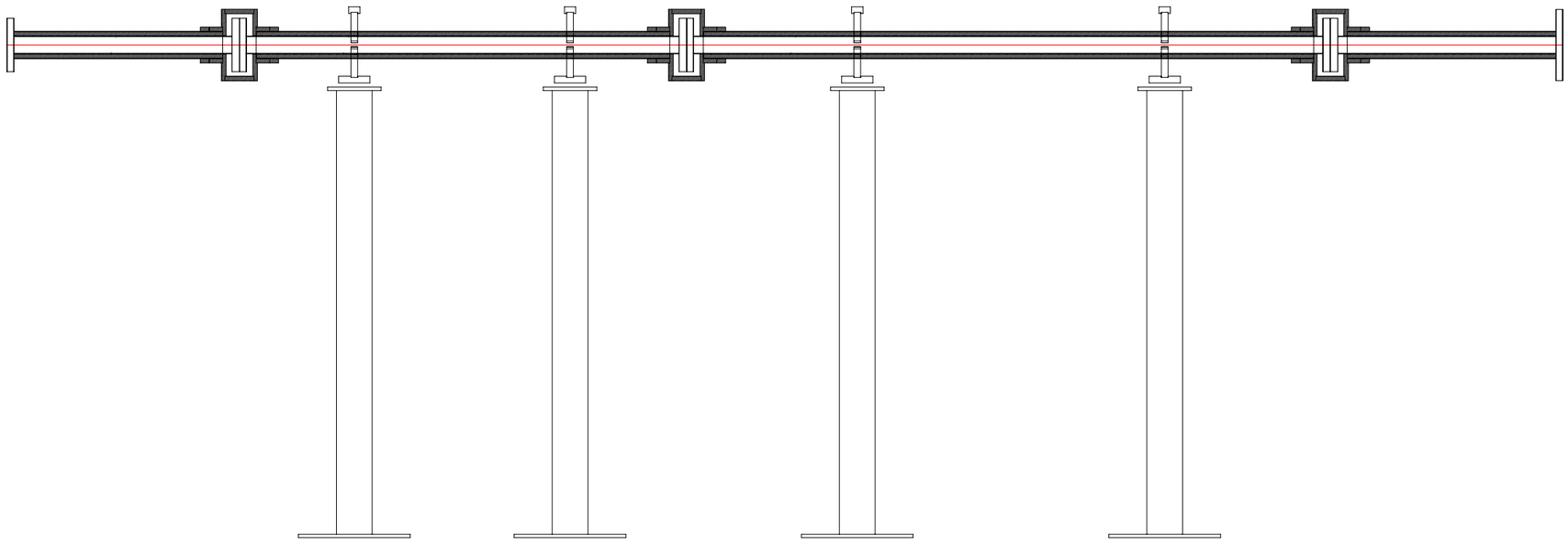
MODEL NAME: 21_SECTOR.ASM
ELECTRONIC FILE NAME:
MATERIAL:

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
SHIELDED PIPE - C TO END STATIONS

SIZE: B — FSCM No: — DRAWING NUMBER: 21-1.5-201000-09 — REV: 9.0
SCALE:

ZONE	REV.	DESCRIPTION	BY	APPROVED	DATE
	1	CHG TO ASSY DWG INCL FLANGE COVERS	BKS		03-27-06



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES TOLERANCES
DECIMALS X ±1 XX ±.03 XXX ±.010
ANGULAR ±.5°

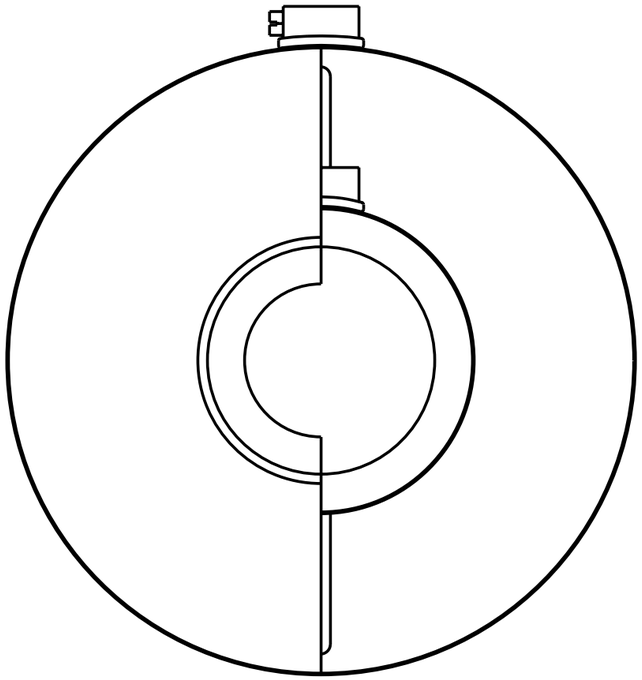
DRAWN BY: B.STILLWELL DATE: 05-15-06
CHECKED BY: DATE:
DESIGNED BY: DATE:
RESPONSIBLE ENGINEER: DATE:
GROUP LEADER: DATE:
APPROVED BY: DATE:

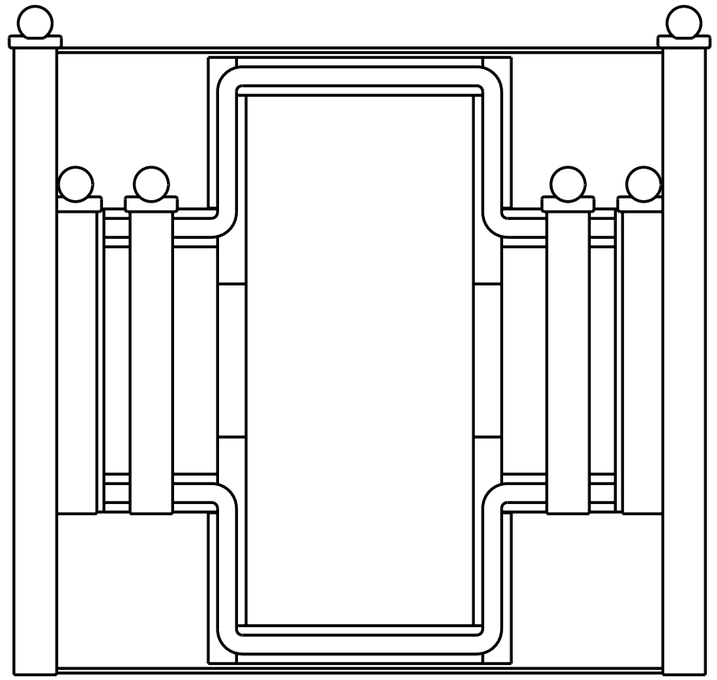
MODEL NAME: 21_SECTOR.ASM
RELEASE LEVEL: MODEL VER: DRVG VER:
ELECTRONIC FILE NAME: MATERIAL:

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE (LS-CAT)

TITLE:
21-ID-HUTCHES
SECTOR 21 BEAMLINE ASSEMBLY
SHIELDED TRANSPORT
ASSEMBLY / LEAD CROSS-SECTION

SIZE: B | FSCM No: | DRAWING NUMBER: 21-1.5-241000-09 | REV.: 9.0
SCALE:

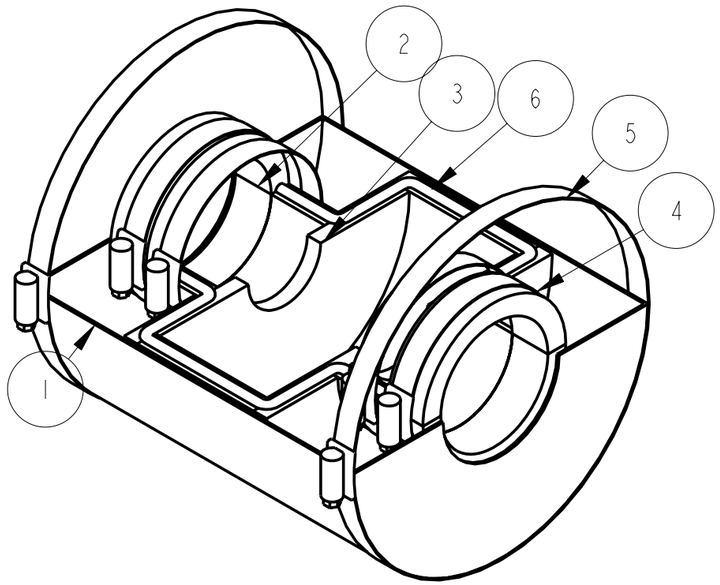




REVISIONS					
ZONE	REV	DESCRIPTION	BY	APPROVED	DATE

1 SOURCE OR EQUIVALENT
McMASTER-CARR
P.O. BOX 4355
CHICAGO, IL 60680-4355
630-883-0300

FORM LEAD ROD TO GASKET SHAPE AS NEEDED.
MATERIAL: LEAD ROD .25 DIA. PER FEDERAL SPEC. NO. QQ-L-201F GRADE 'C' (99.90% PURE LEAD)



ITEM	DRAWING / PART NUMBER	NOMENCLATURE OR DESCRIPTION	MATERIAL / SPEC	QTY
1	21-1.5-242002	COVER	LEAD, GRADE "C"	2
2	21-1.5-242001	END COLLAR	LEAD, GRADE "C"	4
3	21-1.5-242100	HALF SHIELD ASSEMBLY	LEAD, GRADE "C"	2
1 4	45945K35	HOSE CLAMP		4
1 5	45945K41	HOSE CLAMP		2
6		LEAD ROD, D=0.25"	SEE NOTE	2

PARTS LIST / BILL OF MATERIALS

UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES
TOLERANCES

DECIMALS: .X ±.1, .XX ±.03, .XXX ±.010
ANGULAR: ±.5°

SURFACE ROUGHNESS 125

REMOVE ALL BURRS AND BREAK SHARP EDGES .03 MAX. SURFACE TEXTURE IN ACCORDANCE WITH LATEST ASME B46.1-2002 DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH LATEST ASME Y14.5M-1994

MODEL NAME: 21_IDG_SHIELDED_TRANSPORT0000.ASM

ELECTRONIC FILE NAME: 141

DRAWN BY: B.STILLWELL	DATE 3-15-2005	
CHECKED BY:	DATE	
DESIGNED BY: B.STILLWELL	DATE 3-15-2005	
RESPONSIBLE ENGINEER:	DATE	
GROUP LEADER:	DATE	
APPROVED BY:	DATE	
RELEASE LEVEL:	MODEL VER:	DRWG VER:

MATERIAL: SEE PARTS LIST

THIS DRAWING IS THE PROPERTY OF
NORTHWESTERN UNIVERSITY
LIFE SCIENCES COLLABORATIVE

TITLE:
21 ID-G
MONO BEAM TRANSPORT
FLANGE ENCLOSURE

SIZE	FSCM No	DRAWING NUMBER:	REV.
B		21-1.5-242000	0.0

SCALE: 1:2 DO NOT SCALE DRAWING

Appendix L: Beryllium Window Coating

DuraBeryllium™ Windows

Coated and Uncoated Beryllium Windows

There are two types of beryllium windows available: one is the standard beryllium window and the other a proprietary DuraBeryllium™ window. DuraBeryllium is a proprietary coating process in which beryllium foil is coated with our patented DuraCoat™ process to make it vacuum-tight and chemically resistant. This foil is protected with a refractory low-Z coating that makes it resistant to atmospheric moisture and chemicals.

Foil Sizes

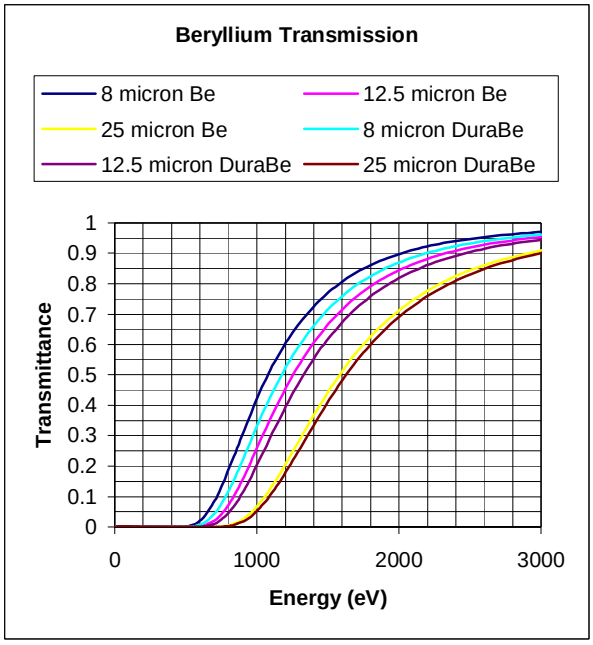
Thickness	Diameter
8 micron	9.2 mm
8 micron	12.0 mm
12.5 micron	12.0 mm
12.5 micron	16.0 mm
25 micron	16.0 mm

Please contact us if these sizes do not meet your needs. Custom sizes are available upon customer request.

Transmission Performance

DuraCoat, our proprietary refractory coating material, consists entirely of elements lighter than sodium, resulting in very low x-ray absorption.

Transmission Performance



Differential Pressure

The mechanical properties of DuraBeryllium are essentially those of beryllium.

Mounting

Please contact MOXTEK for meeting your needs for active area, pressure differential, and x-ray transmission.
We can supply you with coated beryllium sheet or foil or, if you prefer, we can supply you with complete window assemblies.

Temperature Performance

Brazed foil can withstand 650°C in vacuum or 500°C in air while epoxied foils are dependent upon the limits of the epoxy. Epoxy mounted Beryllium and DuraBeryllium windows are capable of exposure to over 100°C at a differential pressure of 1 atm with approved mount designs.

MOXTEK

Orem, Utah USA

DuraBeryllium™ Windows

Cleanability

DuraBeryllium windows are cleanable and the cleaning solvent is not a health hazard. Any ultraclean solvent can be used. Isopropyl, methanol or ethanol are recommended. Contact factory for details.

Vacuum Tightness

DuraBeryllium meets the most demanding leak specifications.

Leak Rate Analysis in mbar L/sec of Helium

Sample Number	MLQ1	MLQ2	MLQ3
Initial Leak Rate	1×10^{-7}	6×10^{-7}	1×10^{-10}
DuraCoat™ Leak Rate	$<1 \times 10^{-10}$	$<1 \times 10^{-10}$	$<1 \times 10^{-10}$
Current Leak Rate	$<1 \times 10^{-10}$	$<1 \times 10^{-10}$	$<1 \times 10^{-10}$

Product Data

Hardness: 2000 Vickers
Electrical Resistivity: <4x104 ohm-cm

Chemical compatibility

The DuraBeryllium windows are resistant to most solvents, acids, and bases while the beryllium windows are not. DuraBeryllium is the perfect material for x-ray detector, source and process control windows. While its x-ray transmission is virtually identical to that of bare beryllium, this high-quality foil is protected by DuraCoat, a proprietary refractory material that makes it not only vacuum tight but also highly resistant to moisture, chemicals, and abrasion, or TCE. Since cleaning products are not contaminated with beryllium, the process creates no hazardous waste. And your clean windows, using DuraBeryllium foil as thin as 5 µm, will increase the consistency and reliability of your x-ray instruments.

Etching Reagents[+]	Nominal pH	Etch Rates (nm/min @25°C)
HF	0.80	negligible
H_2SO_4	1.38	negligible
HNO_3	1.17	1.09
Acetic Acid	1.80	negligible
H_3PO_4	1.17	0.146
Aqua Regia [++]	1.00	0.14
$NH_4OH + H_2O_2$	11.40	negligible
NaOH Solution*	13.70	negligible
Ferricyanide**	13.70	165
Permanganate [#]	13.70	900

+ All solutions are concentrated unless otherwise indicated.
++ Solution is 1M HNO3, 3M HCl, 1M H20
* Solution is 0.5M NaOH, 0.6M H_2O_2
** Solution is 0.6M $K_3Fe(CN)_6$, 0.5M NaOH, 0.02M oxalic acid
[#]Solution is 0.6M $KmNO_4$, 0.5M KOH

Warranty

The standard warranty is 1 year under normal operating conditions. Any imposed damage will not be warranted.

Delivery Lead Time

Standard sizes: 3 weeks a.r.o.
Nonstandard Sizes: 6 weeks a.r.o.

Cost

Contact MOXTEK for a quotation.

Contact

For more information call MOXTEK at 801-225-0930. www.moxtek.com